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PALI: A JOINTLY-SCALED MULTILINGUAL LANGUAGE-IMAGE MODEL

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ABSTRACT

Effective scaling and a flexible task interface enable large language models to excel at many tasks. We present **PaLI** (**Pa**thways **L**anguage and **I**mage model), a model that extends this approach to the joint modeling of language and vision. PaLI generates text based on visual and textual inputs, and with this interface performs many vision, language, and multimodal tasks, in many languages. To train PaLI, we make use of large pre-trained encoder-decoder language models and Vision Transformers (ViTs). This allows us to capitalize on their existing capabilities and leverage the substantial cost of training them. We find that joint scaling of the vision and language components is important. Since existing Transformers for language are much larger than their vision counterparts, we train a large, 4-billion parameter ViT (ViT-e) to quantify the benefits from even larger-capacity vision models. To train PaLI, we create a large multilingual mix of pre-training tasks, based on a new image-text training set containing 10B images and texts in over 100 languages. PaLI achieves state-of-the-art in multiple vision and language tasks (such as captioning, visual question-answering, scene-text understanding), while retaining a simple, modular, and scalable design.

1 INTRODUCTION

Increasing neural network capacity has been a successful trend in the modeling of language and vision tasks. On the language side, models such as T5 (Raffel et al., 2020), GPT-3 (Brown et al., 2020), Megatron-Turing (Shoeybi et al., 2019), GLaM (Du et al., 2022), Chinchilla (Hoffmann et al., 2022), and PaLM (Chowdhery et al., 2022) have shown significant advantages from training large Transformers on large amounts text data. On the vision side, CNNs (Mahajan et al., 2018; Huang et al., 2019; Kolesnikov et al., 2020), Vision Transformers (Dosovitskiy et al., 2021), and other models (Tolstikhin et al., 2021; Riquelme et al., 2021) have seen similar benefits from scale (Zhai et al., 2022a), albeit to a lesser extent than in language. Language-and-vision modeling has followed a similar trend, e.g., SimVLM (Wang et al., 2021), Florence (Yuan et al., 2021), CoCa (Yu et al., 2022), GIT (Wang et al., 2022a), BEiT-3 (Wang et al., 2022c), and Flamingo (Alayrac et al., 2022).

We introduce PaLI, a model that performs image-only, language-only, and image+language tasks across many languages, using a single "image-and-text to text" interface. A key characteristic of PaLI is a more balanced parameter share between the language and vision components, with more capacity to the vision backbone yielding large gains in performance. Another key ingredient to PaLI is the reuse of large unimodal backbones for language and vision modeling, in order to transfer existing capabilities and reduce training cost. On the language side, we reuse the 13B-parameter model mT5-XXL (Xue et al., 2021), which already packages language understanding and generation capabilities. We show that these capabilities are maintained and extended into a multimodal setting. On the vision side, in addition to reusing the 2B-parameter ViT-G model (Zhai et al., 2022a), we

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PALI：一个联合缩放的多语言语言-图像模型

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摘要

有效的规模扩展和灵活的任务接口使大型语言模型能够在许多任务中表现出色。我们提出了 PaLI（Pathways Language and Image model，Pathways 语言与图像模型），这是一个将这种方法扩展到语言与视觉联合建模的模型。PaLI 基于视觉和文本输入生成文本，并通过这一接口执行多种视觉、语言和多模态任务，且支持多种语言。为了训练 PaLI，我们利用了大规模预训练的编码器-解码器语言模型和视觉 Transformer（ViT）。这使我们能够充分利用它们已有的能力，并发挥训练这些模型所投入的大量成本的价值。我们发现，视觉和语言组件的联合规模扩展非常重要。由于现有的语言 Transformer 远大于其视觉对应模型，我们训练了一个拥有 40 亿参数的大型 ViT（ViT-e），以量化更大容量视觉模型所带来的收益。为了训练 PaLI，我们构建了一个大型多语言预训练任务混合体，该混合体基于一个新的图文训练集，其中包含 100 多种语言的 100 亿张图像及文本。PaLI 在多个视觉和语言任务（如图像描述生成、视觉问答、场景文本理解）上达到了最先进水平，同时保持了简单、模块化且可扩展的设计。

1 引言

增加神经网络容量已成为语言和视觉任务建模中的一项成功趋势。在语言方面，T5（Raffel 等，2020）、GPT-3（Brown 等，2020）、Megatron-Turing（Shoeybi 等，2019）、GLaM（Du 等，2022）、Chinchilla（Hoffmann 等，2022）和 PaLM（Chowdhery 等，2022）等模型表明，在大量文本数据上训练大型 Transformer 具有显著优势。在视觉方面，CNN（Mahajan 等，2018；Huang 等，2019；Kolesnikov 等，2020）、Vision Transformers（Dosovitskiy 等，2021）以及其他模型（Tolstikhin 等，2021；Riquelme 等，2021）也从规模扩大中获得了类似收益（Zhai 等，2022a），尽管程度不如语言领域。语言与视觉建模也遵循了类似趋势，例如 SimVLM（Wang 等，2021）、Florence（Yuan 等，2021）、CoCa（Yu 等，2022）、GIT（Wang 等，2022a）、BEiT-3（Wang 等，2022c）和 Flamingo（Alayrac 等，2022）。

我们介绍了 PaLI，这是一种使用单一“图像和文本到文本”接口，在多种语言中执行仅图像、仅语言以及图像+语言任务的模型。PaLI 的一个关键特征是语言组件与视觉组件之间的参数分配更加均衡，并且为视觉主干提供更多容量，从而带来显著的性能提升。PaLI 的另一个关键要素是复用用于语言和视觉建模的大型单模态主干，以迁移现有能力并降低训练成本。在语言方面，我们复用了 130 亿参数的 mT5-XXL 模型（Xue 等，2021），该模型已经具备语言理解和生成能力。我们表明，这些能力在多模态设置中得以保持并扩展。在视觉方面，除了复用 20 亿参数的 ViT-G 模型（Zhai 等，2022a）之外，我们

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train a 4B-parameter model, which we call ViT-e ("enormous"). ViT-e achieves good performance on image-only tasks, such as 90.9% ImageNet fine-tuning, and 84.9% on ObjectNet (Barbu et al., 2019).

We find benefits from jointly scaling both the vision and the language components, with vision providing a better return on investment (accuracy improvement per parameter/FLOP). As a result, the capacity of our largest PaLI model, PaLI-17B, is distributed relatively equitably between the two modalities, with the ViT-e component accounting for about 25% of the total parameter count. This is not always the case for prior work in large-capacity vision and language modeling (Wang et al., 2022a; Alayrac et al., 2022), due to the prior scale mismatch between vision and language backbones. We enable knowledge-sharing between multiple image and/or language tasks by casting them into a generalized VQA-like task. We frame all tasks using an "image+query to answer" modeling interface, in which both the query and answer are expressed as text tokens. This allows PaLI to capitalize on transfer learning across tasks, and enhance language-and-image understanding capabilities in a wide range of vision and language problems: image captioning, visual question-answering, scene-text understanding, and others (Figure 1).

To train PaLI-17B, we build a new high-volume image-and-language dataset WebLI, which consists of 10 billion images and tens of billions of image-text pairs. Importantly, the WebLI dataset contains text in over 100 languages. By training the model to perform multimodal tasks in many languages, we greatly increase the task diversity, and test the model's ability to effectively scale both across tasks and across languages. As a reference for future usage, we provide a data card to report information about the WebLI and its construction.

PaLI-17B achieves state-of-the-art (SOTA) results on multiple benchmarks, outperforming some strong models. Specifically, PaLI outperforms recent and concurrent models on the long-standing COCO Captioning benchmark (Chen et al., 2015), with **149.1** CIDEr score on the Karpathy split (Karpathy & Fei-Fei, 2015). PaLI also achieves a new SOTA of **84.3%** on VQAv2 (Goyal et al., 2017) while using an open-vocabulary text generative setting that is similar to Flamingo (Alayrac et al., 2022). This result outperforms even models evaluated in a fixed-vocabulary classification setting, e.g. CoCa (Yu et al., 2022), SimVLM (Wang et al., 2021), BEiT-3 (Wang et al., 2022c). Last but not least, our work provides a scaling roadmap for future multimodal models. Our results support the conclusion that scaling the components of each modality yields better performance compared to more skewed alternatives. Model scaling is also important for language-image understanding in multiple languages. In summary, our contributions are the following:

- We design a simple, modularized and scalable sequence-to-sequence learning architecture that can be efficiently trained by reusing existing Transformer-based unimodal checkpoints.
- We perform joint scaling on both the language and vision components for a wide range of parameters, and show no saturation of performance on both components for the largest model size we consider, PaLI-17B. More importantly, we show that multimodal performance greatly benefits from scaling the vision component beyond the previous-largest ViT, which provides a scaling roadmap for future vision & language models.
- We empirically validate that a mixture-of-objectives benefits the performance of large vision & language models.
- We scale up pre-training data to include over 100 languages, and train a large-capacity multilingual multimodal model. We show that a properly-scaled model can handle well a large number of languages, while still achieving SOTA performance on English-only tasks.

2 Related Work

Pre-trained models have proven effective in both vision (Dosovitskiy et al., 2021; Zhai et al., 2022a) and language (Raffel et al., 2020; Brown et al., 2020) tasks. Image-text pre-training has also become the default approach to tackle V&L tasks (Tan & Bansal, 2019; Chen et al., 2020; Zhang et al., 2021; Cho et al., 2021; Hu et al., 2022). While benefiting from the text representation and generation capabilities of the Transformer architecture, some of these vision-language models rely on external systems (such as Fast(er) R-CNN (Ren et al., 2015)) to provide detected object names and the related precomputed dense features. Such reliance limited the capability to scale up the model and performance. With the introduction of Vision Transformers (Dosovitskiy et al., 2021), vision and

训练了一个 40 亿参数的模型，我们称之为 ViT-e（“enormous”，意为“巨大”）。ViT-e 在仅图像任务上取得了良好性能，例如 ImageNet 微调达到 90.9%，在 ObjectNet（Barbu 等，2019）上达到 84.9%。我们发现，同时扩展视觉和语言组件能够带来收益，其中视觉组件提供了更高的投资回报（每参数/FLOP 带来的准确率提升）。因此，我们最大的 PaLI 模型 PaLI-17B 的容量在两种模态之间分配得相对均衡，其中 ViT-e 组件约占总参数量的 25%。由于视觉和语言骨干网络之间先前存在规模不匹配，这在以往的大容量视觉与语言建模工作中并非总是如此（Wang et al., 2022a; Alayrac et al., 2022）。我们通过将多个图像和/或语言任务转化为一种广义的类 VQA 任务，实现了它们之间的知识共享。我们使用“图像+查询到答案”的建模接口来表述所有任务，其中查询和答案都表示为文本标记。这使 PaLI 能够利用跨任务迁移学习，并在广泛的视觉与语言问题中增强语言-图像理解能力：图像描述、视觉问答、场景文本理解等（图 1）。

为了训练 PaLI-17B，我们构建了一个新的大规模图像与语言数据集 WebLI，其中包含 100 亿张图像和数百亿个图文对。重要的是，WebLI 数据集包含超过 100 种语言的文本。通过训练模型以多种语言执行多模态任务，我们极大地增加了任务多样性，并测试了模型在跨任务和跨语言方面有效扩展的能力。作为未来使用的参考，我们提供了一份数据卡，用于报告有关 WebLI 及其构建的信息。

PaLI-17B 在多个基准测试中取得了最先进（SOTA）的结果，优于一些强大的模型。具体而言，PaLI 在长期使用的 COCO Captioning 基准测试（Chen 等，2015）上优于近期及同期模型，在 Karpathy 划分（Karpathy & Fei-Fei，2015）上取得了 149.1 的 CIDEr 分数。PaLI 还在 VQAv2（Goyal 等，2017）上取得了 84.3% 的新 SOTA，同时使用了类似于 Flamingo（Alayrac 等，2022）的开放词汇文本生成设置。这一结果甚至优于在固定词汇分类设置下评估的模型，例如 CoCa（Yu 等，2022）、SimVLM（Wang 等，2021）、BEiT-3（Wang 等，2022c）。最后但同样重要的是，我们的工作为未来的多模态模型提供了一条扩展路线图。我们的结果支持这样一个结论：与更偏斜的替代方案相比，扩展各模态的组件能带来更好的性能。模型扩展对于多语言的语言-图像理解也很重要。总之，我们的贡献如下：

- 我们设计了一种简单、模块化且可扩展的序列到序列学习架构，可通过复用现有基于 Transformer 的单模态检查点进行高效训练。
- 我们对语言和视觉组件在广泛的参数范围内进行联合扩展，并表明在我们考虑的最大模型规模 PaLI-17B 上，这两个组件的性能都没有出现饱和。更重要的是，我们表明，将视觉组件扩展到超过此前最大的 ViT，可极大提升多模态性能，这为未来的视觉与语言模型提供了扩展路线图。
- 我们通过实证验证了混合目标有助于提升大型视觉与语言模型的性能。
- 我们将预训练数据扩展到涵盖 100 多种语言，并训练了一个大容量的多语言多模态模型。我们表明，经过适当扩展的模型能够很好地处理大量语言，同时仍然在仅英语任务上取得 SOTA 性能。

2 相关工作

预训练模型已被证明在视觉（Dosovitskiy et al., 2021; Zhai et al., 2022a）和语言（Raffel et al., 2020; Brown et al., 2020）任务中都很有效。图像-文本预训练也已成为处理 V&L 任务的默认方法（Tan & Bansal, 2019; Chen et al., 2020; Zhang et al., 2021; Cho et al., 2021; Hu et al., 2022）。虽然受益于 Transformer 架构的文本表示和生成能力，但其中一些视觉-语言模型依赖外部系统（如 Fast(er) R-CNN（Ren et al., 2015））来提供检测到的对象名称及相关的预计算密集特征。这种依赖限制了模型和性能的扩展能力。随着视觉 Transformer（Dosovitskiy et al., 2021）的引入，视觉和

language modalities can be jointly modeled by transformers in a more scalable fashion (Yuan et al., 2021; Yu et al., 2022; Wang et al., 2022a; Alayrac et al., 2022).

One approach for image-text pre-training is contrastive learning (Radford et al., 2021; Jia et al., 2021). Zhai et al. (2022b) show that with a pre-trained and locked vision model, one needs to train only a paired text encoder model to get good language embeddings. Yuan et al. (2021) extend contrastively pre-trained models to more downstream tasks with task-specific adaptations. Beside image and language, MERLOT (Zellers et al., 2021) has found success in video understanding and reasoning through video-language pretraining. Another approach is to train vision-language models to generate text autoregressively (Donahue et al., 2015; Vinyals et al., 2015). This approach has the advantage of a unified formulation of vision-language tasks as a text generation problem (Cho et al., 2021; Wang et al., 2022b; Piergiovanni et al., 2022b). In Cho et al. (2021), the vision-language model is trained to recover masked text. SimVLM (Wang et al., 2021) propose an image-language pre-training approach leveraging a prefix language modeling objective. The unified framework OFA (Wang et al., 2022b) extends the generation capability to include text to image generation. Concurrent with our work, Unified-IO (Lu et al., 2022) further scaled up the number of objectives and tasks and demonstrated decent performance across the board through only multi-task pre-training without task-specific fine-tuning.

Recent works explore joint vision and language modeling with increased model capacity. CoCa (Yu et al., 2022) pre-trains a 2.1B image-text encoder-decoder model jointly with contrastive loss and generative loss. GIT (Wang et al., 2022a) trains a model consisting of a single image encoder and a text decoder with a captioning (generative) loss, where the image encoder is pre-trained with contrastive loss. In their latest version, GIT2, the model size is scaled up to 5.1B, with the majority of parameters on the vision side (4.8B). BEiT-3 (Wang et al., 2022c) presents an architecture with vision, language, and vision-language experts, operating with a shared multi-head self-attention followed by a switch for "expert" modules, resulting in a 1.9B model trained from scratch on a variety of public image, text and image-text datasets. Flamingo (Alayrac et al., 2022) is built upon a 70B language model (Hoffmann et al., 2022) as a decoder-only model whose majority of parameters are frozen in order to preserve language-generation capabilities, along with a 435M vision encoder.

Vision-language pre-training also benefits from automatically mined and filtered large-scale datasets such as Conceptual Captions (CC3M) and CC12M (Sharma et al., 2018; Changpinyo et al., 2021), with 3 and 12 million image-text pairs, respectively. With more relaxed filtering, LEMON (Hu et al., 2022) collected a larger dataset with 200M examples, which is further expanded to 800M examples in GIT (Wang et al., 2022a). For better scaling the model, larger, noisier datasets such as the ALIGN dataset (1.8B) (Jia et al., 2021) have been constructed, which has benefited SimVLM (Wang et al., 2021) and CoCa (Yu et al., 2022). While these image-text datasets have fueled the foundational V&L models with state-of-the-art performance, they are English-only, and there has been limited attempts to create datasets not English-centric and unlock the multilingual capability of these models.

3 THE PALI MODEL

3.1 ARCHITECTURE

With PaLI, we aim to perform both unimodal (language, vision) and multimodal (language and vision) tasks. Typically, many of these tasks are best handled by different models. For instance, image classification, and many formulations of VQA, require predicting elements from a fixed set, while language-only tasks and image captioning require open-vocabulary text generation. Similar to the recent work OFA (Wang et al., 2022b) and a concurrent work (Lu et al., 2022), we resolve this by using a sufficiently general interface for all tasks considered: the model accepts as input an image and text string, and generates text as output. The same interface is used both during pre-training and fine-tuning. Since all tasks are performed with the same model, i.e. we have no tasks-specific parameters or "heads", we use text-based prompts to indicate to the model which task to perform.

Figure 1 shows a high-level schematic of the model architecture. At its core, PaLI has a text encoder-decoder Transformer (Vaswani et al., 2017). To include vision as input, the text encoder is fed with a sequence of visual "tokens": output patch features of a Vision Transformer which takes as input an image. No pooling is applied to the output of the Vision Transformer before passing the visual tokens to the encoder-decoder model via cross-attention. We reuse previously trained unimodal checkpoints.

语言模态可以通过 Transformer 以更具可扩展性的方式进行联合建模（Yuan 等，2021；Yu 等，2022；Wang 等，2022a；Alayrac 等，2022）。

图文预训练的一种方法是对比学习（Radford et al., 2021；Jia et al., 2021）。Zhai et al.（2022b）表明，在使用预训练且冻结的视觉模型时，只需训练一个配对的文本编码器模型即可获得良好的语言嵌入。Yuan et al.（2021）通过任务特定的适配，将对比式预训练模型扩展到更多下游任务。除图像和语言外，MERLOT（Zellers et al., 2021）通过视频-语言预训练，在视频理解和推理方面取得了成功。另一种方法是训练视觉-语言模型以自回归方式生成文本（Donahue et al., 2015；Vinyals et al., 2015）。这种方法的优势在于可将视觉-语言任务统一表述为文本生成问题（Cho et al., 2021；Wang et al., 2022b；Piergiovanni et al., 2022b）。在 Cho et al.（2021）中，视觉-语言模型被训练用于恢复被遮蔽的文本。SimVLM（Wang et al., 2021）提出了一种利用前缀语言建模目标的图像-语言预训练方法。统一框架 OFA（Wang et al., 2022b）将生成能力扩展到包括文本到图像生成。与我们的工作同期，Unified-IO（Lu et al., 2022）进一步扩大了目标和任务的数量，并展示了仅通过多任务预训练、无需任务特定微调即可在各方面取得不错性能。

近期研究探索了在更大模型容量下的联合视觉与语言建模。CoCa（Yu 等，2022）使用对比损失和生成损失联合预训练了一个 2.1B 的图文编码器-解码器模型。GIT（Wang 等，2022a）训练了一个由单一图像编码器和文本解码器组成的模型，采用图像描述生成损失，其中图像编码器通过对比损失进行预训练。在其最新版本 GIT2 中，模型规模扩展到 5.1B，其中大多数参数位于视觉侧（4.8B）。BEiT-3（Wang 等，2022c）提出了一种包含视觉、语言以及视觉-语言专家的架构，采用共享的多头自注意力机制，随后通过切换机制选择“专家”模块，最终形成一个 1.9B 的模型，并在多种公开图像、文本和图文数据集上从头训练。Flamingo（Alayrac 等，2022）基于一个 70B 的语言模型（Hoffmann 等，2022）构建，是一个仅解码器模型；为保留语言生成能力，其大部分参数被冻结，同时配备一个 435M 的视觉编码器。

视觉-语言预训练也受益于自动挖掘和过滤的大规模数据集，如 Conceptual Captions（CC3M）和 CC12M（Sharma 等，2018；Changpinyo 等，2021），它们分别包含 300 万和 1200 万个图像-文本对。通过更宽松的过滤，LEMON（Hu 等，2022）收集了一个包含 2 亿个样本的更大数据集，并在 GIT（Wang 等，2022a）中进一步扩展到 8 亿个样本。为了更好地扩展模型，人们构建了更大、噪声更多的数据集，如 ALIGN 数据集（18 亿）（Jia 等，2021），这使 SimVLM（Wang 等，2021）和 CoCa（Yu 等，2022）受益。尽管这些图像-文本数据集推动了具有最先进性能的基础 V&L 模型的发展，但它们仅限于英语，而且在创建非英语中心的数据集以及释放这些模型的多语言能力方面，尝试仍然有限。

3 PALI 模型

3.1 架构

借助 PaLI，我们旨在同时执行单模态（语言、视觉）和多模态（语言与视觉）任务。通常，这些任务中的许多都最适合由不同的模型来处理。例如，图像分类以及许多 VQA 形式需要从固定集合中预测元素，而纯语言任务和图像描述生成则需要开放词汇的文本生成。类似于近期工作 OFA（Wang et al., 2022b）以及同期工作（Lu et al., 2022），我们通过为所有考虑的任务使用足够通用的接口来解决这一问题：模型接受一张图像和一个文本字符串作为输入，并生成文本作为输出。在预训练和微调期间都使用相同的接口。由于所有任务都由同一个模型执行，也就是说，我们没有特定于任务的参数或“头”，因此我们使用基于文本的提示来指示模型要执行哪项任务。

图1展示了该模型架构的高层示意图。PaLI 的核心是一个文本编码器-解码器 Transformer（Vaswani 等，2017）。为了将视觉作为输入，文本编码器会接收一系列视觉“词元”：即视觉 Transformer 的输出图像块特征，而该视觉 Transformer 以图像作为输入。在通过交叉注意力将视觉词元传递给编码器-解码器模型之前，不会对视觉 Transformer 的输出应用池化。我们复用了先前训练好的单模态检查点。

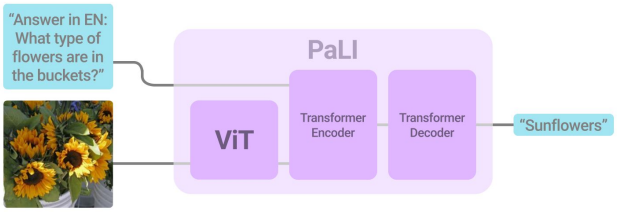


Figure 1: The PaLI main architecture is simple and scalable. It uses an encoder-decoder Transformer model, with a large-capacity ViT component for image processing.

For the text encoder-decoder, we reuse pre-trained mT5 (Xue et al., 2021) models, while for the image encoder, we reuse large vanilla ViT models (Dosovitskiy et al., 2021; Zhai et al., 2022a).

The visual component We introduce and train the largest vanilla ViT architecture to date, named **ViT-e**. ViT-e has the same architecture and uses the same training recipe as the 1.8B parameter ViT-G model (Zhai et al., 2022a), while scaling to 4B parameters. The only other difference is that we apply learning rate cool-down twice, once with and once without inception crop augmentation, and average ("soup") the weights of the two models as in Wortsman et al. (2022). While the scaling laws have been studied in both the vision domain and the language domain, scaling behaviour is less explored in combined vision and language models. Scaling up vision backbones leads to saturating gains on classification tasks such as ImageNet (Zhai et al., 2022a). We further confirm this, observing that ViT-e is only marginally better than ViT-G on ImageNet (Table 17). However, we observe substantial performance improvements from ViT-e on vision-language tasks in PaLI (Section 4). For example, ViT-e yields almost three additional CIDEr points over ViT-G on the COCO captioning task. This hints towards future headroom for vision-language tasks with even larger ViT backbones.

The language component We adopt the mT5 (Xue et al., 2021) backbone as our language component. We experiment using the pre-trained mT5-Large (1B parameters) and mT5-XXL (13B parameters), from which we initialize the language encoder-decoder of PaLI. We train on a mix of many tasks, including pure language understanding tasks (Section A.2). This helps avoid catastrophic forgetting of the mT5's language understanding and generation abilities. As a result, PaLI-17B continues to achieve similar levels of language-understanding accuracy on both the English benchmarks (Wang et al., 2019a) and across languages measured by the XTREME benchmark (Hu et al., 2020) (Section 4).

The overall model Three model sizes are considered (Table 7): 1) PaLI-3B, where the language component is initialized from mT5-Large (Xue et al., 2021) (1B parameters), and the vision component is ViT-G (Zhai et al., 2022a) (1.8B parameters). 2) PaLI-15B, where the language component is initialized from mT5-XXL (Xue et al., 2021) (13B parameters), and the vision component is ViT-G (1.8B parameters). 3) PaLI-17B, where the language model is initialized from mT5-XXL, and the vision component is the newly-trained ViT-e model (4B parameters).

3.2 DATA

WebLI Dataset Scaling studies for deep learning show that larger models require larger datasets to train effectively (Hoffmann et al., 2022; Kaplan et al., 2020; Zhai et al., 2022a). To unlock the potential of multilingual image-language pre-training, we introduce WebLI, a multilingual image-language dataset built from images and texts available on the public web. WebLI scales up the image language data collection from English-only datasets to 109 languages, which enables us to pre-train PaLI multilingually, and perform downstream tasks across many languages. The data collection process is similar to those reported in (Jia et al., 2021; Zhai et al., 2022b). Due to the abundance of multilingual content on the internet, the collection process for the WebLI dataset can be scaled to cover 10 billion images and 12 billion alt-texts. In addition to annotation with web text, we use publicly available automatic service to extract OCR annotations on all images, resulting in 29 billion image-OCR pairs. To balance quality and retain scale, we filter the dataset to the highest quality subset retaining only the top 10% scoring of the original WebLI image-text pairs (about 1B examples), which we use to train PaLI. Examples and statistics for the WebLI corpus and a complete datasheet (Pushkarna et al., 2022) are shown in Appendix B (Figure 4) and G.

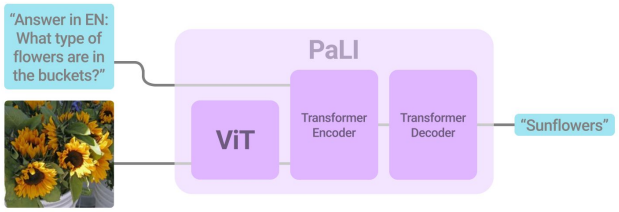


图 1：PaLI 的主要架构简单且可扩展。它使用编码器-解码器 Transformer 模型，并配备大容量 ViT 组件用于图像处理。

对于文本编码器-解码器，我们复用了预训练的 mT5（Xue 等，2021）模型；而对于图像编码器，我们复用了大型原始 ViT 模型（Dosovitskiy 等，2021；Zhai 等，2022a）。

视觉组件 我们引入并训练了迄今为止最大的原生 ViT 架构，命名为 ViT-e。ViT-e 采用与 18 亿参数 ViT-G 模型（Zhai 等，2022a）相同的架构，并使用相同的训练方案，同时将规模扩展到 40 亿参数。唯一的其他区别是，我们进行了两次学习率冷却：一次使用 inception crop 数据增强，另一次不使用；并按照 Wortsman 等（2022）的方法对两个模型的权重进行平均（“soup”）。尽管缩放定律已在视觉领域和语言领域中得到研究，但在视觉与语言结合的模型中，缩放行为的探索仍然较少。扩大视觉骨干网络的规模会导致在 ImageNet 等分类任务上的收益趋于饱和（Zhai 等，2022a）。我们进一步证实了这一点，观察到 ViT-e 在 ImageNet 上仅略优于 ViT-G（表 17）。然而，我们观察到，在 PaLI 的视觉-语言任务中，ViT-e 带来了显著的性能提升（第 4 节）。例如，在 COCO 图像描述任务上，ViT-e 相比 ViT-G 额外提升了近 3 个 CIDEr 分数。这表明，使用更大的 ViT 骨干网络，视觉-语言任务未来仍有进一步提升的空间。

语言组件 我们采用 mT5（Xue 等，2021）主干作为我们的语言组件。我们使用预训练的 mT5-Large（10 亿参数）和 mT5-XXL（130 亿参数）进行实验，并以此初始化 PaLI 的语言编码器-解码器。我们在包含许多任务的混合任务上进行训练，其中包括纯语言理解任务（第 A.2 节）。这有助于避免 mT5 的语言理解和生成能力发生灾难性遗忘。因此，PaLI-17B 在英语基准（Wang 等，2019a）以及 XTREME 基准（Hu 等，2020）衡量的跨语言任务上，仍能达到相近水平的语言理解准确率（第 4 节）。

整体模型 考虑了三种模型规模（表 7）：1) PaLI-3B，其中语言组件从 mT5-Large（Xue et al., 2021）（10 亿参数）初始化，视觉组件为 ViT-G（Zhai et al., 2022a）（18 亿参数）。2) PaLI-15B，其中语言组件从 mT5-XXL（Xue et al., 2021）（130 亿参数）初始化，视觉组件为 ViT-G（18 亿参数）。3) PaLI-17B，其中语言模型从 mT5-XXL 初始化，视觉组件为新训练的 ViT-e 模型（40 亿参数）。

3.2 数据

WebLI 数据集 关于深度学习的规模化研究表明，更大的模型需要更大的数据集才能有效训练（Hoffmann 等，2022；Kaplan 等，2020；Zhai 等，2022a）。为释放多语言图像-语言预训练的潜力，我们引入了 WebLI，这是一个由公共网络上的图像和文本构建的多语言图像-语言数据集。WebLI 将图像语言数据收集从仅英语数据集扩展到 109 种语言，使我们能够以多语言方式预训练 PaLI，并在多种语言上执行下游任务。数据收集过程类似于（Jia 等，2021；Zhai 等，2022b）中报告的过程。由于互联网上存在丰富的多语言内容，WebLI 数据集的收集过程可以扩展到覆盖 100 亿张图像和 120 亿条替代文本。除了使用网页文本进行标注外，我们还使用公开可用的自动服务提取所有图像上的 OCR 标注，从而得到 290 亿个图像-OCR 对。为了在质量与规模之间取得平衡，我们将数据集过滤为最高质量的子集，仅保留原始 WebLI 图像-文本对中得分最高的前 10%（约 10 亿个样本），并用其来训练 PaLI。WebLI 语料库的示例和统计信息以及完整的数据说明表（Pushkarna 等，2022）见附录 B（图 4）和附录 G。

Training mixture To accommodate diverse tasks in the image-language space, we train PaLI using a mixture of eight pre-training tasks. This mixture is designed to span a range of general capabilities useful for downstream tasks. **Span corruption on text-only data** uses the same technique described by Xue et al. (2021) on text-only examples. **Split-captioning on WebLI alt-text data** is inspired by the pre-training objective of Wang et al. (2021), and works by splitting each alt-text string randomly into two parts, $\langle \text{cap}_1 \rangle$ and $\langle \text{cap}_2 \rangle$, used for input and target, respectively. **Captioning on CC3M-35L** with the alt-text string in language $\langle \text{lang} \rangle$ as the target, based on the Conceptual Captions (Sharma et al., 2018) training data and machine translated alt-texts. **OCR on WebLI OCR-text data** uses the concatenation of the annotated OCR texts in language $\langle \text{lang} \rangle$ (Kil et al., 2022) produced by publicly available automatic service for the input image. **English and Cross-Lingual VQA** is VQ^2A-CC3M (Changpinyo et al., 2022a), translated in the same way as CC3M-35L. Note that we use English answers in all instances here, as the English-native answers for VQA are often short and too prone to errors to perform out-of-context automatic translation. **English and Cross-Lingual visual question generation (VQG)** is also based on native and translated VQ^2A-CC3M-35L VQA triplets. Similarly, we use only English answers here. **English-only Object-Aware (OA) VQA** is based on VQA triplets derived from automatically-produced, non-exhaustive object labels, inspired by Piergiovanni et al. (2022a). The QA pairs include listing all the objects in the image and whether a subset of objects are in the image. To create these examples, we require object-level annotations, for which we use Open Images (Kuznetsova et al., 2020). **Object detection** is a generative object-detection task inspired by Chen et al. (2021; 2022).

We specify each task using a training data source and a template-based prompt, and train the model using a language-model–style teacher forcing (Goodfellow et al., 2016) with a standard softmax cross-entropy loss. The coefficients for the training mixture are empirically determined, with 1.6B total examples in the mixture (Appendix A.2). The whole mixture is slightly smaller and designed to be cleaner than the datasets used in SimVLM (1.8B), CoCa (1.8B), and Flamingo (2.3B). However, unlike the aforementioned datasets, examples in our 1.6B dataset follow a long-tailed distribution over the 100+ languages covered. To prevent leakage between the pre-training examples and the downstream benchmarks. WebLI has undergone near de-duplication (Jia et al., 2021) of the images against the train, validation, and test splits of 68 common vision/vision-language datasets. For other datasets in the mixture, we performed the same de-duplication against all the downstream tasks.

3.3 Model Training

All PaLI variants are trained for one epoch over the entire pre-training dataset (1.6B) with 224×224 image resolution. Only the parameters of the language component are updated, the vision component is frozen, which is beneficial (Sec. 4.6). For the largest model, PaLI-17B, we perform an additional high-res (588×588) phase similar to previous works (Radford et al., 2021; Yuan et al., 2021; Yu et al., 2022). This phase is only for 10k steps, covering 10M examples in total, with all the parameters of PaLI updated. More details for training PaLI and the ViT-e backbone are in Appendix A.1.

4 Experiments

We fine-tune and evaluate PaLI-3B and PaLI-15B checkpoints at 490×490 resolutions. For PaLI-17B, unless otherwise stated, the checkpoint produced by the two-phase pre-training is fine-tuned and evaluated at 588×588 resolution. For all the benchmarks, cross-entropy loss is used for fine-tuning.

4.1 Image Captioning

We fine-tune on **COCO Captions** (Chen et al., 2015) on the widely adopted Karpathy split (Karpathy & Fei-Fei, 2015). PaLI outperforms the latest SOTA trained with cross-entropy loss (Wang et al., 2022c), and establishes a new high of CIDEr score (Vedantam et al., 2015) at 149.1 (Table 1) for models without CIDEr-optimization. **NoCaps** (Agrawal et al., 2019) is an evaluation benchmark for image captioning that has similar style to COCO, but targets many more visual concepts than those included in the COCO. We follow previous works by evaluating NoCaps using a model fine-tuned on COCO. PaLI-17B achieves a 124.4 CIDEr score on test, comparable to the recent result of 124.8 from GIT2 (Wang et al., 2022a). GIT2 achieves 124.2, 125.5, 122.3 on in-domain, near-domain, and out-of-domain splits of the NoCaps test set, respectively. PaLI-17B achieves 121.1,

训练混合 为了适应图像-语言空间中的多样化任务，我们使用八种预训练任务的混合来训练 PaLI。该混合旨在涵盖一系列对下游任务有用的通用能力。仅文本数据上的片段破坏采用 Xue 等人（2021）在仅文本样本中描述的相同技术。WebLI 替代文本数据上的分割式描述生成受到 Wang 等人（2021）预训练目标的启发，其做法是将每个替代文本字符串随机拆分为两部分，$\langle \text{cap}_1 \rangle$ 和 $\langle \text{cap}_2 \rangle$，分别用作输入和目标。CC3M-35L 上的描述生成以语言 ⟨lang⟩ 的替代文本字符串作为目标，基于 Conceptual Captions（Sharma 等人，2018）训练数据以及机器翻译的替代文本。WebLI OCR 文本数据上的 OCR 使用输入图像中由公开可用的自动服务生成的、语言为 ⟨lang⟩ (的标注 OCR 文本的拼接（Kil 等人，2022)）。英语和跨语言 VQA 是 VQ^2A- CC3M（Changpinyo 等人，2022a），其翻译方式与 CC3M-35L 相同。请注意，这里我们在所有实例中都使用英语答案，因为 VQA 的英语原生答案通常很短，并且在脱离上下文的自动翻译中太容易出错。英语和跨语言视觉问题生成（VQG）也基于原生和翻译后的 VQ^2A-CC3M-35L VQA 三元组。同样，我们在这里仅使用英语答案。仅英语的对象感知（OA）VQA 基于从自动生成的、非穷尽对象标签中派生的 VQA 三元组，受到 Piergiovanni 等人（2022a）的启发。问答对包括列出图像中的所有对象，以及判断某一对象子集是否出现在图像中。为了创建这些样本，我们需要对象级标注，为此我们使用 Open Images（Kuznetsova 等人，2020）。目标检测是一项生成式目标检测任务，受到 Chen 等人（2021；2022）的启发。

我们使用训练数据源和基于模板的提示来指定每个任务，并采用语言模型式的教师强制（Goodfellow 等，2016）和标准 softmax 交叉熵损失来训练模型。训练混合数据的系数是通过经验确定的，混合数据中共有 16 亿个样本（附录 A.2）。整个混合数据集略小于 SimVLM（18 亿）、CoCa（18 亿）和 Flamingo（23 亿）所使用的数据集，并且旨在更加干净。然而，与上述数据集不同，我们 16 亿数据集中的样本在所覆盖的 100+ 种语言上呈长尾分布。为防止预训练样本与下游基准之间发生泄漏，WebLI 已对图像与 68 个常见视觉/视觉-语言数据集的训练、验证和测试划分进行了近似去重（Jia 等，2021）。对于混合数据中的其他数据集，我们也针对所有下游任务进行了相同的去重处理。

3.3 模型训练

所有 PaLI 变体都在整个预训练数据集（1.6B）上训练一个 epoch，图像分辨率为 224×224。仅更新语言组件的参数，视觉组件被冻结，这样做是有益的（第 4.6 节）。对于最大的模型 PaLI-17B，我们执行了一个额外的高分辨率（588×588）阶段，类似于以往工作（Radford 等，2021；Yuan 等，2021；Yu 等，2022）。该阶段仅进行 10k 步，总共覆盖 10M 个样本，并更新 PaLI 的所有参数。有关训练 PaLI 和 ViT-e 骨干网络的更多细节见附录 A.1。

4 实验

我们在 490×490 分辨率下对 PaLI-3B 和 PaLI-15B 检查点进行微调和评估。对于 PaLI-17B，除非另有说明，由两阶段预训练产生的检查点会在 588×588 分辨率下进行微调和评估。对于所有基准测试，微调均使用交叉熵损失。

4.1 图像描述

我们在广泛采用的 Karpathy 划分（Karpathy & Fei-Fei, 2015）上，基于 COCO Captions（Chen et al., 2015）进行微调。PaLI 优于使用交叉熵损失训练的最新 SOTA（Wang et al., 2022c），并在未进行 CIDEr 优化的模型中取得了 149.1 的 CIDEr 分数（Vedantam et al., 2015）新高（表 1）。NoCaps（Agrawal et al., 2019）是一个用于图像描述的评测基准，其风格与 COCO 相似，但目标涵盖的视觉概念远多于 COCO 中包含的概念。我们遵循以往工作，使用在 COCO 上微调的模型来评估 NoCaps。PaLI-17B 在测试集上取得了 124.4 的 CIDEr 分数，与 GIT2（Wang et al., 2022a）近期取得的 124.8 结果相当。GIT2 在 NoCaps 测试集的域内、近域和域外划分上分别取得 124.2、125.5、122.3。PaLI-17B 取得 121.1,

124.4 and 126.7, respectively. This suggests that for PaLI-17B, the domain transfer from COCO to NoCaps is slightly sub-optimal compared with models pre-trained with English only. Nevertheless, PaLI-17B outperforms all prior models on recognizing and describing long-tail objects outside of COCO's domain. **TextCaps** (Sidorov et al., 2020) focuses on captioning for images containing text. **VizWiz-Cap** (Gurari et al., 2020) contains images taken by people who are blind, which also involves scene-text understanding. We fine-tune on TextCaps and VizWiz-Cap using OCR strings generated by publicly available automatic service, similar to the protocol used in (Yang et al., 2021). Further details, including results evaluating PaLI-17B without OCR as input, are provided in Appendix C.5.

Table 1: CIDEr results for image captioning over the English benchmarks COCO Captions (Karpathy split), NoCaps, TextCaps, and VizWiz-Cap.

	COCO	NoCaps		TextCaps		VizWiz-Cap	
Model	Karpathy-test	val	test	val	test	test-dev	test-std
LEMON (0.7B)	139.1	117.3	114.3	-	-	-	-
SimVLM	143.3	112.2	110.3	-	-	-	-
CoCa (2.1B)	143.6	122.4	120.6	-	-	-	-
GIT (0.7B)	144.8	125.5	123.4	143.7	138.2	113.1	114.4
GIT2 (5.1B)	145.0	**126.9**	**124.8**	148.6	145.0	119.4	120.8
OFA (0.9B)	145.3	-	-	-	-	-	-
Flamingo (80B)	138.1	-	-	-	-	-	-
BEiT-3 (1.9B)	147.6	-	-	-	-	-	-
PaLI-3B	145.4	121.1	-	143.6	-	117.2	-
PaLI-15B	146.2	121.2	-	150.1	-	121.7	-
PaLI-17B	**149.1**	**127.0**	**124.4**	**160.0**	**160.4**	**123.0**	**124.7**

Multilingual captioning on Crossmodal-3600 Following Thapliyal et al. (2022), we fine-tune PaLI models on COCO-35L, which is COCO captions translated into 35 languages similar to CC3M-35L, before evaluating on Crossmodal-3600. We used the checkpoints pre-trained at 224×224 resolution and fine-tuned on COCO-35L at the same resolution. We normalize the unicode, tokenize, and remove all punctuation before calculating CIDEr scores. For languages without word boundaries such as Chinese, Japanese, Korean and Thai, a neural model is used for segmenting the text. To illustrate the range of improvements over a variety of language families with different scripts and different resources, we use seven languages in Table 2 to show their exact CIDEr scores, in addition to the 35-language average score. PaLI outperforms previous SOTA by large margins. Note that due to different linguistic structures, the variance of CIDEr scores across different languages does not indicate lower quality of prediction on certain languages. In Appendix C.2, we back-translate the non-English predictions to English, and demonstrated that the capability of PaLI on both English and other languages is rather consistent.

Table 2: CIDEr scores on image captioning for the Crossmodal-3600 benchmark for seven diverse languages (English, French, Hindi, Hebrew, Romanian, Thai, and Chinese), as well as the average of the 35 languages covered by the benchmark.

Model	en	fr	hi	iw	ro	th	zh	35-lang avg.
Thapliyal et al. (2022) (0.8B)	57.6	40.9	20.6	16.1	13.9	35.5	19.8	28.9
PaLI-3B	92.8	68.6	30.3	39.2	30.3	65.9	32.2	47.0
PaLI-17B	**98.1**	**75.5**	**31.3**	**46.8**	**35.8**	**72.1**	**36.5**	**53.6**

4.2 Visual Question Answering

All the VQA fine-tuning experiments in this paper are performed in the open-vocabulary setting using the 250k mT5 (Xue et al., 2021) vocabulary (Table 3). Most prior works, e.g. SimVLM (Wang et al., 2021), CoCa (Yu et al., 2022) and BEiT-3 (Wang et al., 2022c), use the VQA-as-classification setting, where the best answer among a predefined set (usually of size 3k) needs to be selected. Note that the VQA-as-open-generation setting is challenging because: (1) The generated text is directly compared to the desired answer and only an exact match is counted as accurate. (2) The PaLI vocabulary covers 100+ languages and is significantly larger than both those used in the classification setting, and those used by previous single-language open-generation models (Alayrac et al., 2022; Wang et al., 2022a).

分别为 124.4 和 126.7。这表明，对于 PaLI-17B，从 COCO 到 NoCaps 的领域迁移相比仅使用英语预训练的模型略显次优。尽管如此，PaLI-17B 在识别和描述 COCO 领域之外的长尾对象方面优于所有先前模型。TextCaps（Sidorov 等，2020）专注于包含文本图像的字幕生成。VizWiz-Cap（Gurari 等，2020）包含由盲人拍摄的图像，也涉及场景文本理解。我们使用公开可用的自动服务生成的 OCR 字符串在 TextCaps 和 VizWiz-Cap 上进行微调，类似于（Yang 等，2021）中使用的协议。更多细节，包括评估 PaLI-17B 在不将 OCR 作为输入时的结果，见附录 C.5。

表 1：在英语基准 COCO Captions（Karpathy 划分）、NoCaps、TextCaps 和 VizWiz-Cap 上进行图像描述的 CIDEr 结果。

	COCO	NoCaps		TextCaps		VizWiz-Cap	
Model	Karpathy-test	val	test	val	test	test-dev	test-std
LEMON (0.7B)	139.1	117.3	114.3	-	-	-	-
SimVLM	143.3	112.2	110.3	-	-	-	-
CoCa (2.1B)	143.6	122.4	120.6	-	-	-	-
GIT (0.7B)	144.8	125.5	123.4	143.7	138.2	113.1	114.4
GIT2 (5.1B)	145.0	**126.9**	**124.8**	148.6	145.0	119.4	120.8
OFA (0.9B)	145.3	-	-	-	-	-	-
Flamingo (80B)	138.1	-	-	-	-	-	-
BEiT-3 (1.9B)	147.6	-	-	-	-	-	-
PaLI-3B	145.4	121.1	-	143.6	-	117.2	-
PaLI-15B	146.2	121.2	-	150.1	-	121.7	-
PaLI-17B	**149.1**	**127.0**	**124.4**	**160.0**	**160.4**	**123.0**	**124.7**

Crossmodal-3600 上的多语言字幕生成 遵循 Thapliyal 等人（2022）的做法，我们先在 COCO-35L 上对 PaLI 模型进行微调，然后再在 Crossmodal-3600 上进行评估；COCO-35L 是将 COCO 字幕翻译成 35 种语言的数据集，类似于 CC3M-35L。我们使用以 224×224 分辨率预训练的检查点，并在相同分辨率下于 COCO-35L 上进行微调。在计算 CIDEr 分数之前，我们会对 Unicode 进行规范化、分词，并移除所有标点符号。对于中文、日文、韩文和泰文等没有词边界的语言，我们使用神经模型对文本进行分词。为了展示在具有不同文字体系和不同资源水平的多种语系上的改进范围，除了 35 种语言的平均分外，我们还在表 2 中使用七种语言来展示其确切的 CIDEr 分数。PaLI 以大幅优势超越了以往的 SOTA。需要注意的是，由于不同语言结构存在差异，不同语言之间 CIDEr 分数的方差并不表示某些语言的预测质量较低。在附录 C.2 中，我们将非英语预测结果反向翻译为英语，并证明 PaLI 在英语和其他语言上的能力相当一致。

表 2：在 Crossmodal-3600 基准上，七种不同语言（英语、法语、印地语、希伯来语、罗马尼亚语、泰语和中文）的图像描述 CIDEr 得分，以及该基准涵盖的 35 种语言的平均得分。

Model	en	fr	hi	iw	ro	th	zh	35-lang avg.
Thapliyal et al. (2022) (0.8B)	57.6	40.9	20.6	16.1	13.9	35.5	19.8	28.9
PaLI-3B	92.8	68.6	30.3	39.2	30.3	65.9	32.2	47.0
PaLI-17B	**98.1**	**75.5**	**31.3**	**46.8**	**35.8**	**72.1**	**36.5**	**53.6**

4.2 视觉问答

本文中的所有 VQA 微调实验均在开放词汇设定下进行，使用 250k 的 mT5（Xue 等，2021）词汇表（表 3）。大多数先前工作，例如 SimVLM（Wang 等，2021）、CoCa（Yu 等，2022）和 BEiT-3（Wang 等，2022c），使用将 VQA 视为分类的设定，其中需要从预定义集合（通常大小为 3k）中选择最佳答案。需要注意的是，将 VQA 视为开放生成的设定具有挑战性，因为：（1）生成的文本会直接与期望答案进行比较，只有完全匹配才计为正确。（2）PaLI 词汇表覆盖 100+ 种语言，并且显著大于分类设定中使用的词汇表，也大于以往单语言开放生成模型所使用的词汇表（Alayrac 等，2022；Wang 等，2022a）。

Table 3: VQA Accuracy results on VQAv2, OKVQA, TextVQA, VizWiz-QA, and ANLS result on ST-VQA. PaLI models are evaluated in the open-vocabulary generation setting, and still outperform previous models that use closed-vocabulary classification (SimVLM, CoCa, BEiT-3, OFA). The result on OKVQA by Flamingo (with "*") is obtained in a 32-shot learning setup. Mia (Qiao et al., 2021) (with "†") is the winning model of TextVQA Challenge 2021, based on fine-tuning T5-XL (Raffel et al., 2020). Numbers shown in gray are from models using closed-vocabulary classification.

	VQAv2		OKVQA	TextVQA		VizWiz-QA		ST-VQA	
Method	test-dev	test-std	val	val	test	test-dev	test	val	test
SimVLM	80.03	80.34	-	-	-	-	-	-	-
CoCa (2.1B)	82.3	82.3	-	-	-	-	-	-	-
GIT (0.7B)	78.56	78.81	-	59.93	59.75	68.0	67.5	69.1	69.6
GIT2 (5.1B)	81.74	81.92	-	68.38	67.27	70.97	70.1	75.1	75.8
OFA (0.9B)	82.0	82.0	-	-	-	-	-	-	-
Flamingo (80B)	82.0	82.1	57.8*	57.1	54.1	65.7	65.4	-	-
BEiT-3 (1.9B)	84.2	84.0	-	-	-	-	-	-	-
KAT	-	-	54.4	-	-	-	-	-	-
Mia	-	-	-	-	**73.67**†	-	-	-	-
PaLI-3B	81.4	-	52.4	60.12	-	67.5	-	67.5	69.7
PaLI-15B	82.9	-	56.5	65.49	-	71.1	-	73.2	76.5
PaLI-17B	**84.3**	**84.3**	**64.5**	**71.81**	73.06	**74.4**	**73.3**	**77.1**	**79.9**

On **VQAv2**, PaLI achieves 84.3 accuracy on VQAv2, and outperforms previous SOTA as follows: (1) By +2.2 accuracy points on the open-vocabulary generation setting, compared to Flamingo (Alayrac et al., 2022). (2) By +0.3 accuracy points when compared against the best result on the closed-vocabulary classification setting, BEiT-3 (Wang et al., 2022c). **OKVQA** requires external knowledge to answer its questions, that is, knowledge not directly present in the image input, and instead needs to be indirectly inferred by the model. PaLI-17B achieves 64.5 accuracy, pushing SOTA for the pretrain-finetune setup higher by 10.1 accuracy points, compared to KAT (Gui et al., 2021) at 54.4 accuracy. The best result for the 32-shot learning setup is from Flamingo (Alayrac et al., 2022) at 57.8 accuracy. The results from Flamingo and PaLI-17B suggest that leveraging external knowledge does not necessarily require specific training, and instead can be achieved with generic large-capacity models trained on large amounts of data. **TextVQA** (Singh et al., 2019), **VizWiz-QA** (Gurari et al., 2018) and **ST-VQA** (Biten et al., 2019) require the ability to perform question answering in the presence of text in the input image. We fine-tune using OCR strings generated by publicly available automatic service, similar to the protocol in TAP (Yang et al., 2021) and Mia (Qiao et al., 2021). Evaluation on TextVQA and VizWiz-QA without OCR as input is provided in Appendix C.5.

Cross-lingual and Multilingual VQA on xGQA and MaXM Both xGQA (Pfeiffer et al., 2022) and MaXM (Changpinyo et al., 2022b) are test-only VQA benchmarks that require multilingual understanding of visual questions. The setting in xGQA is cross-lingual (English-answers only), whereas for MaXM it is multilingual (answer in the same language as the question). We evaluate PaLI-17B pre-trained at 224 image resolution and fine-tuned on the native and translated VQAv2 (Goyal et al., 2017) (the Karpathy train split) in the 13 languages covered by xGQA and MaXM (VQAv2-13L) at 378 resolution. Table 4 shows significant gains on both benchmarks across all languages.

Table 4: Cross-lingual VQA results on xGQA (Pfeiffer et al., 2022) (left) and multilingual VQA results on MaXM (Changpinyo et al., 2022b) (right). All models are fine-tuned on translated VQAv2 in 13 languages. Exact-match accuracy is reported. Referenced MPT results are from (Changpinyo et al., 2022b)

	xGQA								MaXM						
Model	en	bn	de	id	ko	pt	ru	zh	en	fr	hi	iw	ro	th	zh
MPT	41.5	38.6	40.5	39.5	38.7	39.8	39.5	39.5	36.6	36.2	55.1	40.6	42.3	50.0	30.3
PaLI-17B	**54.2**	**50.0**	**52.2**	**50.6**	**50.4**	**51.3**	**50.3**	**50.6**	**56.4**	**46.4**	**67.3**	**60.0**	**57.4**	**65.6**	**46.9**

4.3 Language-understanding Capabilities

Since PaLI is pre-trained with a diverse mixture of multimodal tasks with image and text data, it raises the question on whether it would "forget" its language modeling capability, and therefore

表 3：在 VQAv2、OKVQA、TextVQA、VizWiz-QA 上的 VQA 准确率结果，以及在 ST-VQA 上的 ANLS 结果。PaLI 模型在开放词汇生成设置下进行评估，并且仍然优于以往使用封闭词汇分类的模型（SimVLM、CoCa、BEiT-3、OFA）。Flamingo 在 OKVQA 上的结果（带“*”）是在 32-shot 学习设置下获得的。Mia（Qiao 等，2021）（带“†”）是 TextVQA Challenge 2021 的获胜模型，基于对 T5-XL（Raffel 等，2020）进行微调。灰色显示的数字来自使用封闭词汇分类的模型。

	VQAv2		OKVQA	TextVQA		VizWiz-QA		ST-VQA	
Method	test-dev	test-std	val	val	test	test-dev	test	val	test
SimVLM	80.03	80.34	-	-	-	-	-	-	-
CoCa (2.1B)	82.3	82.3	-	-	-	-	-	-	-
GIT (0.7B)	78.56	78.81	-	59.93	59.75	68.0	67.5	69.1	69.6
GIT2 (5.1B)	81.74	81.92	-	68.38	67.27	70.97	70.1	75.1	75.8
OFA (0.9B)	82.0	82.0	-	-	-	-	-	-	-
Flamingo (80B)	82.0	82.1	57.8*	57.1	54.1	65.7	65.4	-	-
BEiT-3 (1.9B)	84.2	84.0	-	-	-	-	-	-	-
KAT	-	-	54.4	-	-	-	-	-	-
Mia	-	-	-	-	**73.67**†	-	-	-	-
PaLI-3B	81.4	-	52.4	60.12	-	67.5	-	67.5	69.7
PaLI-15B	82.9	-	56.5	65.49	-	71.1	-	73.2	76.5
PaLI-17B	**84.3**	**84.3**	**64.5**	**71.81**	73.06	**74.4**	**73.3**	**77.1**	**79.9**

在 VQAv2 上，PaLI 达到了 84.3 的准确率，并且相较于此前的 SOTA 表现如下：（1）在开放词汇生成设置下，与 Flamingo（Alayrac 等，2022）相比，高出 +2.2 个准确率百分点。（2）在与封闭词汇分类设置下的最佳结果 BEiT-3（Wang 等，2022c）相比时，高出 +0.3 个准确率百分点。OKVQA 需要外部知识来回答其问题，也就是说，需要并非直接存在于图像输入中的知识，而是需要模型间接推断出来。PaLI-17B 达到 64.5 的准确率，与准确率为 54.4 的 KAT（Gui 等，2021）相比，将预训练-微调设置下的 SOTA 提高了 10.1 个准确率百分点。32-shot 学习设置下的最佳结果来自 Flamingo（Alayrac 等，2022），准确率为 57.8。Flamingo 和 PaLI-17B 的结果表明，利用外部知识并不一定需要专门训练，而是可以通过在大量数据上训练的通用大容量模型来实现。TextVQA（Singh 等，2019）、VizWiz-QA（Gurari 等，2018）和 ST-VQA（Biten 等，2019）要求模型具备在输入图像中存在文本的情况下进行问答的能力。我们使用由公开可用的自动服务生成的 OCR 字符串进行微调，类似于 TAP（Yang 等，2021）和 Mia（Qiao 等，2021）中的协议。在不将 OCR 作为输入的情况下，对 TextVQA 和 VizWiz-QA 的评估见附录 C.5。

xGQA 和 MaXM 上的跨语言与多语言 VQA。xGQA（Pfeiffer 等，2022）和 MaXM（Changpinyo 等，2022b）都是仅用于测试的 VQA 基准，要求对视觉问题进行多语言理解。xGQA 的设置是跨语言的（仅英文答案），而 MaXM 则是多语言的（以与问题相同的语言作答）。我们评估了在 224 图像分辨率下预训练的 PaLI-17B，并在 xGQA 和 MaXM 所覆盖的 13 种语言的原生与翻译版 VQAv2（Goyal 等，2017）（Karpathy 训练划分）上以 378 分辨率进行微调（VQAv2-13L）。表 4 显示了在所有语言上两个基准均取得了显著提升。

表4：xGQA（Pfeiffer 等，2022）上的跨语言 VQA 结果（左）以及 MaXM（Changpinyo 等，2022b）上的多语言 VQA 结果（右）。所有模型均在翻译成 13 种语言的 VQAv2 上进行微调。报告的是精确匹配准确率。引用的 MPT 结果来自（Changpinyo 等，2022b）

	xGQA								MaXM						
Model	en	bn	de	id	ko	pt	ru	zh	en	fr	hi	iw	ro	th	zh
MPT	41.5	38.6	40.5	39.5	38.7	39.8	39.5	39.5	36.6	36.2	55.1	40.6	42.3	50.0	30.3
PaLI-17B	**54.2**	**50.0**	**52.2**	**50.6**	**50.4**	**51.3**	**50.3**	**50.6**	**56.4**	**46.4**	**67.3**	**60.0**	**57.4**	**65.6**	**46.9**

4.3 语言理解能力

由于 PaLI 使用图像和文本数据的多种多模态任务混合进行预训练，因此这引出了一个问题：它是否会“忘记”其语言建模能力，从而

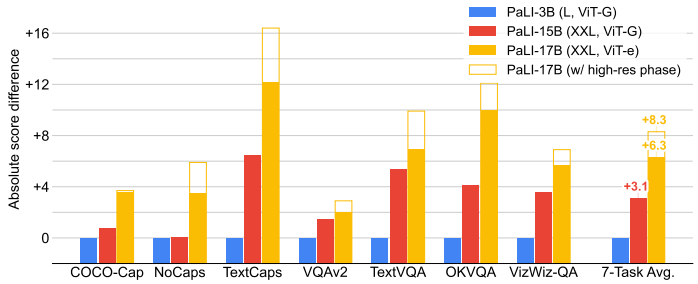


Figure 2: PaLI scaling for a number of tasks. We report CIDEr scores for captioning tasks, and accuracy scores for VQA tasks. Both scaling the language side (from 1B to 13B parameters) and the vision side of the model (from 2B to 4B parameters) yield improvements across all tasks. The results represented by solid bars are from the standard 224×224 resolution pre-training. The empty orange bars correspond to PaLI-17B checkpoints with the high resolution pre-training phase.

exhibit inferior performance on language-understanding tasks compared to its unimodal starting checkpoint (mT5-XXL in the case of PaLI-17B). Therefore, we compare mT5-XXL and PaLI-17B on a range of language understanding benchmarks, including the English-only SuperGLUE benchmark (Wang et al., 2019a), as well as three multilingual benchmarks from the XTREME (Hu et al., 2020): XNLI (Conneau et al., 2018), which is a textual entailment task covering 14 languages, XQuAD (Artetxe et al., 2020) and TyDiQA-GoldP (Clark et al., 2020), which are both question-answering tasks covering 10 and 11 languages, respectively. For the three XTREME benchmarks, we evaluate in the zero-shot (ZS) transfer setting, whereas for SuperGLUE the models are fine-tuned (FT). Table 11 in Appendix C.1 summarizes the results. Despite the pre-training mixture heavily favoring the V&L tasks, PaLI-17B is able to maintain a high-level of language-understanding capabilities for English, and it is on-par with the state-of-the-art mT5-XXL checkpoint on the XTREME benchmarks.

4.4 Zero-shot Image Classification

We evaluate the PaLI checkpoints (without high-res phase) at 224×224 resolution on ImageNet and ImageNet OOD evaluation sets: ImageNet (Deng et al., 2009), ImageNet-R (Hendrycks et al., 2021a), ImageNet-A (Hendrycks et al., 2021b), ImageNet-Sketch (Wang et al., 2019b), ImageNet-v2 (Recht et al., 2019) and ObjectNet (Barbu et al., 2019). We use the same interface as for all other tasks. Instead of training a classifier on top of PaLI, we condition on the image and use PaLI's decoder to score strings corresponding to each class directly. (See Appendix C.8 for details) The top-1 accuracies are presented in Table 5, where it clearly shows that PaLI-17B is significantly better than smaller variants. We are not aware of any previous work for large scale zero-shot evaluation on ImageNet with a generative model. However, PaLI with a zero-shot setting outperforms the 1-shot learning result from Flamingo (Alayrac et al., 2022).

Table 5: Top 1 accuracy results of 0-shot image classification on ImageNet, ImageNet-R, ImageNet-A, ImageNet-Sketch, Imagenet-v2, and ObjectNet. Top-5 results are in the Appendix (Table 22).

Model (ImageNet data)	INet	INet-R	INet-A	INet-Sketch	INet-v2	ObjNet
Flamingo-80B (**1-shot**)	71.9	-	-	-	-	-
Flamingo-80B (**5-shot**)	77.3	-	-	-	-	-
PaLI-3B (**0-shot**)	70.06	80.15	37.92	61.11	62.55	38.87
PaLI-15B (**0-shot**)	70.27	81.21	41.16	61.03	62.81	39.51
PaLI-17B (**0-shot**)	**72.11**	**81.97**	**44.70**	**63.83**	**64.46**	**42.62**

4.5 Model Scaling

Due to the modular architecture, the image and language components of PaLI can be scaled independently. We demonstrate that jointly scaling the capacity of both components leads to performance improvements. Figure 2 quantifies this improvement across seven V&L benchmarks where we

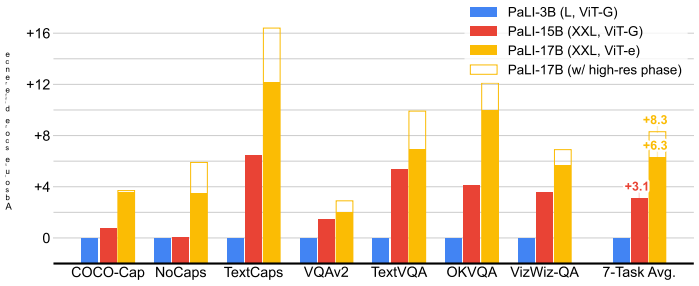


图 2：PaLI 在多项任务上的扩展。我们报告了图像描述任务的 CIDEr 分数，以及 VQA 任务的准确率分数。无论是扩展模型的语言侧（从 1B 到 13B 参数），还是扩展视觉侧（从 2B 到 4B 参数），都能在所有任务上带来提升。实心柱表示的结果来自标准 224×224 分辨率预训练。空心橙色柱对应于经过高分辨率预训练阶段的 PaLI-17B 检查点。

在语言理解任务上的表现相比其单模态起始检查点（在 PaLI-17B 的情况下为 mT5-XXL）更差。因此，我们在一系列语言理解基准上比较了 mT5-XXL 和 PaLI-17B，包括仅英语的 SuperGLUE 基准（Wang 等，2019a），以及来自 XTREME（Hu 等，2020）的三个多语言基准：XNLI（Conneau 等，2018），这是一个覆盖 14 种语言的文本蕴含任务；XQuAD（Artetxe 等，2020）和 TyDiQA-GoldP（Clark 等，2020），二者都是问答任务，分别覆盖 10 种和 11 种语言。对于这三个 XTREME 基准，我们在零样本（ZS）迁移设置下进行评估，而对于 SuperGLUE，则对模型进行微调（FT）。附录 C.1 中的表 11 总结了结果。尽管预训练混合数据严重偏向 V&L 任务，PaLI-17B 仍能够保持高水平的英语语言理解能力，并且在 XTREME 基准上与最先进的 mT5-XXL 检查点表现相当。

4.4 零样本图像分类

我们在 224×224 分辨率下，在 ImageNet 和 ImageNet OOD 评估集上评估 PaLI 检查点（不包含高分辨率阶段）：ImageNet（Deng 等，2009）、ImageNet-R（Hendrycks 等，2021a）、ImageNet-A（Hendrycks 等，2021b）、ImageNet-Sketch（Wang 等，2019b）、ImageNet-v2（Recht 等，2019）和 ObjectNet（Barbu 等，2019）。我们使用与所有其他任务相同的接口。我们没有在 PaLI 之上训练分类器，而是以图像为条件，并使用 PaLI 的解码器直接为对应于每个类别的字符串打分。（详情见附录 C.8）表 5 给出了 top-1 准确率，其中清楚地显示 PaLI-17B 显著优于较小的变体。我们尚未发现此前有使用生成式模型在 ImageNet 上进行大规模零样本评估的工作。然而，在零样本设置下，PaLI 的表现优于 Flamingo（Alayrac 等，2022）的 1-shot 学习结果。

表 5：在 ImageNet、ImageNet-R、ImageNet-A、ImageNet-Sketch、Imagenet-v2 和 ObjectNet 上进行零样本图像分类的 Top-1 准确率结果。Top-5 结果见附录（表 22）。

Model (ImageNet data)	INet	INet-R	INet-A	INet-Sketch	INet-v2	ObjNet
Flamingo-80B (**1-shot**)	71.9	-	-	-	-	-
Flamingo-80B (**5-shot**)	77.3	-	-	-	-	-
PaLI-3B (**0-shot**)	70.06	80.15	37.92	61.11	62.55	38.87
PaLI-15B (**0-shot**)	70.27	81.21	41.16	61.03	62.81	39.51
PaLI-17B (**0-shot**)	**72.11**	**81.97**	**44.70**	**63.83**	**64.46**	**42.62**

4.5 模型扩展

由于采用模块化架构，PaLI 的图像和语言组件可以独立扩展。我们证明，同时扩展这两个组件的容量会带来性能提升。图 2 量化了这一提升，涵盖了七个 V&L 基准测试，其中我们

have also evaluated the PaLI-17B checkpoint without the high resolution pre-training phase for fair comparison. These improvements are noticeable both when scaling the language-model capacity (from L to XXL), and the vision-model capacity (from ViT-G to ViT-e). Figure 2 also shows that scaling the visual component is important: when scaling from a ViT-G to a ViT-e model, although the overall model size is increased by only about 13% (+2B parameters), the average performance improvement over all seven benchmarks (additional +3.2) is larger than the one obtained with much larger increases in the capacity of the language model (+3.1) which takes more parameters (+12B). The high-resolution pre-training phase at 588×588 resolution brings an additional +2.0 points, which also indicates the potential of scaling up the vision component of the model. This observation also resonates with the significant improvement from PaLI-15B to 17B on generative ImageNet zero-shot classification (Table 5). Table 12 shows the results of a 5B version of PaLI with mT5-L and ViT-e on two benchmarks, which also resonates with the finding of the benefit of joint scaling. For context, in prior work, V&L scaling is usually conducted at lower model capacity: for instance, CoCa (Yu et al., 2022) scales up to 2.1B parameters, or scaling is done primarily via the language-modeling backbone, e.g. Flamingo (Alayrac et al., 2022) scales the text backbone to 80B but the image backbone remains at 435M. Finally, on the Crossmodal-3600 benchmark, we show that scale has a large impact on multilingual performance as well (Figure 5 in the Appendix).

4.6 Ablation studies

We examine the composition of the task mixture and demonstrate the effectiveness of our multiple-objective mixture design. To this end, we pre-train a PaLI-3B model with 200M data coverage for each setting, before fine-tuning on a combination of English and multilingual V&L tasks (Table 6). Aside from the four tasks from our main evaluation for PaLI, we also add a VQAv2-based VQG benchmark (Akula et al., 2021). The relative weight of each components remains the same as the full mixture (Table 9). As a first observation, the split-cap objective on WebLI appears to be the most critical, across all benchmarks. Second, the object-related components also boost performance on all benchmarks. Third, the captioning objective on CC3M-35L helps on COCO; on XM-3600, its positive contribution for non-EN languages and the slight degradation for English is a reflection of CC3M-35L having a much higher non-EN example ratio (34/35) compared to WebLI alt-text (60% English, Figure 4). Fourth, adding VQA helps TextVQA; in addition, the VQG objective improves the model's VQG capability without impacting the performance on other benchmarks. Last but not least, the OCR objective positively impacts OCR-related tasks such as TextVQA, at a slight negative impact on captioning performance. We also note that VQAv2, due to its large training set size, is much less sensitive to the change in pre-training mixture. In addition, we perform ablations to quantify the positive impact of initializing from uni-modal checkpoints, as opposed to from-scratch training (Table 14); the minor accuracy improvement from freezing the ViT backbone during pre-training (Table 15); the effect of pretraining with non-English WebLI examples on multi-(cross-)lingual performance (Table 16).

Table 6: Mixture of objectives (PaLI-3B). TextVQA is fine-tuned with 490×490 resolution, while all other benchmarks are fine-tuned with 224×224. Results for VQAv2 are on the Karpathy validation set. XM-3600 denotes Crossmodal-3600, and "6L" is the average of the six non-English languages in Table 2. The order in which the components are ablated follows the presented order in Sec. 3.2, and "object-related" refers to the object-aware QA and generative object detection components together. TextVQA is fine-tuned without detected OCR string to better showcase the model's OCR capability

Component	COCO	TextVQA	VQAv2	XM-3600 (EN / 6L)	VQG (ZS / FT)
Full mixture	141.4	41.6	76.0	93.8 / 42.5	96.7 / 194.0
w/o split-cap	140.4 (-1.0)	38.8 (-2.8)	75.5 (-0.5)	87.5 (-6.3) / 41.5 (-1.0)	86.3 (-10.4) / 190.5 (-3.5)
w/o captioning	140.5 (-0.9)	41.2 (-0.4)	75.9 (-0.1)	94.9 (+1.1) / 39.9 (-2.6)	101.3 (+4.6) / 193.3 (-0.7)
w/o OCR	142.3 (+0.9)	39.9 (-1.7)	75.9 (-0.1)	95.4 (+1.6) / 43.6 (+1.1)	92.5 (-4.2) / 193.7 (-0.3)
w/o VQA	140.9 (-0.5)	40.0 (-1.6)	75.9 (-0.1)	93.9 (+0.1) / 42.7 (+0.2)	94.1 (-2.6) / 193.2 (-0.8)
w/o VQG	141.4 (+0.0)	41.3 (-0.3)	75.8 (-0.2)	95.1 (+1.3) / 42.0 (-0.5)	17.9 (-78.8) / 188.2 (-5.8)
w/o object-related	140.9 (-0.5)	40.2 (-1.4)	75.4 (-0.6)	90.9 (-2.9) / 41.8 (-0.7)	81.7 (-15.0) / 189.1 (-4.9)

还评估了未经过高分辨率预训练阶段的 PaLI-17B 检查点，以便进行公平比较。无论是在扩展语言模型容量（从 L 到 XXL）还是视觉模型容量（从 ViT-G 到 ViT-e）时，这些改进都很明显。图 2 还表明，扩展视觉组件非常重要：当从 ViT-G 扩展到 ViT-e 模型时，尽管整体模型规模仅增加约 13%（+2B 参数），但在全部七个基准上的平均性能提升（额外 +3.2）大于通过大幅增加语言模型容量（+3.1）所获得的提升，而后者需要更多参数（+12B）。在 588×588 分辨率下进行的高分辨率预训练阶段又带来了额外 +2.0 分，这也表明了扩展模型视觉组件的潜力。这一观察结果也与 PaLI-15B 到 17B 在生成式 ImageNet 零样本分类上的显著提升相呼应（表 5）。表 12 展示了使用 mT5-L 和 ViT-e 的 5B 版本 PaLI 在两个基准上的结果，这同样印证了联合扩展带来收益的发现。作为背景，在以往工作中，V&L 扩展通常是在较低模型容量下进行的：例如，CoCa（Yu 等，2022）扩展到 2.1B 参数；或者扩展主要通过语言建模主干完成，例如 Flamingo（Alayrac 等，2022）将文本主干扩展到 80B，但图像主干仍为 435M。最后，在 Crossmodal-3600 基准上，我们表明规模对多语言性能同样具有很大影响（附录中的图 5）。

4.6 消融实验

我们考察了任务混合的组成，并展示了我们的多目标混合设计的有效性。为此，我们在每种设置下使用 200M 数据覆盖量预训练一个 PaLI-3B 模型，然后在英语和多语言 V&L 任务的组合上进行微调（表 6）。除了 PaLI 主评测中的四个任务外，我们还加入了一个基于 VQA v2 的 VQG 基准（Akula et al., 2021）。各组成部分的相对权重与完整混合相同（表 9）。首先观察到，在所有基准上，WebLI 上的 split-cap 目标似乎最为关键。其次，与对象相关的组成部分也提升了所有基准上的性能。第三，CC3M-35L 上的图像描述目标有助于 COCO；在 XM-3600 上，它对非英语语言的积极贡献以及对英语的轻微性能下降，反映出与 WebLI alt-text（60% 为英语，图 4）相比，CC3M-35L 具有高得多的非英语样本比例（34/35）。第四，加入 VQA 有助于 TextVQA；此外，VQG 目标提升了模型的 VQG 能力，而不影响其他基准上的性能。最后但同样重要的是，OCR 目标对与 OCR 相关的任务（如 TextVQA）产生了积极影响，但对图像描述性能有轻微负面影响。我们还注意到，由于 VQAv2 的训练集规模较大，它对预训练混合变化的敏感性要低得多。此外，我们进行了消融实验，以量化从单模态检查点初始化相较于从头训练的积极影响（表 14）；预训练期间冻结 ViT 主干带来的轻微准确率提升（表 15）；以及使用非英语 WebLI 样本进行预训练对多（跨）语言性能的影响（表 16）。

表 6：目标混合（PaLI-3B）。TextVQA 以 490×490 分辨率进行微调，而所有其他基准均以 224×224 进行微调。VQAv2 的结果基于 Karpathy 验证集。XM-3600 表示 Crossmodal-3600，“6L”是表 2 中六种非英语语言的平均值。组件消融的顺序遵循第 3.2 节中呈现的顺序，“object-related” 指的是目标感知问答和生成式目标检测组件的组合。TextVQA 在微调时未使用检测到的 OCR 字符串，以更好地展示模型的 OCR 能力。

Component	COCO	TextVQA	VQAv2	XM-3600 (EN / 6L)	VQG (ZS / FT)
Full mixture	141.4	41.6	76.0	93.8 / 42.5	96.7 / 194.0
w/o split-cap	140.4 (-1.0)	38.8 (-2.8)	75.5 (-0.5)	87.5 (-6.3) / 41.5 (-1.0)	86.3 (-10.4) / 190.5 (-3.5)
w/o captioning	140.5 (-0.9)	41.2 (-0.4)	75.9 (-0.1)	94.9 (+1.1) / 39.9 (-2.6)	101.3 (+4.6) / 193.3 (-0.7)
w/o OCR	142.3 (+0.9)	39.9 (-1.7)	75.9 (-0.1)	95.4 (+1.6) / 43.6 (+1.1)	92.5 (-4.2) / 193.7 (-0.3)
w/o VQA	140.9 (-0.5)	40.0 (-1.6)	75.9 (-0.1)	93.9 (+0.1) / 42.7 (+0.2)	94.1 (-2.6) / 193.2 (-0.8)
w/o VQG	141.4 (+0.0)	41.3 (-0.3)	75.8 (-0.2)	95.1 (+1.3) / 42.0 (-0.5)	17.9 (-78.8) / 188.2 (-5.8)
w/o object-related	140.9 (-0.5)	40.2 (-1.4)	75.4 (-0.6)	90.9 (-2.9) / 41.8 (-0.7)	81.7 (-15.0) / 189.1 (-4.9)

ETHICS STATEMENT AND BROADER IMPACTS

Large models may have broader societal impact. While such models have demonstrated strong performance on public benchmarks, they might contain unknown biases or stereotypes, or propagate inaccurate or otherwise distorted information. While we have made efforts to measure some of these issues, such models need to be re-assessed carefully before being used for specific purposes. The dataset used for pre-training is automatically harvested, and filtering of the data is automatic. That process may leave undesirable images or text annotations, descriptions or concepts to be incorporated into the model. We have also attempted to train the model to operate in more than 100 languages, which we believe is an important step forward for image-language models. However, languages have various levels of data presence and coverage, so the language-generated text varies in quality depending on the language, and might contain inaccurate or undesirable outputs.

REPRODUCIBILITY STATEMENTS

Our model is based on open sourced components - ViT and mT5 (Dosovitskiy et al., 2021; Xue et al., 2021). Model architecture details for each component is in Section 3.1. The configuration of ViT-e when scaling is provided in Table 7 and Section A.1. We have provided training and fine-tuning details in Section 3.3 and in Section A in the Appendix. Data and model cards are also provided in the Appendix.

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REFERENCES

Manoj Acharya, Kushal Kafle, and Christopher Kanan. Tallyqa: Answering complex counting questions. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pp. 8076–8084, 2019.

Harsh Agrawal, Karan Desai, Yufei Wang, Xinlei Chen, Rishabh Jain, Mark Johnson, Dhruv Batra, Devi Parikh, Stefan Lee, and Peter Anderson. nocaps: Novel object captioning at scale. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 8948–8957, 2019.

Arjun Akula, Soravit Changpinyo, Boqing Gong, Piyush Sharma, Song-Chun Zhu, and Radu Soricut. Crossvqa: Scalably generating benchmarks for systematically testing vqa generalization. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 2148–2166, 2021.

Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *arXiv preprint arXiv:2204.14198*, 2022.

Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4623–4637, 2020.

Andrei Barbu, David Mayo, Julian Alverio, William Luo, Christopher Wang, Dan Gutfreund, Joshua Tenenbaum, and Boris Katz. ObjectNet: a large-scale bias-controlled dataset for pushing the limits of object recognition models. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, pp. 9453–9463, 2019.

伦理声明和更广泛影响

大型模型可能具有更广泛的社会影响。尽管此类模型在公开基准测试中表现出很强的性能，但它们可能包含未知的偏见或刻板印象，或传播不准确或以其他方式失真的信息。虽然我们已努力衡量其中一些问题，但在将此类模型用于特定目的之前，仍需对其进行谨慎的重新评估。用于预训练的数据集是自动采集的，数据过滤也是自动完成的。该过程可能会导致不良图像或文本标注、描述或概念被纳入模型。我们还尝试训练该模型，使其能够以 100 多种语言运行，我们认为这是图像-语言模型向前迈出的重要一步。然而，不同语言的数据存在量和覆盖范围各不相同，因此语言生成文本的质量会因语言而异，并且可能包含不准确或不良的输出。

可复现性声明

我们的模型基于开源组件——ViT 和 mT5（Dosovitskiy 等，2021；Xue 等，2021）。每个组件的模型架构细节见第 3.1 节。ViT-e 在扩展时的配置见表 7 和第 A.1 节。我们在第 3.3 节以及附录 A 中提供了训练和微调细节。数据卡和模型卡也在附录中提供。

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参考文献

Manoj Acharya、Kushal Kafle 和 Christopher Kanan。TallyQA：回答复杂计数问题。收录于 *Proceedings of the AAAI conference on artificial intelligence*，第 33 卷，第 8076–8084 页，2019 年。

Harsh Agrawal、Karan Desai、Yufei Wang、Xinlei Chen、Rishabh Jain、Mark Johnson、Dhruv Batra、Devi Parikh、Stefan Lee 和 Peter Anderson。nocaps：大规模新颖物体图像描述。载于 *Proceedings of the IEEE/CVF International Conference on Computer Vision*，第 8948–8957 页，2019 年。

Arjun Akula、Soravit Changpinyo、Boqing Gong、Piyush Sharma、Song-Chun Zhu 和 Radu Soricut。Crossvqa：可扩展地生成用于系统测试 VQA 泛化能力的基准。在 *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*，第 2148–2166 页，2021 年。

Jean-Baptiste Alayrac、Jeff Donahue、Pauline Luc、Antoine Miech、Iain Barr、Yana Hasson、Karel Lenc、Arthur Mensch、Katie Millican、Malcolm Reynolds 等。Flamingo：一种用于少样本学习的视觉语言模型。*arXiv preprint arXiv:2204.14198*，2022。

Mikel Artetxe、Sebastian Ruder 和 Dani Yogatama。关于单语表示的跨语言可迁移性。载于 *Proceedings of the 58th Annual Meeting of the Association for Computa- tional Linguistics*，第 4623–4637 页，2020 年。

Andrei Barbu、David Mayo、Julian Alverio、William Luo、Christopher Wang、Dan Gutfreund、Joshua Tenenbaum 和 Boris Katz。ObjectNet：一个用于推动目标识别模型极限的大规模偏差控制数据集。收录于 *Proceedings of the 33rd International Conference on Neural Information Processing Systems*，第 9453–9463 页，2019 年。

Lucas Beyer, Olivier J Hénaff, Alexander Kolesnikov, Xiaohua Zhai, and Aäron van den Oord. Are we done with ImageNet? *arXiv preprint arXiv:2006.07159*, 2020.

Lucas Beyer, Xiaohua Zhai, and Alexander Kolesnikov. Big Vision. `https://github.com/google-research/big_vision`, 2022.

Ali Furkan Biten, Rubèn Tito, Andrés Mafla, Lluis Gomez, Marçal Rusiñol, C.V. Jawahar, Ernest Valveny, and Dimosthenis Karatzas. Scene text visual question answering. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4290–4300, 2019. doi: 10.1109/ICCV.2019.00439.

James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs, 2018. URL `http://github.com/google/jax`.

Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, pp. 1877–1901, 2020.

Soravit Changpinyo, Piyush Sharma, Nan Ding, and Radu Soricut. Conceptual 12m: Pushing web-scale image-text pre-training to recognize long-tail visual concepts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3558–3568, 2021.

Soravit Changpinyo, Doron Kukliansky, Idan Szpektor, Xi Chen, Nan Ding, and Radu Soricut. All you may need for VQA are image captions. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 1947–1963, Jul 2022a.

Soravit Changpinyo, Linting Xue, Idan Szpektor, Ashish V. Thapliyal, Julien Amelot, Michal Yarom, Xi Chen, and Radu Soricut. MaXM: Towards multilingual visual question answering. *arXiv preprint arXiv:2209.05401*, 2022b.

Ting Chen, Saurabh Saxena, Lala Li, David J Fleet, and Geoffrey Hinton. Pix2seq: A language modeling framework for object detection. *arXiv preprint arXiv:2109.10852*, 2021.

Ting Chen, Saurabh Saxena, Lala Li, Tsung-Yi Lin, David J Fleet, and Geoffrey Hinton. A unified sequence interface for vision tasks. *arXiv preprint arXiv:2206.07669*, 2022.

Xinlei Chen, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam, Saurabh Gupta, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO Captions: Data collection and evaluation server. *arXiv preprint arXiv:1504.00325*, 2015.

Yen-Chun Chen, Linjie Li, Licheng Yu, Ahmed El Kholy, Faisal Ahmed, Zhe Gan, Yu Cheng, and Jingjing Liu. Uniter: Universal image-text representation learning. In *European conference on computer vision*, pp. 104–120, 2020.

Jaemin Cho, Jie Lei, Hao Tan, and Mohit Bansal. Unifying vision-and-language tasks via text generation. In *International Conference on Machine Learning*, pp. 1931–1942, 2021.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.

Jonathan H Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages. *Transactions of the Association for Computational Linguistics*, 8: 454–470, 2020.

Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel Bowman, Holger Schwenk, and Veselin Stoyanov. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2475–2485, 2018.

Lucas Beyer、Olivier J Hénaff、Alexander Kolesnikov、Xiaohua Zhai 和 Aäron van den Oord。我们完成 ImageNet 了吗？*arXiv preprint arXiv:2006.07159*，2020。Lucas Beyer、Xiaohua Zhai 和 Alexander Kolesnikov。Big Vision。https://github.com/ google-research/big_vision，2022。Ali Furkan Biten、Rubèn Tito、Andrés Mafla、Lluis Gomez、Marçal Rusiñol、C.V. Jawahar、Ernest Valveny 和 Dimosthenis Karatzas。场景文本视觉问答。载于 *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*，第 4290–4300 页，2019。doi: 10.1109/ICCV. 2019.00439。James Bradbury、Roy Frostig、Peter Hawkins、Matthew James Johnson、Chris Leary、Dougal Maclaurin、George Necula、Adam Paszke、Jake VanderPlas、Skye Wanderman-Milne 和 Qiao Zhang。JAX：Python+NumPy 程序的可组合变换，2018。URL http://github.com/google/jax。Tom B Brown、Benjamin Mann、Nick Ryder、Melanie Subbiah、Jared Kaplan、Prafulla Dhariwal、Arvind Neelakantan、Pranav Shyam、Girish Sastry、Amanda Askell 等。语言模型是少样本学习者。载于 *Proceedings of the 34th International Conference on Neural Information Processing Systems*，第 1877–1901 页，2020。Soravit Changpinyo、Piyush Sharma、Nan Ding 和 Radu Soricut。Conceptual 12m：推动网络规模图像-文本预训练以识别长尾视觉概念。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 3558–3568 页，2021。Soravit Changpinyo、Doron Kukliansky、Idan Szpektor、Xi Chen、Nan Ding 和 Radu Soricut。VQA 可能只需要图像描述。载于 *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*，第 1947–1963 页，2022 年 7 月。Soravit Changpinyo、Linting Xue、Idan Szpektor、Ashish V. Thapliyal、Julien Amelot、Michal Yarom、Xi Chen 和 Radu Soricut。MaXM：迈向多语言视觉问答。*arXiv preprint arXiv:2209.05401*，2022b。Ting Chen、Saurabh Saxena、Lala Li、David J Fleet 和 Geoffrey Hinton。Pix2seq：用于目标检测的语言建模框架。*arXiv preprint arXiv:2109.10852*，2021。Ting Chen、Saurabh Saxena、Lala Li、Tsung-Yi Lin、David J Fleet 和 Geoffrey Hinton。视觉任务的统一序列接口。*arXiv preprint arXiv:2206.07669*，2022。Xinlei Chen、Hao Fang、Tsung-Yi Lin、Ramakrishna Vedantam、Saurabh Gupta、Piotr Dollár 和 C. Lawrence Zitnick。Microsoft COCO Captions：数据收集与评估服务器。*arXiv preprint arXiv:1504.00325*，2015。Yen-Chun Chen、Linjie Li、Licheng Yu、Ahmed El Kholy、Faisal Ahmed、Zhe Gan、Yu Cheng 和 Jingjing Liu。Uniter：通用图像-文本表示学习。载于 *European conference on computer vision*，第 104–120 页，2020。Jaemin Cho、Jie Lei、Hao Tan 和 Mohit Bansal。通过文本生成统一视觉与语言任务。载于 *International Conference on Machine Learning*，第 1931–1942 页，2021。Aakanksha Chowdhery、Sharan Narang、Jacob Devlin、Maarten Bosma、Gaurav Mishra、Adam Roberts、Paul Barham、Hyung Won Chung、Charles Sutton、Sebastian Gehrmann 等。Palm：使用 Pathways 扩展语言建模。*arXiv preprint arXiv:2204.02311*，2022。Jonathan H Clark、Eunsol Choi、Michael Collins、Dan Garrette、Tom Kwiatkowski、Vitaly Nikolaev 和 Jennimaria Palomaki。TyDi QA：面向类型多样语言的信息寻求型问答基准。*Transactions of the Association for Computational Linguistics*，8：454–470，2020。Alexis Conneau、Ruty Rinott、Guillaume Lample、Adina Williams、Samuel Bowman、Holger Schwenk 和 Veselin Stoyanov。XNLI：评估跨语言句子表示。载于 *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*，第 2475–2485 页，2018。

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255, 2009.

Jeffrey Donahue, Lisa Anne Hendricks, Sergio Guadarrama, Marcus Rohrbach, Subhashini Venugopalan, Kate Saenko, and Trevor Darrell. Long-term recurrent convolutional networks for visual recognition and description. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2625–2634, 2015.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *9th International Conference on Learning Representations, ICLR 2021,*, 2021.

Nan Du, Yanping Huang, Andrew M Dai, Simon Tong, Dmitry Lepikhin, Yuanzhong Xu, Maxim Krikun, Yanqi Zhou, Adams Wei Yu, Orhan Firat, et al. GLaM: Efficient scaling of language models with mixture-of-experts. In *International Conference on Machine Learning*, pp. 5547–5569, 2022.

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé Iii, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12): 86–92, 2021.

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016. `http://www.deeplearningbook.org`.

Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6904–6913, 2017.

Liangke Gui, Borui Wang, Qiuyuan Huang, Alex Hauptmann, Yonatan Bisk, and Jianfeng Gao. KAT: A knowledge augmented transformer for vision-and-language. *arXiv preprint arXiv:2112.08614*, 2021.

Danna Gurari, Qing Li, Abigale J Stangl, Anhong Guo, Chi Lin, Kristen Grauman, Jiebo Luo, and Jeffrey P Bigham. VizWiz grand challenge: Answering visual questions from blind people. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3608–3617, 2018.

Danna Gurari, Yinan Zhao, Meng Zhang, and Nilavra Bhattacharya. Captioning images taken by people who are blind. In *European Conference on Computer Vision*, pp. 417–434. Springer, 2020.

Jonathan Heek, Anselm Levskaya, Avital Oliver, Marvin Ritter, Bertrand Rondepierre, Andreas Steiner, and Marc van Zee. Flax: A neural network library and ecosystem for JAX, 2020. URL `http://github.com/google/flax`.

Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, et al. The many faces of robustness: A critical analysis of out-of-distribution generalization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 8340–8349, 2021a.

Dan Hendrycks, Kevin Zhao, Steven Basart, Jacob Steinhardt, and Dawn Song. Natural adversarial examples. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15262–15271, 2021b.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

Junjie Hu, Sebastian Ruder, Aditya Siddhant, Graham Neubig, Orhan Firat, and Melvin Johnson. XTREME: A massively multilingual multi-task benchmark for evaluating cross-lingual generalisation. In *International Conference on Machine Learning*, pp. 4411–4421, 2020.

Jia Deng、Wei Dong、Richard Socher、Li-Jia Li、Kai Li 和 Li Fei-Fei。ImageNet：一个大规模层次化图像数据库。载于 *2009 IEEE conference on computer vision and pattern recognition*，第 248–255 页，2009。Jeffrey Donahue、Lisa Anne Hendricks、Sergio Guadarrama、Marcus Rohrbach、Subhashini Venu-gopalan、Kate Saenko 和 Trevor Darrell。用于视觉识别与描述的长期循环卷积网络。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 2625–2634 页，2015。Alexey Dosovitskiy、Lucas Beyer、Alexander Kolesnikov、Dirk Weissenborn、Xiaohua Zhai、Thomas Unterthiner、Mostafa Dehghani、Matthias Minderer、Georg Heigold、Sylvain Gelly、Jakob Uszkoreit 和 Neil Houlsby。一张图像胜过 16x16 个词：用于大规模图像识别的 Transformer。载于 *9th International Conference on Learning Representations, ICLR 2021,*，2021。Nan Du、Yanping Huang、Andrew M Dai、Simon Tong、Dmitry Lepikhin、Yuanzhong Xu、Maxim Krikun、Yanqi Zhou、Adams Wei Yu、Orhan Firat 等。GLaM：通过专家混合实现语言模型的高效扩展。载于 *International Conference on Machine Learning*，第 5547–5569 页，2022。Timnit Gebru、Jamie Morgenstern、Briana Vecchione、Jennifer Wortman Vaughan、Hanna Wallach、Hal Daumé Iii 和 Kate Crawford。数据集的数据表。*Communications of the ACM*，64(12)：86–92，2021。Ian Goodfellow、Yoshua Bengio 和 Aaron Courville。*Deep Learning*。MIT Press，2016。http: //www.deeplearningbook.org。Yash Goyal、Tejas Khot、Douglas Summers-Stay、Dhruv Batra 和 Devi Parikh。让 VQA 中的 V 发挥作用：提升图像理解在视觉问答中的作用。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 6904–6913 页，2017。Liangke Gui、Borui Wang、Qiuyuan Huang、Alex Hauptmann、Yonatan Bisk 和 Jianfeng Gao。KAT：一种用于视觉与语言的知识增强型 Transformer。*arXiv preprint arXiv:2112.08614*，2021。Danna Gurari、Qing Li、Abigale J Stangl、Anhong Guo、Chi Lin、Kristen Grauman、Jiebo Luo 和 Jeffrey P Bigham。VizWiz 大挑战：回答盲人提出的视觉问题。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 3608–3617 页，2018。Danna Gurari、Yinan Zhao、Meng Zhang 和 Nilavra Bhattacharya。为盲人拍摄的图像生成描述。载于 *European Conference on Computer Vision*，第 417–434 页。Springer，2020。Jonathan Heek、Anselm Levskaya、Avital Oliver、Marvin Ritter、Bertrand Rondepierre、Andreas Steiner 和 Marc van Zee。Flax：用于 JAX 的神经网络库与生态系统，2020。URL http: //github.com/google/flax。Dan Hendrycks、Steven Basart、Norman Mu、Saurav Kadavath、Frank Wang、Evan Dorundo、Rahul Desai、Tyler Zhu、Samyak Parajuli、Mike Guo 等。鲁棒性的多面性：对分布外泛化的批判性分析。载于 *Proceedings of the IEEE/CVF International Conference on Computer Vision*，第 8340–8349 页，2021a。Dan Hendrycks、Kevin Zhao、Steven Basart、Jacob Steinhardt 和 Dawn Song。自然对抗样本。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 15262–15271 页，2021b。Jordan Hoffmann、Sebastian Borgeaud、Arthur Mensch、Elena Buchatskaya、Trevor Cai、Eliza Rutherford、Diego de Las Casas、Lisa Anne Hendricks、Johannes Welbl、Aidan Clark 等。训练计算最优的大型语言模型。*arXiv preprint arXiv:2203.15556*，2022。Junjie Hu、Sebastian Ruder、Aditya Siddhant、Graham Neubig、Orhan Firat 和 Melvin Johnson。XTREME：用于评估跨语言泛化的大规模多语言多任务基准。载于 *International Conference on Machine Learning*，第 4411–4421 页，2020。

Xiaowei Hu, Zhe Gan, Jianfeng Wang, Zhengyuan Yang, Zicheng Liu, Yumao Lu, and Lijuan Wang. Scaling up vision-language pre-training for image captioning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17980–17989, 2022.

Yanping Huang, Youlong Cheng, Ankur Bapna, Orhan Firat, Mia Xu Chen, Dehao Chen, HyoukJoong Lee, Jiquan Ngiam, Quoc V Le, Yonghui Wu, et al. GPipe: efficient training of giant neural networks using pipeline parallelism. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, pp. 103–112, 2019.

Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision. In *International Conference on Machine Learning*, pp. 4904–4916, 2021.

Norman P Jouppi, Doe Hyun Yoon, George Kurian, Sheng Li, Nishant Patil, James Laudon, Cliff Young, and David Patterson. A domain-specific supercomputer for training deep neural networks. *Communications of the ACM*, 63(7):67–78, 2020.

Da-Cheng Juan, Chun-Ta Lu, Zhen Li, Futang Peng, Aleksei Timofeev, Yi-Ting Chen, Yaxi Gao, Tom Duerig, Andrew Tomkins, and Sujith Ravi. Graph-rise: Graph-regularized image semantic embedding. *arXiv preprint arXiv:1902.10814*, 2019.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Andrej Karpathy and Li Fei-Fei. Deep visual-semantic alignments for generating image descriptions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3128–3137, 2015.

Jihyung Kil, Soravit Changpinyo, Xi Chen, Hexiang Hu, Sebastian Goodman, Wei-Lun Chao, and Radu Soricut. PreSTU: Pre-training for scene-text understanding. *arXiv preprint arXiv:2209.05534*, 2022.

Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big Transfer (BiT): General visual representation learning. *Lecture Notes in Computer Science*, pp. 491–507, 2020. ISSN 1611-3349.

Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. Visual Genome: Connecting language and vision using crowdsourced dense image annotations. *International journal of computer vision*, 123(1):32–73, 2017.

Alina Kuznetsova, Hassan Rom, Neil Alldrin, Jasper Uijlings, Ivan Krasin, Jordi Pont-Tuset, Shahab Kamali, Stefan Popov, Matteo Malloci, Alexander Kolesnikov, et al. The Open Images dataset v4. *International Journal of Computer Vision*, 128(7):1956–1981, 2020.

Jiasen Lu, Christopher Clark, Rowan Zellers, Roozbeh Mottaghi, and Aniruddha Kembhavi. Unified-io: A unified model for vision, language, and multi-modal tasks. *arXiv preprint arXiv:2206.08916*, 2022.

Dhruv Mahajan, Ross Girshick, Vignesh Ramanathan, Kaiming He, Manohar Paluri, Yixuan Li, Ashwin Bharambe, and Laurens Van Der Maaten. Exploring the limits of weakly supervised pretraining. In *Proceedings of the European conference on computer vision (ECCV)*, pp. 181–196, 2018.

Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the conference on fairness, accountability, and transparency*, pp. 220–229, 2019.

Jonas Pfeiffer, Gregor Geigle, Aishwarya Kamath, Jan-Martin Steitz, Stefan Roth, Ivan Vulić, and Iryna Gurevych. xGQA: Cross-lingual visual question answering. In *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 2497–2511, 2022.

Xiaowei Hu, Zhe Gan, Jianfeng Wang, Zhengyuan Yang, Zicheng Liu, Yumao Lu, and Lijuan Wang. 扩展用于图像描述生成的视觉-语言预训练。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17980–17989, 2022. Yanping Huang, Youlong Cheng, Ankur Bapna, Orhan Firat, Mia Xu Chen, Dehao Chen, HyoukJoong Lee, Jiquan Ngiam, Quoc V Le, Yonghui Wu, et al. GPipe：使用流水线并行高效训练巨型神经网络。载于 *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, pp. 103–112, 2019. Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. 通过带噪文本监督扩展视觉与视觉-语言表征学习。载于 *International Conference on Machine Learning*, pp. 4904–4916, 2021. Norman P Jouppi, Doe Hyun Yoon, George Kurian, Sheng Li, Nishant Patil, James Laudon, Cliff Young, and David Patterson. 用于训练深度神经网络的领域专用超级计算机。*Communications of the ACM*, 63(7): 67–78, 2020. Da-Cheng Juan, Chun-Ta Lu, Zhen Li, Futang Peng, Aleksei Timofeev, Yi-Ting Chen, Yaxi Gao, Tom Duerig, Andrew Tomkins, and Sujith Ravi. Graph-rise：图正则化的图像语义嵌入。*arXiv preprint arXiv:1902.10814*, 2019. Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 神经语言模型的缩放定律。*arXiv preprint arXiv:2001.08361*, 2020. Andrej Karpathy and Li Fei-Fei. 用于生成图像描述的深度视觉-语义对齐。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3128–3137, 2015. Jihyung Kil, Soravit Changpinyo, Xi Chen, Hexiang Hu, Sebastian Goodman, Wei-Lun Chao, and Radu Soricut. PreSTU：用于场景文本理解的预训练。*arXiv preprint arXiv:2209.05534*, 2022. Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big Transfer（BiT）：通用视觉表征学习。*Lecture Notes in Computer Science*, pp. 491–507, 2020. ISSN 1611-3349. Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. Visual Genome：使用众包密集图像标注连接语言与视觉。*International journal of computer vision*, 123(1):32–73, 2017. Alina Kuznetsova, Hassan Rom, Neil Alldrin, Jasper Uijlings, Ivan Krasin, Jordi Pont-Tuset, Shahab Kamali, Stefan Popov, Matteo Malloci, Alexander Kolesnikov, et al. Open Images 数据集 v4。*International Journal of Computer Vision*, 128(7):1956–1981, 2020. Jiasen Lu, Christopher Clark, Rowan Zellers, Roozbeh Mottaghi, and Aniruddha Kembhavi. Unified-io：用于视觉、语言和多模态任务的统一模型。*arXiv preprint arXiv:2206.08916*, 2022. Dhruv Mahajan, Ross Girshick, Vignesh Ramanathan, Kaiming He, Manohar Paluri, Yixuan Li, Ashwin Bharambe, and Laurens Van Der Maaten. 探索弱监督预训练的极限。载于 *Proceedings of the European conference on computer vision (ECCV)*, pp. 181–196, 2018. Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 用于模型报告的模型卡。载于 *Proceedings of the conference on fairness, accountability, and transparency*, pp. 220–229, 2019. Jonas Pfeiffer, Gregor Geigle, Aishwarya Kamath, Jan-Martin Steitz, Stefan Roth, Ivan Vulić, and Iryna Gurevych. xGQA：跨语言视觉问答。载于 *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 2497–2511, 2022.

Hieu Pham, Zihang Dai, Golnaz Ghiasi, Hanxiao Liu, Adams Wei Yu, Minh-Thang Luong, Mingxing Tan, and Quoc V Le. Combined scaling for zero-shot transfer learning. *arXiv preprint arXiv:2111.10050*, 2021.

AJ Piergiovanni, Weicheng Kuo, and Anelia Angelova. Pre-training image-language transformers for open-vocabulary tasks. In *T4V: Transformers for Vision Workshop, Conference on Computer Vision and Pattern Recognition*, 2022a.

AJ Piergiovanni, Wei Li, Weicheng Kuo, Mohammad Saffar, Fred Bertsch, and Anelia Angelova. Answer-Me: Multi-task learning for generalization to many question-answering tasks. *arXiv preprint arXiv:2205.00949*, 2022b.

Mahima Pushkarna, Andrew Zaldivar, and Oddur Kjartansson. Data Cards: Purposeful and transparent dataset documentation for responsible AI. In *FAccT '22: 2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 1776–1826, 2022.

Yixuan Qiao, Hao Chen, Jun Wang, Yihao Chen, Xianbin Ye, Ziliang Li, Xianbiao Qi, Peng Gao, and Guotong Xie. Winner team Mia at TextVQA challenge 2021: Vision-and-language representation learning with pre-trained sequence-to-sequence model. *arXiv preprint arXiv:2106.15332*, 2021.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pp. 8748–8763, 2021.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *International Conference on Machine Learning*, pp. 5389–5400, 2019.

Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems*, 28, 2015.

Carlos Riquelme, Joan Puigcerver, Basil Mustafa, Maxim Neumann, Rodolphe Jenatton, André Susano Pinto, Daniel Keysers, and Neil Houlsby. Scaling vision with sparse mixture of experts. *Advances in Neural Information Processing Systems*, 34:8583–8595, 2021.

Adam Roberts, Hyung Won Chung, Anselm Levskaya, Gaurav Mishra, James Bradbury, Daniel Andor, Sharan Narang, Brian Lester, Colin Gaffney, Afroz Mohiuddin, et al. Scaling up models and data with `t5x` and `seqio`. *arXiv preprint arXiv:2203.17189*, 2022.

Candice Schumann, Susanna Ricco, Utsav Prabhu, Vittorio Ferrari, and Caroline Pantofaru. A step toward more inclusive people annotations for fairness. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, AIES '21, pp. 916–925, 2021. ISBN 9781450384735.

Shuai Shao, Zeming Li, Tianyuan Zhang, Chao Peng, Gang Yu, Xiangyu Zhang, Jing Li, and Jian Sun. Objects365: A large-scale, high-quality dataset for object detection. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 8430–8439, 2019.

Piyush Sharma, Nan Ding, Sebastian Goodman, and Radu Soricut. Conceptual Captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, pp. 2556–2565, 2018.

Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. In *International Conference on Machine Learning*, pp. 4596–4604, 2018.

Mohammad Shoeybi, Mostofa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. Megatron-LM: Training multi-billion parameter language models using model parallelism. *arXiv preprint arXiv:1909.08053*, 2019.

Hieu Pham、Zihang Dai、Golnaz Ghiasi、Hanxiao Liu、Adams Wei Yu、Minh-Thang Luong、Mingxing Tan 和 Quoc V Le。用于零样本迁移学习的组合缩放。*arXiv preprint arXiv:2111.10050*，2021。AJ Piergiovanni、Weicheng Kuo 和 Anelia Angelova。面向开放词汇任务的图像-语言 Transformer 预训练。载于 *T4V: Transformers for Vision Workshop, Conference on Computer Vision and Pattern Recognition*，2022a。AJ Piergiovanni、Wei Li、Weicheng Kuo、Mohammad Saffar、Fred Bertsch 和 Anelia Angelova。Answer-Me：用于泛化到多种问答任务的多任务学习。*arXiv preprint arXiv:2205.00949*，2022b。Mahima Pushkarna、Andrew Zaldivar 和 Oddur Kjartansson。数据卡：面向负责任 AI 的有目的且透明的数据集文档。载于 *FAccT '22: 2022 ACM Conference on Fairness, Accountability, and Transparency*，第 1776–1826 页，2022。Yixuan Qiao、Hao Chen、Jun Wang、Yihao Chen、Xianbin Ye、Ziliang Li、Xianbiao Qi、Peng Gao 和 Guotong Xie。TextVQA 挑战赛 2021 冠军团队 Mia：使用预训练序列到序列模型的视觉与语言表示学习。*arXiv preprint arXiv:2106.15332*，2021。Alec Radford、Jong Wook Kim、Chris Hallacy、Aditya Ramesh、Gabriel Goh、Sandhini Agarwal、Girish Sastry、Amanda Askell、Pamela Mishkin、Jack Clark 等。通过自然语言监督学习可迁移的视觉模型。载于 *International Conference on Machine Learning*，第 8748–8763 页，2021。Colin Raffel、Noam Shazeer、Adam Roberts、Katherine Lee、Sharan Narang、Michael Matena、Yanqi Zhou、Wei Li 和 Peter J. Liu。使用统一的文本到文本 Transformer 探索迁移学习的极限。*Journal of Machine Learning Research*，21(140):1–67，2020。Benjamin Recht、Rebecca Roelofs、Ludwig Schmidt 和 Vaishaal Shankar。ImageNet 分类器能泛化到 ImageNet 吗？载于 *International Conference on Machine Learning*，第 5389–5400 页，2019。Shaoqing Ren、Kaiming He、Ross Girshick 和 Jian Sun。Faster R-CNN：利用区域提议网络实现实时目标检测。*Advances in neural information processing systems*，28，2015。Carlos Riquelme、Joan Puigcerver、Basil Mustafa、Maxim Neumann、Rodolphe Jenatton、André Susano Pinto、Daniel Keysers 和 Neil Houlsby。用稀疏专家混合扩展视觉模型。*Advances in Neural Information Processing Systems*，34:8583–8595，2021。Adam Roberts、Hyung Won Chung、Anselm Levskaya、Gaurav Mishra、James Bradbury、Daniel Andor、Sharan Narang、Brian Lester、Colin Gaffney、Afroz Mohiuddin 等。使用 t5x 和 seqio 扩展模型和数据。*arXiv preprint arXiv:2203.17189*，2022。Candice Schumann、Susanna Ricco、Utsav Prabhu、Vittorio Ferrari 和 Caroline Pantofaru。迈向更具包容性的人物标注以促进公平性的一步。载于 *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*，AIES '21，第 916–925 页，2021。ISBN 9781450384735。Shuai Shao、Zeming Li、Tianyuan Zhang、Chao Peng、Gang Yu、Xiangyu Zhang、Jing Li 和 Jian Sun。Objects365：用于目标检测的大规模高质量数据集。载于 *Proceedings of the IEEE/CVF international conference on computer vision*，第 8430–8439 页，2019。Piyush Sharma、Nan Ding、Sebastian Goodman 和 Radu Soricut。Conceptual Captions：用于自动图像字幕生成的清洗后、上位词化的图像替代文本数据集。载于 *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*，第 2556–2565 页，2018。Noam Shazeer 和 Mitchell Stern。Adafactor：具有次线性内存成本的自适应学习率。载于 *International Conference on Machine Learning*，第 4596–4604 页，2018。Mohammad Shoeybi、Mostofa Patwary、Raul Puri、Patrick LeGresley、Jared Casper 和 Bryan Catanzaro。Megatron-LM：使用模型并行训练数十亿参数语言模型。*arXiv preprint arXiv:1909.08053*，2019。

Oleksii Sidorov, Ronghang Hu, Marcus Rohrbach, and Amanpreet Singh. TextCaps: a dataset for image captioning with reading comprehension. In *European conference on computer vision*, pp. 742–758, 2020.

Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards VQA models that can read. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8317–8326, 2019.

Hao Tan and Mohit Bansal. LXMERT: Learning cross-modality encoder representations from transformers. *arXiv preprint arXiv:1908.07490*, 2019.

Ashish V. Thapliyal, Jordi Pont-Tuset, Xi Chen, and Radu Soricut. Crossmodal-3600: A massively multilingual multimodal evaluation dataset. *arXiv preprint arXiv:2205.12522*, 2022.

Ilya O Tolstikhin, Neil Houlsby, Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Thomas Unterthiner, Jessica Yung, Andreas Steiner, Daniel Keysers, Jakob Uszkoreit, et al. MLP-Mixer: An all-MLP architecture for vision. *Advances in Neural Information Processing Systems*, 34: 24261–24272, 2021.

Hugo Touvron, Andrea Vedaldi, Matthijs Douze, and Herve Jegou. Fixing the train-test resolution discrepancy. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, pp. 8252–8262, 2019.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, pp. 6000–6010, 2017.

Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. CIDEr: Consensus-based image description evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4566–4575, 2015.

Oriol Vinyals, Alexander Toshev, Samy Bengio, and Dumitru Erhan. Show and tell: A neural image caption generator. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3156–3164, 2015.

Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R Bowman. SuperGLUE: a stickier benchmark for general-purpose language understanding systems. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, pp. 3266–3280, 2019a.

Haohan Wang, Songwei Ge, Zachary Lipton, and Eric P Xing. Learning robust global representations by penalizing local predictive power. In *Advances in Neural Information Processing Systems*, pp. 10506–10518, 2019b.

Jianfeng Wang, Zhengyuan Yang, Xiaowei Hu, Linjie Li, Kevin Lin, Zhe Gan, Zicheng Liu, Ce Liu, and Lijuan Wang. GIT: A generative image-to-text transformer for vision and language. *arXiv preprint arXiv:2205.14100*, 2022a.

Peng Wang, An Yang, Rui Men, Junyang Lin, Shuai Bai, Zhikang Li, Jianxin Ma, Chang Zhou, Jingren Zhou, and Hongxia Yang. Unifying architectures, tasks, and modalities through a simple sequence-to-sequence learning framework. *arXiv preprint arXiv:2202.03052*, 2022b.

Wenhui Wang, Hangbo Bao, Li Dong, Johan Bjorck, Zhiliang Peng, Qiang Liu, Kriti Aggarwal, Owais Khan Mohammed, Saksham Singhal, Subhojit Som, et al. Image as a foreign language: BEiT pretraining for all vision and vision-language tasks. *arXiv preprint arXiv:2208.10442*, 2022c.

Zirui Wang, Jiahui Yu, Adams Wei Yu, Zihang Dai, Yulia Tsvetkov, and Yuan Cao. SimVLM: Simple visual language model pretraining with weak supervision. *arXiv preprint arXiv:2108.10904*, 2021.

Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, pp. 23965–23998, 2022.

Oleksii Sidorov、Ronghang Hu、Marcus Rohrbach 和 Amanpreet Singh。TextCaps：一个用于具备阅读理解的图像描述生成数据集。载于 *European conference on computer vision*，第 742–758 页，2020。Amanpreet Singh、Vivek Natarajan、Meet Shah、Yu Jiang、Xinlei Chen、Dhruv Batra、Devi Parikh 和 Marcus Rohrbach。迈向能够阅读的 VQA 模型。载于 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*，第 8317–8326 页，2019。Hao Tan 和 Mohit Bansal。LXMERT：从 Transformer 学习跨模态编码器表示。*arXiv preprint arXiv:1908.07490*，2019。Ashish V. Thapliyal、Jordi Pont-Tuset、Xi Chen 和 Radu Soricut。Crossmodal-3600：一个大规模多语言多模态评估数据集。*arXiv preprint arXiv:2205.12522*，2022。Ilya O Tolstikhin、Neil Houlsby、Alexander Kolesnikov、Lucas Beyer、Xiaohua Zhai、Thomas Un- terthiner、Jessica Yung、Andreas Steiner、Daniel Keysers、Jakob Uszkoreit 等。MLP-Mixer：一种用于视觉的全 MLP 架构。*Advances in Neural Information Processing Systems*，34：24261–24272，2021。Hugo Touvron、Andrea Vedaldi、Matthijs Douze 和 Herve Jegou。修正训练-测试分辨率不一致问题。载于 *Proceedings of the 33rd International Conference on Neural Information Processing Systems*，第 8252–8262 页，2019。Ashish Vaswani、Noam Shazeer、Niki Parmar、Jakob Uszkoreit、Llion Jones、Aidan N Gomez、 ukasz Kaiser 和 Illia Polosukhin。注意力就是你所需要的一切。载于 *Proceedings of the 31st International Conference on Neural Information Processing Systems*，第 6000–6010 页，2017。Ramakrishna Vedantam、C Lawrence Zitnick 和 Devi Parikh。CIDEr：基于共识的图像描述评估。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 4566–4575 页，2015。Oriol Vinyals、Alexander Toshev、Samy Bengio 和 Dumitru Erhan。Show and tell：一种神经图像描述生成器。载于 *Proceedings of the IEEE conference on computer vision and pattern recognition*，第 3156–3164 页，2015。Alex Wang、Yada Pruksachatkun、Nikita Nangia、Amanpreet Singh、Julian Michael、Felix Hill、Omer Levy 和 Samuel R Bowman。SuperGLUE：用于通用语言理解系统的更具挑战性的基准。载于 *Proceedings of the 33rd International Conference on Neural Information Processing Systems*，第 3266–3280 页，2019a。Haohan Wang、Songwei Ge、Zachary Lipton 和 Eric P Xing。通过惩罚局部预测能力来学习鲁棒的全局表示。载于 *Advances in Neural Information Processing Systems*，第 10506–10518 页，2019b。Jianfeng Wang、Zhengyuan Yang、Xiaowei Hu、Linjie Li、Kevin Lin、Zhe Gan、Zicheng Liu、Ce Liu 和 Lijuan Wang。GIT：用于视觉与语言的生成式图像到文本 Transformer。*arXiv preprint arXiv:2205.14100*，2022a。Peng Wang、An Yang、Rui Men、Junyang Lin、Shuai Bai、Zhikang Li、Jianxin Ma、Chang Zhou、Jingren Zhou 和 Hongxia Yang。通过简单的序列到序列学习框架统一架构、任务和模态。*arXiv preprint arXiv:2202.03052*，2022b。Wenhui Wang、Hangbo Bao、Li Dong、Johan Bjorck、Zhiliang Peng、Qiang Liu、Kriti Aggarwal、Owais Khan Mohammed、Saksham Singhal、Subhojit Som 等。将图像视为外语：用于所有视觉及视觉-语言任务的 BEiT 预训练。*arXiv preprint arXiv:2208.10442*，2022c。Zirui Wang、Jiahui Yu、Adams Wei Yu、Zihang Dai、Yulia Tsvetkov 和 Yuan Cao。SimVLM：基于弱监督的简单视觉语言模型预训练。*arXiv preprint arXiv:2108.10904*，2021。Mitchell Wortsman、Gabriel Ilharco、Samir Ya Gadre、Rebecca Roelofs、Raphael Gontijo-Lopes、Ari S Morcos、Hongseok Namkoong、Ali Farhadi、Yair Carmon、Simon Kornblith 等。模型汤：对多个微调模型的权重取平均可在不增加推理时间的情况下提高准确率。载于 *International Conference on Machine Learning*，第 23965–23998 页，2022。

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 483–498, Jun 2021.

Linting Xue, Aditya Barua, Noah Constant, Rami Al-Rfou, Sharan Narang, Mihir Kale, Adam Roberts, and Colin Raffel. ByT5: Towards a token-free future with pre-trained byte-to-byte models. *Transactions of the Association for Computational Linguistics*, 10:291–306, 2022.

Zhengyuan Yang, Yijuan Lu, Jianfeng Wang, Xi Yin, Dinei Florencio, Lijuan Wang, Cha Zhang, Lei Zhang, and Jiebo Luo. TAP: Text-aware pre-training for Text-VQA and Text-Caption. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8751–8761, 2021.

Jiahui Yu, Zirui Wang, Vijay Vasudevan, Legg Yeung, Mojtaba Seyedhosseini, and Yonghui Wu. CoCa: Contrastive captioners are image-text foundation models. *arXiv preprint arXiv:2205.01917*, 2022.

Lu Yuan, Dongdong Chen, Yi-Ling Chen, Noel Codella, Xiyang Dai, Jianfeng Gao, Houdong Hu, Xuedong Huang, Boxin Li, Chunyuan Li, et al. Florence: A new foundation model for computer vision. *arXiv preprint arXiv:2111.11432*, 2021.

Rowan Zellers, Ximing Lu, Jack Hessel, Youngjae Yu, Jae Sung Park, Jize Cao, Ali Farhadi, and Yejin Choi. Merlot: Multimodal neural script knowledge models. *Advances in Neural Information Processing Systems*, 34:23634–23651, 2021.

Xiaohua Zhai, Joan Puigcerver, Alexander Kolesnikov, Pierre Ruyssen, Carlos Riquelme, Mario Lucic, Josip Djolonga, Andre Susano Pinto, Maxim Neumann, Alexey Dosovitskiy, et al. A large-scale study of representation learning with the visual task adaptation benchmark. *arXiv preprint arXiv:1910.04867*, 2019.

Xiaohua Zhai, Alexander Kolesnikov, Neil Houlsby, and Lucas Beyer. Scaling vision transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12104–12113, 2022a.

Xiaohua Zhai, Xiao Wang, Basil Mustafa, Andreas Steiner, Daniel Keysers, Alexander Kolesnikov, and Lucas Beyer. LiT: Zero-shot transfer with locked-image text tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18123–18133, 2022b.

Pengchuan Zhang, Xiujun Li, Xiaowei Hu, Jianwei Yang, Lei Zhang, Lijuan Wang, Yejin Choi, and Jianfeng Gao. VinVL: Revisiting visual representations in vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5579–5588, 2021.

薛林婷、Noah Constant、Adam Roberts、Mihir Kale、Rami Al-Rfou、Aditya Siddhant、Aditya Barua 和 Colin Raffel。mT5：一种大规模多语言预训练文本到文本 Transformer。载于 *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*，第 483–498 页，2021 年 6 月。薛林婷、Aditya Barua、Noah Constant、Rami Al-Rfou、Sharan Narang、Mihir Kale、Adam Roberts 和 Colin Raffel。ByT5：迈向无标记未来的预训练字节到字节模型。*Transactions of the Association for Computational Linguistics*，10:291–306，2022。Zhengyuan Yang、Yijuan Lu、Jianfeng Wang、Xi Yin、Dinei Florencio、Lijuan Wang、Cha Zhang、Lei Zhang 和 Jiebo Luo。TAP：用于 Text-VQA 和 Text-Caption 的文本感知预训练。载于 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*，第 8751–8761 页，2021。Jiahui Yu、Zirui Wang、Vijay Vasudevan、Legg Yeung、Mojtaba Seyedhosseini 和 Yonghui Wu。CoCa：对比式字幕生成器是图像-文本基础模型。*arXiv preprint arXiv:2205.01917*，2022。Lu Yuan、Dongdong Chen、Yi-Ling Chen、Noel Codella、Xiyang Dai、Jianfeng Gao、Houdong Hu、Xuedong Huang、Boxin Li、Chunyuan Li 等。Florence：一种新的计算机视觉基础模型。*arXiv preprint arXiv:2111.11432*，2021。Rowan Zellers、Ximing Lu、Jack Hessel、Youngjae Yu、Jae Sung Park、Jize Cao、Ali Farhadi 和 Yejin Choi。Merlot：多模态神经脚本知识模型。*Advances in Neural Information Processing Systems*，34:23634–23651，2021。Xiaohua Zhai、Joan Puigcerver、Alexander Kolesnikov、Pierre Ruyssen、Carlos Riquelme、Mario Lucic、Josip Djolonga、Andre Susano Pinto、Maxim Neumann、Alexey Dosovitskiy 等。基于视觉任务适应基准的表征学习大规模研究。*arXiv preprint arXiv:1910.04867*，2019。Xiaohua Zhai、Alexander Kolesnikov、Neil Houlsby 和 Lucas Beyer。扩展视觉 Transformer。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 12104–12113 页，2022a。Xiaohua Zhai、Xiao Wang、Basil Mustafa、Andreas Steiner、Daniel Keysers、Alexander Kolesnikov 和 Lucas Beyer。LiT：通过锁定图像的文本调优实现零样本迁移。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 18123–18133 页，2022b。Pengchuan Zhang、Xiujun Li、Xiaowei Hu、Jianwei Yang、Lei Zhang、Lijuan Wang、Yejin Choi 和 Jianfeng Gao。VinVL：重新审视视觉-语言模型中的视觉表示。载于 *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*，第 5579–5588 页，2021。

A PaLI MODEL ADDITIONAL INFORMATION

A.1 PaLI MODEL DETAILS

Figure 3 visualizes some examples of PaLI on several tasks, such as image captioning, visual question answering, OCR-oriented captioning and question answering. Examples in multiple languages are shown as well.

Below, we show more specifics about the PaLI model and its components.

Model variants Table 7 lists the main PaLI models used where the largest is PaLI-17B of 17B parameters.

Model	Components	Image Encoder	Multimodal Encoder-Decoder	Total
PaLI-3B	ViT-G, mT5-L	1.8B	1.2B	3.0B
PaLI-15B	ViT-G, mT5-XXL	1.8B	13B	14.8B
PaLI-17B	ViT-e, mT5-XXL	3.9B	13B	16.9B

Table 7: The size in terms of number of parameters for the trained PaLI model versions.

ViT-e Backbone We show ViT-e's configuration in Table 8 alongside ViT-g and ViT-G for reference. Width, depth and MLP dimensions are all further scaled up in ViT-e, resulting in a model with 4B parameters. The model training setup is copied from the ViT-G model (Zhai et al., 2022a), on the JFT-3B dataset (Zhai et al., 2022a), with $16,384$ batch size, 224×224 resolution. We train the model for 1M steps using 0.0008 initial learning rate, with an inverse square-root learning rate decay, and a linear cool-down to zero for the final 100k steps. The only additional technique added is model souping (Wortsman et al., 2022): we run the 900K to 1M cool-down twice, once with inception cropping and once with resizing only. Thus, the final ViT-e model consists of the average weights of these two cool-downs. ViT-e is pretrained using the `big_vision` codebase (Beyer et al., 2022).

Name	Width	Depth	MLP	Heads	Params (M)	GFLOPs	
						224^2	384^2
g/14	1408	40	6144	16	1011	533.1	1596.4
G/14	1664	48	8192	16	1843	965.3	2859.9
e/14	1792	56	15360	16	3926	1980	5777

Table 8: ViT-e architecture details.

The overall model The overall PaLI models are implemented in `JAX/Flax` (Bradbury et al., 2018) using the open-source `T5X` (Roberts et al., 2022) and `Flaxformer` (Heek et al., 2020) frameworks. For the learning rate, we use a 1k-step linear warmup, followed by inverse square-root decay. For PaLI-3B, we use a peak learning rate of 1e-2. For larger models, PaLI-15B and PaLI-17B, we use a peak learning rate of 5e-3. We use the Adafactor (Shazeer & Stern, 2018) optimizer with $\beta_1 = 0$ and second-moment exponential decay set to 0.8.

The largest model, PaLI-17B, is pretrained using 1,024 GCP-TPUv4 chips for 7 days. It uses a four-way model partitioning (Roberts et al., 2022) and a batch size of 4,096. This is slightly less TPU resources than used to train other large vision and language models on TPUs. SimVLM used 2,048 GCP-TPUv3 for 5 days (Wang et al., 2021), while CoCa used 2,048 GCP-TPUv4 chips for 5 days (Yu et al., 2022). Flamingo used 1,536 GCP-TPUv4 chips for 15 days (Alayrac et al., 2022).

During training, the model passes over 1.6B images, one epoch over the entire pretraining dataset. The image resolution for this pass is 224×224. During training, only the parameters of the language component are updated and the vision component is frozen, which provides a boost in performance (Sec. 4.6).

Continuation of pretraining at higher image resolution For the largest model, PaLI-17B, we perform a further high-resolution (588×588) pre-finetuning for the multilingual tasks. When scaling

A PALI 模型补充信息

A.1 PALI 模型细节

图 3 可视化展示了 PaLI 在多项任务上的一些示例，例如图像描述、视觉问答、面向 OCR 的描述和问答。同时也展示了多种语言的示例。

下面，我们展示有关 PaLI 模型及其组件的更多具体信息。

模型变体 表 7 列出了所使用的主要 PaLI 模型，其中最大的是拥有 17B 参数的 PaLI-17B。

Model	Components	Image Encoder	Multimodal Encoder-Decoder	Total
PaLI-3B	ViT-G, mT5-L	1.8B	1.2B	3.0B
PaLI-15B	ViT-G, mT5-XXL	1.8B	13B	14.8B
PaLI-17B	ViT-e, mT5-XXL	3.9B	13B	16.9B

表 7：已训练的 PaLI 模型版本以参数数量衡量的规模。

ViT-e 骨干网络 我们在表 8 中展示了 ViT-e 的配置，并同时列出 ViT-g 和 ViT-G 以供参考。ViT-e 进一步扩大了宽度、深度和 MLP 维度，从而得到一个拥有 4B 参数的模型。模型训练设置沿用 ViT-G 模型（Zhai et al., 2022a），在 JFT-3B 数据集（Zhai et al., 2022a）上进行训练，批量大小为 16, 384，分辨率为 224×224。我们使用 0.0008 的初始学习率训练模型 1M 步，采用反平方根学习率衰减，并在最后 100k 步线性冷却至零。唯一额外加入的技术是模型汤（model souping）（Wortsman et al., 2022）：我们将 900K 到 1M 的冷却阶段运行两次，一次使用 inception 裁剪，另一次仅使用 resizing。因此，最终的 ViT-e 模型由这两次冷却所得权重的平均值组成。ViT-e 使用 big_vision 代码库（Beyer et al., 2022）进行预训练。

Name	**Width**	**Depth**	**MLP**	**Heads**	**Params (M)**	**GFLOPs**	
						224^2	384^2
g/14	1408	40	6144	16	1011	533.1	1596.4
G/14	1664	48	8192	16	1843	965.3	2859.9
e/14	1792	56	15360	16	3926	1980	5777

表 8：ViT-e 架构详情。

整体模型 整体 PaLI 模型使用开源 T5X（Roberts 等，2022）和 Flaxformer（Heek 等，2020）框架，在 JAX/Flax（Bradbury 等，2018）中实现。对于学习率，我们采用 1k 步线性预热，随后进行反平方根衰减。对于 PaLI-3B，我们使用 1e-2 的峰值学习率。对于更大的模型 PaLI-15B 和 PaLI-17B，我们使用 5e-3 的峰值学习率。我们使用 Adafactor（Shazeer & Stern, 2018）优化器，其中 $\beta_1 =$ 为 0，二阶矩指数衰减设为 0.8。

最大的模型 PaLI-17B 使用 1,024 个 GCP-TPUv4 芯片预训练了 7 天。它采用四路模型划分（Roberts 等，2022）和 4,096 的批量大小。与在 TPU 上训练其他大型视觉与语言模型所使用的 TPU 资源相比，这略少一些。SimVLM 使用 2,048 个 GCP-TPUv3 训练了 5 天（Wang 等，2021），而 CoCa 使用 2,048 个 GCP-TPUv4 芯片训练了 5 天（Yu 等，2022）。Flamingo 使用 1,536 个 GCP-TPUv4 芯片训练了 15 天（Alayrac 等，2022）。

在训练期间，模型会处理超过 16 亿张图像，即对整个预训练数据集进行一个 epoch。此次处理的图像分辨率为 224×224。在训练期间，仅更新语言组件的参数，视觉组件被冻结，这带来了性能提升（第 4.6 节）。

以更高图像分辨率继续预训练 对于最大的模型 PaLI-17B，我们针对多语言任务进一步进行高分辨率（588×588）预微调。在扩展时



Input: Generate the alt_text in EN
Output: A cellar filled with barrels of wine



Input: Generate the alt_text in EN
Output: a clock on a building that says 'lyvania' on it



Input: Generate the alt_text in EN
Output: Two helicopters are flying in the sky and one has a yellow stripe on the tail



Input: Generate the alt_text in FR
Output: Un arbre debout dans un champ avec un ciel violet
(A tree standing in a field with a purple sky)



Input: Generate the alt_text in TH
Output: ลา สี เทา เดิน ไป ตาม ถนน
(A gray donkey walks down the street)



Input: Generate the alt_text in ZH
Output: 一辆 电动 汽车 停 在 充电 桩 上 。
(An electric car parking on a charging station)



Input: Answer in EN: what time is it according to this radio
Output: 1254



Input: Answer in EN: what website is on the wall in back
Output: arsenaldirect.com



Input: Answer in EN: what is the brand of this watch
Output: seiko

Figure 3: PaLI addresses a variety of vision and language tasks across many languages, for example, image captioning, visual question answering, scene-text understanding, etc. Images from the publicly-available TextVQA (Singh et al., 2019) and TextCaps (Sidorov et al., 2020) datasets are shown, together with PaLI inputs and outputs.

up image resolution, the patch size is kept the same, and the number of patches are increased with higher resolution. We perform a 2D bilinear upsampling of the positional embedding to match the increased number of patches. This second stage of training is only for 10k steps at batch size 1024 (10M examples in total) and is performed on a subset of the full training mix. We simplify the mixture of data in this stage to focus on VQA, captioning and OCR capabilities, by including only the OCR, CC3M-35L and VQ^2A in the training mixture and making them equally weighted. In this high-resolution finetuning phase, all of the parameters of PaLI are updated. This high resolution phase was performed using 512 GCP-TPUv4 chips for an additional 3 days.

A.2 The Pretraining Task Mixture

Below are detailed descriptions of each component of our task mixture.



输入：生成英文替代文本 输出：一个装满葡萄酒桶的酒窖



输入：生成英文 alt_text 输出：一栋建筑上的钟，上面写着“lyvania”



输入：用英语生成 alt_text 输出：两架直升机正在天空中飞行，其中一架的尾部有一条黄色条纹



输入：生成法语的 alt_text 输出：一棵树立在田野中，天空呈紫色

(A tree standing in a field with a purple sky)



输入：生成泰语的 alt_text 输出：一头灰色的驴沿着道路行走

(A gray donkey walks down the street)



输入：生成中文替代文本 输出：一辆电动汽车停在充电桩上。

(An electric car parking on a charging station)



输入：用英语回答：根据这台收音机，现在几点 输出：1254



输入：用英语回答：后面墙上的网站是什么 输出：arsenaldirect.com



输入：用英语回答：这块手表是什么品牌 输出：seiko

图 3：PaLI 处理多种语言中的各种视觉和语言任务，例如图像描述、视觉问答、场景文本理解等。图中展示了来自公开可用的 TextVQA（Singh 等，2019）和 TextCaps（Sidorov 等，2020）数据集的图像，以及 PaLI 的输入和输出。

提高图像分辨率时，patch 大小保持不变，而 patch 数量会随着分辨率的提高而增加。我们对位置嵌入进行二维双线性上采样，以匹配增加后的 patch 数量。第二阶段训练仅进行 1 万步，批量大小为 1024（总计 1000 万个样本），并在完整训练混合数据的一个子集上进行。我们在此阶段简化数据混合，以专注于 VQA、图像描述和 OCR 能力：训练混合中仅包含 OCR、CC3M-35L 和 VQ^2A，并使它们具有相同权重。在这一高分辨率微调阶段，PaLI 的所有参数都会被更新。该高分辨率阶段使用 512 个 GCP-TPUv4 芯片额外进行了 3 天。

A.2 预训练任务混合

以下　是我们任务混合中各个组成部分的详细描述　　ure.

- **Span corruption on text-only data** uses the same technique described by Xue et al. (2021), corrupting 15% of the tokens from a given text-only example and using "sentinels" of the form $\langle \text{extra_id_}k \rangle$ for each corrupted span; the text-only examples are using a sample of 100M of text-only examples.
- **Split-captioning (SplitCap) on WebLI alt-text data** is inspired by the pretraining objective of Wang et al. (2021), and works by splitting each alt-text string randomly into two parts, $\langle \text{cap}_1 \rangle$ and $\langle \text{cap}_2 \rangle$. It uses the prompt "*Generate the alt_text in* $\langle \text{lang} \rangle$ *at* $\langle \text{pos} \rangle$*:* $\langle \text{cap}_1 \rangle$ $\langle \text{extra_id_0} \rangle$" (where $\langle \text{lang} \rangle$ is the language code of the alt-text string, and $\langle \text{pos} \rangle$ is the number of words in $\langle \text{cap}_1 \rangle$), with $\langle \text{cap}_2 \rangle$ as the target.
- **Captioning (Cap) on CC3M-35L on native and translated alt-text data** using the prompt "*Generate the alt_text in* $\langle \text{lang} \rangle$ *at 0:* $\langle \text{extra_id_0} \rangle$", with the alt-text string in language $\langle \text{lang} \rangle$ as the target. CC3M-35L is Conceptual Captions (Sharma et al., 2018) training data, translated into an additional 34 languages (the same as the non-English ones covered by Crossmodal-3600 (Thapliyal et al., 2022), except for Cusco-Quechua), for a total of 100M examples.
- **OCR on WebLI OCR-text data** using the prompt "*Generate the ocr_text in* $\langle \text{lang} \rangle$*:* $\langle \text{extra_id_0} \rangle$", with $\langle \text{OCR_text} \rangle$ as the target, where $\langle \text{OCR_text} \rangle$ is the concatenation of the annotated OCR texts in language $\langle \text{lang} \rangle$ (Kil et al., 2022) produced by the publicly available automatic service for the input image.
- **English and Cross-Lingual VQA on native and translated** VQ^2A**-CC3M-35L-100M VQA triplets** using, for a given $\langle image, [question], [answer] \rangle$ VQA triple, the prompt: "*Answer in EN: [question]* $\langle \text{extra_id_0} \rangle$", with $[answer]$ for the target. VQ^2A-CC3M-35L-100M is a 100M random subset of VQ^2A-CC3M (Changpinyo et al., 2022a), translated into the same additional 34 languages as mentioned above. Note that we use English answers in all instances here, as the English-native answers for VQA are often short and too prone to errors to perform out-of-context automatic translation.
- **English and Cross-Lingual visual question generation (VQG) on native and translated** VQ^2A**-CC3M-35L-100M VQA triplets** using, for a given $\langle image, [question], [answer] \rangle$ VQA triple, the prompt: "*Generate a question in* $\langle \text{lang} \rangle$ *for [answer]:* $\langle \text{extra_id_0} \rangle$", with $[question]$ in language $\langle \text{lang} \rangle$ as the target. Similarly, we use only English answers here.
- **English-only Object-Aware (OA) VQA** is based on VQA triplets derived from automatically-produced, non-exhaustive object labels, inspired by Piergiovanni et al. (2022a). We automatically generate 4 different prompt types, based on the available object labels, as follows. (1) Prompt: "*Answer in EN: List the objects present:* $\langle \text{extra_id_0} \rangle$", with the target: $\langle \text{object}_1 \rangle, \ldots, \langle \text{object}_\text{N} \rangle$. (2) Prompt: "*Answer in EN: Is* $\langle \text{object}_\text{k} \rangle$ *in the image?* $\langle \text{extra_id_0} \rangle$", with the target "Yes" or "No". (3) Prompt: "*Answer in EN: Is* $\langle \text{object}_1 \rangle$*,* $\ldots$*,* $\langle \text{object}_\text{N} \rangle$ *in the image?* $\langle \text{extra_id_0} \rangle$", with the target "Yes" or "No". (4) Prompt: "*Answer in EN: Which of* $\langle \text{object}_1 \rangle, \ldots, \langle \text{object}_\text{N} \rangle$ *are in the image?* $\langle \text{extra_id_0} \rangle$", with the target made of the list of object labels present. To create these examples, we require object-level annotations, for which we use Open Images (Kuznetsova et al., 2020), from which we create 50M examples.
- **Object detection** is a generative object-detection task inspired by Chen et al. (2021; 2022). The target sequence describes bounding-box coordinates and object labels, e.g. "*10 20 90 100 cat 20 30 100 100 dog*". The coordinates are in the y_{min} x_{min} y_{max} x_{max} order, and range between 0 and 999. Unlike Chen et al. (2021), the prompt used contains a set of positive and negative class labels, i.e. object classes that are present and not present in the image (e.g. "*detect cat and dog and leopard*"). The prompt is prefixed with the word "*detect*". For the datasets that do not have negative class labels explicitly defined, we randomly sample non-positive class labels. Since WebLI does not contain bounding box annotations, we train on a mixture of public datasets, totalling 16M images: Open Images (Kuznetsova et al., 2020), Visual Genome (Krishna et al., 2017), and Object365 (Shao et al., 2019). The datasets are de-duplicated against evaluation tasks. These examples are included to increase object awareness capabilities of the model.

Dataset mixing ratio for pretraining Table 9 provides the data mixing ratio for pretraining all PaLI variants.

- 纯文本数据上的跨度损坏使用 Xue 等人（2021）描述的相同技术，即从给定的纯文本示例中损坏 15% 的词元，并为每个损坏的跨度使用形式为 ⟨extra_id_k⟩ 的“哨兵”；纯文本示例使用的是 1 亿个纯文本示例的样本。
- WebLI 替代文本数据上的分割式说明文字生成（Split-captioning，SplitCap）受到 Wang 等人（2021）预训练目标的启发，其做法是将每个替代文本字符串随机分成两部分：⟨cap_1⟩ 和 ⟨cap_2⟩。它使用提示词“*Generate the alt_text in* ⟨lang⟩ *at* ⟨pos⟩*:* ⟨cap_1⟩ ⟨extra_id_0⟩”（其中 ⟨lang⟩ 是替代文本字符串的语言代码，⟨pos⟩ 是 ⟨cap_1⟩ 中的词数），并以 ⟨cap_2⟩ 作为目标。
- 在 CC3M-35L 上使用原生和翻译的替代文本数据进行图像描述（Cap），采用提示 "*Generate the alt_text in* ⟨lang⟩ *at 0:* ⟨extra_id_0⟩"，并将语言 ⟨lang⟩ 的替代文本字符串作为目标。CC3M-35L 是 Conceptual Captions（Sharma 等，2018）的训练数据，被翻译成另外 34 种语言（与 Crossmodal-3600（Thapliyal 等，2022）涵盖的非英语语言相同，但不包括库斯科克丘亚语），总计 1 亿个样本。
- 在 WebLI OCR-text 数据上进行 OCR，使用提示 “*Generate the ocr_text in* ⟨lang⟩*:* ⟨extra_id_0⟩”，并以 ⟨OCR_text⟩ 作为目标，其中 ⟨OCR_text⟩ 是由公开可用的自动服务针对输入图像生成的、语言为 ⟨lang⟩ (Kil et al., 2022) 的标注 OCR 文本的拼接。
- 在原生和翻译的 VQ^2A-CC3M-35L-100M VQA 三元组上进行英语和跨语言 VQA，对于给定的 $\langle image, [question], [answer] \rangle$ VQA 三元组，使用提示：“*Answer in EN: [question]* ⟨extra_id_0⟩”，并将 $[answer]$ 作为目标。VQ^2A-CC3M-35L - 100M 是 VQ^2A-CC3M（Changpinyo 等，2022a）的 1 亿随机子集，已翻译成上述相同的另外 34 种语言。请注意，我们在这里的所有实例中都使用英语答案，因为 VQA 的英语原生答案通常很短，而且在脱离上下文的自动翻译中太容易出错。
- 在原生和翻译后的 VQ^2A-CC3M-35L-100M VQA 三元组上进行英语和跨语言视觉问题生成（VQG），对于给定的 $\langle image, [question], [answer] \rangle$ VQA 三元组，使用提示：“*Generate a question in* ⟨lang⟩ *for [answer]:* ⟨extra_id_0⟩”，并以语言 ⟨lang⟩ 中的 $[question]$ 作为目标。同样，我们在这里也只使用英语答案。
- 仅英语的对象感知（OA）VQA 基于由自动生成的、非穷尽的对象标签派生出的 VQA 三元组，灵感来自 Piergiovanni 等人（2022a）。我们根据可用的对象标签自动生成 4 种不同的提示类型，如下所示。(1) 提示："*Answer in EN: List the objects present:* ⟨extra_id_0⟩"，目标为：⟨object_1⟩, . . . , ⟨object_N⟩。(2) 提示："*Answer in EN: Is* ⟨object_k⟩ *in the image?* ⟨extra_id_0⟩"，目标为“Yes”或“No”。(3) 提示："*Answer in EN: Is* ⟨object_1⟩*, . . . ,* ⟨object_N⟩ *in the image?* ⟨extra_id_0⟩"，目标为“Yes”或“No”。(4) 提示："*Answer in EN: Which of* ⟨object_1⟩*, . . . ,* ⟨object_N⟩ *are in the image?* ⟨extra_id_0⟩"，目标由存在的对象标签列表构成。为了创建这些示例，我们需要对象级标注，为此我们使用 Open Images（Kuznetsova 等人，2020），并从中创建了 5000 万个示例。
- 目标检测是一项生成式目标检测任务，灵感来自 Chen 等人（2021；2022）。目标序列描述边界框坐标和对象标签，例如“*10 20 90 100 cat 20 30 100 100 dog*”。坐标采用 y_{min} x_{min} y_{max} x_{max} 顺序，范围在 0 到 999 之间。与 Chen 等人（2021）不同，所使用的提示包含一组正类和负类标签，即图像中存在和不存在的对象类别（例如“*detect cat and dog and leopard*”）。提示以前缀词“*detect*”开头。对于未明确定义负类标签的数据集，我们随机采样非正类标签。由于 WebLI 不包含边界框标注，我们在多个公共数据集的混合数据上进行训练，总计 1600 万张图像：Open Images（Kuznetsova 等人，2020）、Visual Genome（Krishna 等人，2017）和 Object365（Shao 等人，2019）。这些数据集已针对评估任务进行去重。包含这些示例是为了增强模型的对象感知能力。

预训练的数据集混合比例 表 9 提供了用于预训练所有 PaLI 变体的数据混合比例。

	Text-only	WebLI alt-text	OCR	CC3M-35L	VQA	VQG	OA	Detection	Total
Amount (M)	100	1000	100	100	100	100	50	16	1566

Table 9: Mixing ratio of each task for pretraining

A.3 FINE-TUNING DETAILS

Hyperparameters for finetuning the V&L tasks We performed limited hyperparameter search for finetuning. The train steps is mostly selected based on dataset size. The batch size is selected among {128, 256, 512}, and the initial learning rate among {1e-5, 3e-5, 1e-4}. The optimizer setting for finetuning is the same as the setting for pretraining. Note that we did not perform the hyperparameter sweep over all possible combinations. Table 10 summarizes the hyperparameters corresponding to the main results.

Hyper-parameter	COCO & NoCaps	TextCaps	VizWiz-Cap	VQAv2	TextVQA	VizWiz-QA	OKVQA	ST-VQA
Dropout	0.1							
LR decay schedule	linear decay to zero							
Train	20k	10k	5k	20k	5k	5k	5k	5k
Batch size	256							
Initial (peak) LR	3e-5	1e-4	1e-4	1e-4	1e-4	1e-4	3e-5	1e-4

Table 10: Hyper-parameters used in fine-tuning experiments.

Setup for zero-shot image classification For each image, each class is scored using the prompt "*Generate alt_text in EN at 2: Photo of* ⟨extra_id_0⟩", scoring against all 1,000 classes with a target "⟨*en_class_name*⟩", where "⟨*en_class_name*⟩" stands for a classification label in English, such as *"goldfish"*, *"great white shark"*, etc.

B WEBLI DATASET DETAILS

The WebLI dataset covers about 10 billion images and 12 billion alt-texts in 109 languages. We further apply a publicly available automatic service to extract OCR annotations on all images, producing additional 29 billion image-OCR pairs. Examples and statistics for the WebLI corpus are shown in Figure 4.

Due to the scale of WebLI, to mitigate train-to-test leakage, we perform near de-duplication of the images against the train, validation, and test splits of 68 common vision/vision-language datasets. Eliminating these images from the WebLI dataset does not result in any significant shrinkage (0.36%), and avoids any potential "leakage" of examples from the pretraining setup to the downstream evaluation tasks.

To improve the data quality in terms of image-text alignment, we score image and alt-text pairs based on their cross-modal similarity. This score is measured with cosine similarity between embedding representations from each modality, computed as follows. The image embeddings are trained with a graph-based, semi-supervised representation learning approach, as described in Juan et al. (2019). Then, the text embeddings are learned using the frozen image embeddings, based on a contrastive approach using a Transformer encoder for the text, which forces both modality representations to the same embedding space.

We tune a threshold on the image and alt-text pairs' score, and retain only the top 10% best scoring of the original WebLI image-text pairs (about 1B examples), which we use to train PaLI.

[1]The second image is by jopradier (original), used under the CC BY-NC-SA 2.0 license. Remaining images are also used with permissions.

	Text-only	WebLI alt-text	OCR	CC3M-35L	VQA	VQG	OA	Detection	Total
Amount (M)	100	1000	100	100	100	100	50	16	1566

表 9：预训练中各任务的混合比例

A.3 微调细节

V&L 任务微调的超参数 我们对微调进行了有限的超参数搜索。训练步数主要根据数据集大小来选择。批大小从 {128, 256, 512} 中选择，初始学习率从 {1e-5, 3e-5, 1e-4} 中选择。微调的优化器设置与预训练的设置相同。请注意，我们并未对所有可能的组合进行超参数遍历。表 10 总结了与主要结果对应的超参数。

Hyper-parameter	COCO & NoCaps	TextCaps	VizWiz-Cap	VQAv2	TextVQA	VizWiz-QA	OKVQA	ST-VQA
Dropout	0.1							
LR decay schedule	linear decay to zero							
Train	20k	10k	5k	20k	5k	5k	5k	5k
Batch size	256							
Initial (peak) LR	3e-5	1e-4	1e-4	1e-4	1e-4	1e-4	3e-5	1e-4

表10：微调实验中使用的超参数。

零样本图像分类设置 对于每张图像，使用提示词“*Generate alt_text in EN at 2: Photo of* ⟨extra_id_0⟩”对每个类别进行评分，并针对所有 1,000 个类别以目标“⟨*en_class_name*⟩”进行评分，其中“⟨*en_class_name*⟩”表示英文分类标签，例如 *"goldfish"*、*"great white shark"* 等。

B WEBLI 数据集详情

WebLI 数据集涵盖约 100 亿张图像和 120 亿条 109 种语言的替代文本。我们进一步应用一项公开可用的自动化服务，对所有图像提取 OCR 标注，额外生成了 290 亿个图像-OCR 对。WebLI 语料库的示例和统计信息如图 4 所示。

鉴于 WebLI 的规模，为了减少训练到测试的泄漏，我们对图像与 68 个常见视觉/视觉-语言数据集的训练、验证和测试划分进行近似去重。从 WebLI 数据集中移除这些图像并不会导致显著缩减（0.36%），并且避免了从预训练设置到下游评估任务的样本发生任何潜在“泄漏”。

为了提升图像-文本对齐方面的数据质量，我们基于跨模态相似度对图像和替代文本配对进行评分。该分数通过每种模态的嵌入表示之间的余弦相似度来衡量，计算方式如下。图像嵌入采用基于图的半监督表征学习方法进行训练，如 Juan 等人（2019）所述。随后，文本嵌入基于冻结的图像嵌入进行学习，采用一种对比学习方法，并为文本使用 Transformer 编码器，从而迫使两种模态的表示进入同一嵌入空间。

我们根据图像与替代文本配对的得分来调整阈值，并仅保留原始 WebLI 图像-文本配对中得分最高的前 10%（约 10 亿个样本），用于训练 PaLI。

[1]The second image is by jopradier (original), used under the CC BY-NC-SA 2.0 license. Remaining images are also used with permissions.

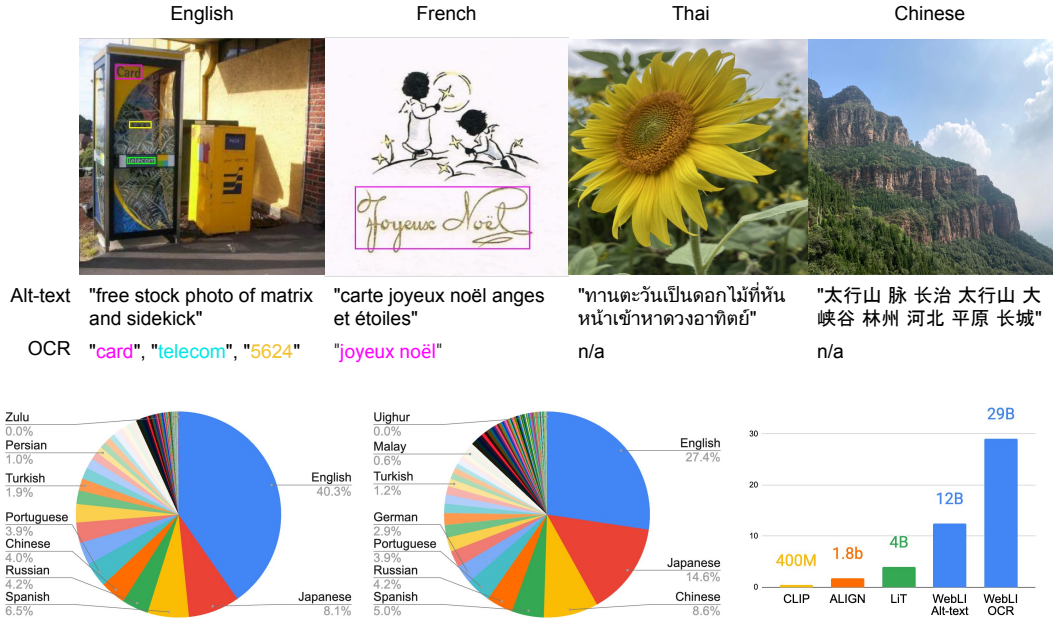


Figure 4: The WebLI dataset. Top: Sampled images[1] associated with multilingual alt-text (available) and OCR (computed using publicly available API). Bottom left/middle: Statistics of recognized languages from alt-text/OCR. Bottom right: Image-text pair counts, compared against other large-scale vision-language datasets.

C Additional Experimental Results

C.1 Language-only evaluation

In Table 11, we evaluate te performance of PaLI on a range of language understanding benchmarks, in order to verify that the language-only capabilities of the model have been preserved. More specifically we compare mT5-XXL and PaLI-17B, evaluating on the English-only SuperGLUE benchmark (Wang et al., 2019a), and on three multilingual benchmarks from the XTREME (Hu et al., 2020): XNLI (Conneau et al., 2018), which is a textual entailment task covering 14 languages, XQuAD (Artetxe et al., 2020) and TyDiQA-GoldP (Clark et al., 2020), which are both question-answering tasks covering 10 and 11 languages, respectively.

Model Method	SuperGLUE FT	XNLI ZS	XQuAD ZS	TyDiQA-GoldP ZS
Metric	Avg. Score	Accuracy	F1/EM	F1/EM
mT5-XXL (Xue et al., 2021)	89.2	85.0	82.5 / 66.8	80.8 / 65.9
mT5-XXL (our setting)	89.3	84.5	82.6 / 66.6	81.6 / 66.3
PaLI-17B	88.2	84.9	81.8 / 66.0	81.2 / 66.5

Table 11: Results on SuperGLUE and three XTREME tasks. The first row is the result reported by mT5 (Xue et al., 2021) and ByT5 (Xue et al., 2022) paper. The second row is our repetition using the publicly available mT5-XXL checkpoint, which is also the starting point for PaLI-17B. The third row results are using the trained PaLI-17B model.

C.2 Additional scaling results

Figure 5 shows that the model scaling impacts significantly the performance for multiple languages. We can see that PaLI-17B improves substantially over PaLI-3B across languages. We also include a plot where for a subset of 600 examples, we back-translate the predictions from six languages,

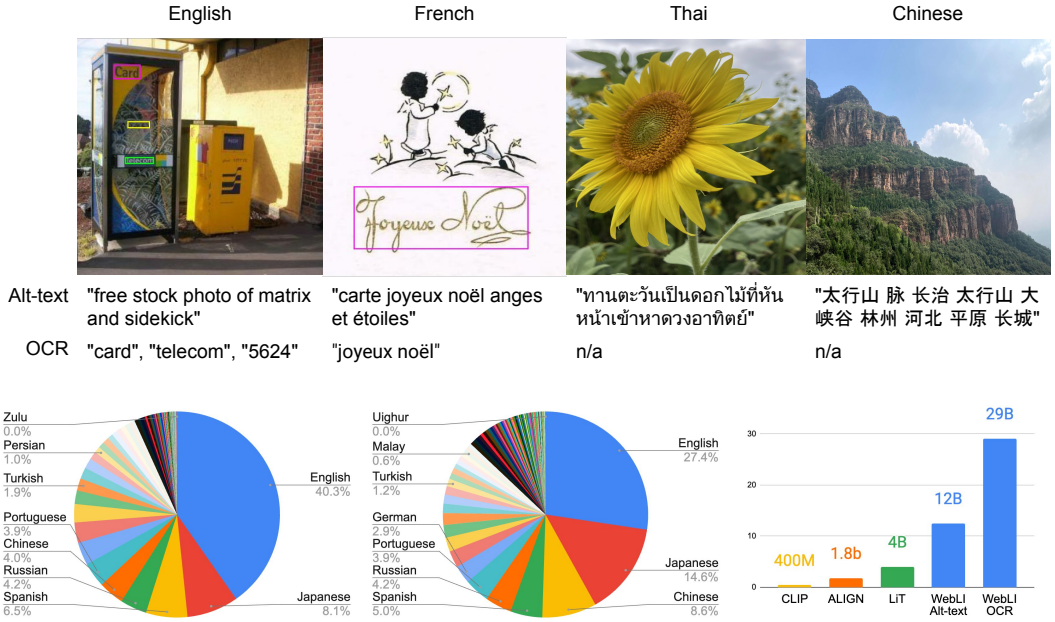


图 4：WebLI 数据集。上方：采样图像[1]，关联有多语言替代文本（可用）和 OCR（使用公开可用的 API 计算）。左下/中下：从替代文本/OCR 中识别出的语言统计。右下：图像-文本对数量，并与其他大规模视觉-语言数据集进行比较。

C 补充实验结果

C.1 仅语言评估

在表 11 中，我们在一系列语言理解基准上评估了 PaLI 的性能，以验证该模型的纯语言能力得到了保留。更具体地说，我们比较了 mT5-XXL 和 PaLI-17B，并在仅英语的 SuperGLUE 基准（Wang 等，2019a）以及来自 XTREME（Hu 等，2020）的三个多语言基准上进行评估：XNLI（Conneau 等，2018），这是一项涵盖 14 种语言的文本蕴含任务；XQuAD（Artetxe 等，2020）和 TyDiQA-GoldP（Clark 等，2020），二者均为问答任务，分别涵盖 10 种和 11 种语言。

Model Method	SuperGLUE FT	XNLI ZS	XQuAD ZS	TyDiQA-GoldP ZS
Metric	Avg. Score	Accuracy	F1/EM	F1/EM
mT5-XXL (Xue et al., 2021)	89.2	85.0	82.5 / 66.8	80.8 / 65.9
mT5-XXL (our setting)	89.3	84.5	82.6 / 66.6	81.6 / 66.3
PaLI-17B	88.2	84.9	81.8 / 66.0	81.2 / 66.5

表 11：在 SuperGLUE 和三个 XTREME 任务上的结果。第一行是 mT5（Xue 等，2021）和 ByT5（Xue 等，2022）论文报告的结果。第二行是我们使用公开可用的 mT5-XXL 检查点进行复现的结果，该检查点也是 PaLI-17B 的起点。第三行结果使用的是训练后的 PaLI-17B 模型。

C.2 其他缩放结果

图5显示，模型规模扩展显著影响多种语言的性能。我们可以看到，PaLI-17B 在各语言上的表现相比 PaLI-3B 有大幅提升。我们还包含了一张图，其中针对 600 个示例的子集，我们将来自六种语言的预测结果进行回译，

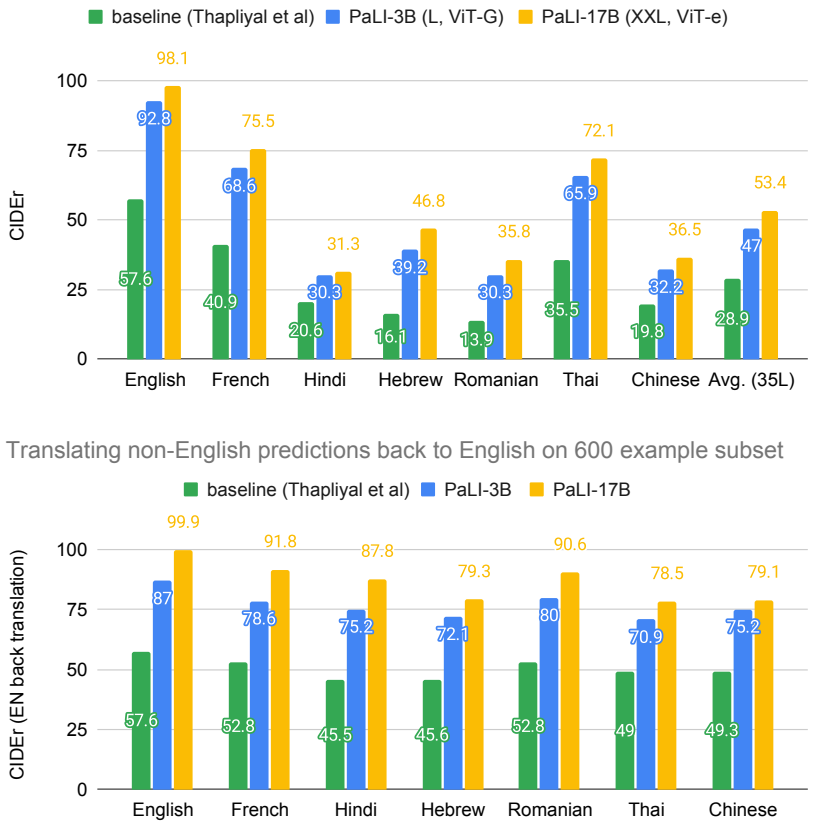


Figure 5: PaLI Scaling performance across multiple languages (See Table 2), using the Crossmodal-3600 benchmark. Larger scale models are important for better performance in these languages, especially low resource ones. (Top) CIDEr scores computed using predictions in each language. (Bottom) For the six languages French, Hindi, Hebrew, Romanian, Thai and Chinese, we sample a 600-example subset and back-translate the non-English predictions to English, and computed the CIDEr score vs. the same English references.

including French, Hindi, Hebrew, Romanian, Thai and Chinese to English and compute the CIDEr score against English references for a better comparison to the English quality. The result shows that the captioning quality across languages is fairly consistent.

We also trained a 5B PaLI model consisting of mT5-Large and ViT-e for additional datapoints. We evaluated this 5B model on two representative captioning and VQA benchmarks, COCO-Cap and OKVQA, and the results are shown in Table 12. We note that the training mixture and hyperparameters of this PaLI-5B checkpoint are slightly different from other PaLI sizes, but the results are still indicative and supportive of our conclusions regarding the value of joint scaling.

On COCO, the improvement from PaLI-3B to 5B (+2.1 CIDEr points) is slightly smaller than the improvement from PaLI-15B to 17B (+2.8). On OKVQA, it is likely that the benefit of having ViT-e cannot be exploited by the mT5-Large enc-dec as much as that by the mT5-XXL on VQA tasks, which require stronger language-understanding capabilities than Image Captioning tasks. In general, it is clear that scaling ViT still has much better return on investment (see the last column in Table 12), even for PaLI-5B where the ViT model is much larger than the encoder-decoder backbone. Note that we computed RoI as "improvement per 1B parameter", using COCO and OKVQA numbers as performance indicators.

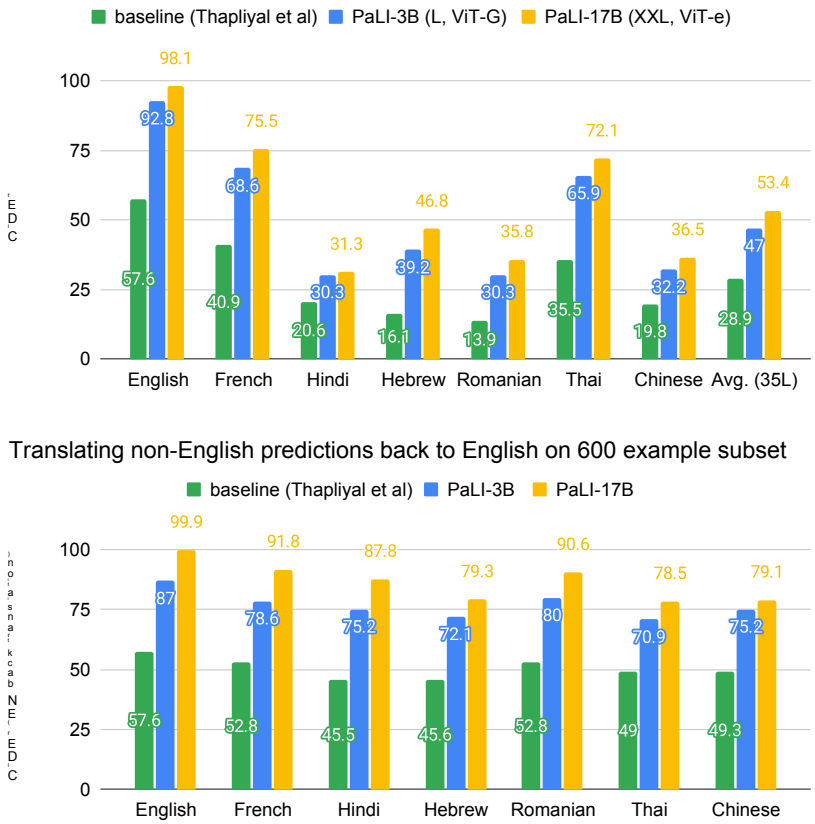


图 5：PaLI 在多种语言上的扩展性能（见表 2），使用 Crossmodal-3600 基准。更大规模的模型对于提升这些语言的性能很重要，尤其是低资源语言。（上）使用各语言的预测结果计算 CIDEr 分数。（下）对于法语、印地语、希伯来语、罗马尼亚语、泰语和中文这六种语言，我们抽取了一个包含 600 个样本的子集，并将非英语预测结果反向翻译成英语，然后与相同的英语参考答案计算 CIDEr 分数。

包括法语、印地语、希伯来语、罗马尼亚语、泰语和中文到英语的翻译，并根据英语参考计算 CIDEr 分数，以便更好地与英语质量进行比较。结果表明，各语言的字幕生成质量相当一致。

我们还训练了一个由 mT5-Large 和 ViT-e 组成的 5B PaLI 模型，以获得额外的数据点。我们在两个具有代表性的图像描述生成和 VQA 基准 COCO-Cap 与 OKVQA 上评估了该 5B 模型，结果如表 12 所示。我们注意到，该 PaLI-5B 检查点的训练混合数据和超参数与其他规模的 PaLI 略有不同，但其结果仍然具有指示性，并支持我们关于联合缩放价值的结论。

在 COCO 上，从 PaLI-3B 到 5B 的提升（+2.1 个 CIDEr 点）略小于从 PaLI-15B 到 17B 的提升（+2.8）。在 OKVQA 上，拥有 ViT-e 的收益很可能无法被 mT5-Large 编码器-解码器像 mT5-XXL 在 VQA 任务中那样充分利用，因为 VQA 任务比图像描述任务需要更强的语言理解能力。总体而言，很明显，扩展 ViT 仍然具有更高的投资回报率（见表 12 最后一列），即使对于 PaLI-5B 也是如此，其中 ViT 模型远大于编码器-解码器主干。请注意，我们使用 COCO 和 OKVQA 的数值作为性能指标，将 RoI 计算为“每 10 亿参数带来的提升”。

Model	Component	COCO-Cap @490 res	OKVQA @490 res	Improvement per 1B more params
PaLI-3B	mT5-Large & ViT-G	145.4	52.4	-
PaLI-5B	mT5-Large & ViT-e	147.5	53.8	+0.9 per 1B more **ViT** params (vs PaLI-3B)
PaLI-15B	mT5-XXL & ViT-G	146.2	56.5	+0.2 per 1B more **mT5** params (vs PaLI-3B)
PaLI-17B	mT5-XXL & ViT-e	149.0	62.4	+2.2 per 1B more **ViT** params (vs PaLI-15B) +0.4 per 1B more **mT5** params (vs PaLI-5B)

Table 12: Result on a 5B version of PaLI consisting of mT5-Large and ViT-e. Results on COCO-Cap and OKVQA with 490×490 are shown together with other sizes.

Model	TallyQA simple	TallyQA complex	Weighted average
PaLI-700M	66.9	55.6	62.4
PaLI-3B	72.0	56.7	65.9
PaLI-17B	76.2	65.5	71.9

Table 13: Result of 700M, 3B and 17B versions of PaLI on TallyQA (Acharya et al., 2019).

C.3 Additional ablations

Table 14 shows that initializing from unimodal checkpoint plays a critical role in PaLI's quality. Table 15 shows that freezing ViT during pretraining leads to an improvement in downstream finetuning on COCO.

Table 16 shows the effect of the non-English part of WebLI data. The table shows two sets of comparison for the pretraining data. 1) Using only the English subset of WebLI vs using only the whole WebLI. 2) Taking out the non-EN part of WebLI from the full mix vs using the full mix. This set of comparison results is performed with a 1.5B version of PaLI model, consisting of mT5-Large and ViT-L (with 300M parameters). This model has a similar parameter ratio (20% for ViT) compared with PaLI-17B (23%). Each model is pretrained to cover 200M of the data. All downstream benchmarks are fine-tuned and evaluated at 224×224 image resolution. The six non-En languages (6L) for XM-3600 are fr, hi, iw, ro, th and zh, and "7L" for xGQA are en, bn, de, id, ko, pt, ru, zh, both are the same as those included in Table 2 and Table 4. The takeaways are as follows:

- (comparison 1, row #1 vs row #2) With only the English portion of WebLI, the model's multilingual captioning capability remains very low (as measured on XM-3600), even with further finetuning on COCO-35L. There is also a clear drop in cross-lingual VQA performance on xGQA.
- (comparison 2, row #3 vs row #4) Taking away the multilingual part of WebLI from the full mixture, which still contains other translated multilingual/cross-lingual datasets (CC3M-35L, VQ2A-CC3M-35L, VQG-CC3M-35L), still has a significant impact on XM-3600 performance. On xGQA, because of the cross-lingual training source VQ2A-CC3M-35L, the impact of removing non-EN WebLI data is reduced but still apparent. With the non-EN WebLI data in the full mix, xGQA performance improves by +0.4 overall and is better than or equal to with only the WebLI-EN in every language.
- Last but definitely not least, there is an interesting result: when training with all the languages of WebLI, the model is performing better on (English) COCO captions, compared to training with English-only WebLI (about +2 CIDEr points). This suggests that 1) the multilingual WebLI may contain extra images with richer objects and their descriptions compared with the English-only subset 2) the model may be able to exploit the shared linguistic structure across languages, benefiting from transfer learning across languages.

Model	Component	COCO-Cap @490 res	OKVQA @490 res	Improvement per 1B more params
PaLI-3B	mT5-Large & ViT-G	145.4	52.4	-
PaLI-5B	mT5-Large & ViT-e	147.5	53.8	+0.9 per 1B more **ViT** params (vs PaLI-3B)
PaLI-15B	mT5-XXL & ViT-G	146.2	56.5	+0.2 per 1B more **mT5** params (vs PaLI-3B)
PaLI-17B	mT5-XXL & ViT-e	149.0	62.4	+2.2 per 1B more **ViT** params (vs PaLI-15B) +0.4 per 1B more **mT5** params (vs PaLI-5B)

表12：由 mT5-Large 和 ViT-e 组成的 PaLI 5B 版本的结果。展示了在 490×490 下 COCO-Cap 和 OKVQA 的结果，并与其他尺寸的结果一起列出。

Model	TallyQA simple	TallyQA complex	Weighted average
PaLI-700M	66.9	55.6	62.4
PaLI-3B	72.0	56.7	65.9
PaLI-17B	76.2	65.5	71.9

表 13：PaLI 的 700M、3B 和 17B 版本在 TallyQA（Acharya 等，2019）上的结果。

C.3 额外消融实验

表 14 表明，从单模态检查点初始化在 PaLI 的质量中起着关键作用。表 15 表明，在预训练期间冻结 ViT 会提升在 COCO 上的下游微调效果。

表16展示了 WebLI 数据中非英语部分的影响。该表展示了两组针对预训练数据的比较：1）仅使用 WebLI 的英语子集与仅使用整个 WebLI。2）从完整混合数据中去除 WebLI 的非英语部分与使用完整混合数据。该组比较结果是在 PaLI 模型的 1.5B 版本上进行的，该模型由 mT5-Large 和 ViT-L（含 300M 参数）组成。与 PaLI-17B（23%）相比，该模型具有相似的参数比例（ViT 占 20%）。每个模型都进行预训练，覆盖 200M 数据。所有下游基准都在 224×224 的图像分辨率下进行微调和评估。XM-3600 的六种非英语语言（6L）为 fr、hi、iw、ro、th 和 zh，而 xGQA 的“7L”为 en、bn、de、id、ko、pt、ru、zh，二者与表2和表4中包含的语言相同。主要结论如下：

- (比较 1，第 #1 行与第 #2 行) 仅使用 WebLI 的英语部分时，模型的多语言图像描述生成能力仍然很低（在 XM-3600 上测得），即使在 COCO-35L 上进一步微调也是如此。在 xGQA 上的跨语言视觉问答性能也出现了明显下降。
- (比较 2，第 #3 行 vs 第 #4 行) 从完整混合数据中移除 WebLI 的多语言部分后，尽管其中仍包含其他翻译的多语言/跨语言数据集（CC3M-35L、VQ2A-CC3M-35L、VQG-CC3M-35L），仍会对 XM-3600 性能产生显著影响。在 xGQA 上，由于存在跨语言训练来源 VQ2A-CC3M-35L，移除非英语 WebLI 数据的影响有所降低，但仍然明显。在完整混合数据中包含非英语 WebLI 数据时，xGQA 的整体性能提升了 +0.4，并且在每种语言上都优于或等同于仅使用 WebLI-EN 的情况。
- 最后但同样重要的是，还有一个有趣的结果：与仅使用英文 WebLI 进行训练相比，使用 WebLI 的所有语言进行训练时，模型在（英文）COCO 图像描述上的表现更好（约高出 +2 个 CIDEr 分数点）。这表明：1）与仅英文子集相比，多语言 WebLI 可能包含更多具有更丰富对象及其描述的图像；2）模型可能能够利用跨语言的共享语言结构，从跨语言迁移学习中受益。

Model	Initialization	COCO (Karp. test)	XM-3600	TextVQA
PaLI-3B	From mT5-Large and ViT-G	141.4	93.8 (EN) / 42.5 (6L)	41.6
	From scratch	72.8	22.1 (EN) / 10.1 (6L)	12.8

Table 14: Comparison between PaLI's initializing from existing unimodal checkpoints and initializing the parameter from scratch. The setup is the same as the main ablation result Table 6.

Model	ViT during finetuning	ViT during pretraining	COCO (Karp. test)
PaLI-3B	Fine-tuned	Frozen	139.3
		Fine-tuned	138.8
PaLI-15B	Fine-tuned	Frozen	141.4
		Fine-tuned	140.1

Table 15: Comparison of performance on COCO for Frozen versus fine-tuned ViT during a short period of pretraining. In this comparison, finetuning of COCO is performed at resolution 224×224.

C.4 Evaluation of PaLI's Visual Component: ViT-e

Table 17 compares the ViT-e architecture with the smaller ViT-G and ViT-g architectures on vision only and vision-language tasks. The results suggest that V&L tasks could benefit more from scaling up the vision backbone, even on the high end. In Table 18, we fine-tune the pretrained ViT-e model on the ImageNet dataset, and then report the evaluation scores on several out-of-distribution test variants: ImageNet-v2, ObjectNet, and ReaL (Beyer et al., 2020). We follow the finetuning protocol of Zhai et al. (2022a), but use a 560×560 resolution. We evaluate the fine-tuned model at 644×644 (Touvron et al., 2019) (chosen according to a held-out 2% of the training set), results are reported in Table 18. ViT-e achieves 90.9% top-1 accuracy on ImageNet and shows clear benefits on the OOD benchmarks.

Since ViT-e is new and has not been evaluated in the prior work, we evaluate its standalone performance. For this, we perform supervised fine-tuning on standard classification tasks. Additionally, we perform LiT transfer (Zhai et al., 2022b) to evaluate the frozen representation quality in a zero-shot setup.

We follow LiT (Zhai et al., 2022b) to add zero-shot transfer capabilities to the (frozen) ViT-e model, the visual component of PaLI. More specifically, we tune a text encoder, when the ViT image encoder is frozen. We use the English subset of the WebLI dataset for the text encoder training, since all evaluation tasks in Table 19 are in English. These results highlight that going from ViT-g to ViT-e provides consistently better results. Notably, LiT with ViT-e achieves 84.9% zero-shot accuracy on the challenging out-of-distribution ObjectNet test set, setting the new state-of-the-art. The VTAB-Natural benchmark (Zhai et al., 2019) consists of seven diverse natural image datasets, for which LiT also benefits from ViT-e over ViT-g. Detailed results on each VTAB-Natural task are in Appendix C.6.

We also test multilingual performance using WebLI in this setting. We further perform LiT transfer using the same multilingual WebLI dataset as used to train PaLI, and use Crossmodal-3600 to evaluate the cross-lingual image-text retrieval performance. Figure 6 shows that LiT ViT-e pretrained on the English subset substantially outperforms the same model pretrained on the multilingual dataset. The same observation applies to a few languages that are similar to English, e.g. Spanish (es), French (fr), Italian (it). However, the multilingual model performs much better on most other languages, especially those with a non-latin script such as Chinese (zh), Japanese (ja), Korean (ko), and Hebrew (iw). On average (avg), the multilingual LiT ViT-e outperforms the English-only model by a large margin. More results could be found in Table 23. These results highlight the importance of having good multilingual benchmarks to measure the benefits of training models on diverse datasets such as WebLI.

C.5 Results on TextCaps, TextVQA and VizWiz-QA without Detected OCR as Input

In the main text, we presented results on TextCaps, TextVQA, VizWiz-Cap, VizWiz-QA and ST-VQA with detected OCR strings as input. Following Kil et al. (2022), we order the OCR items based

Model	Initialization	COCO (Karp. test)	XM-3600	TextVQA
PaLI-3B	From mT5-Large and ViT-G	141.4	93.8 (EN) / 42.5 (6L)	41.6
	From scratch	72.8	22.1 (EN) / 10.1 (6L)	12.8

表14：PaLI 从现有单模态检查点初始化与从零开始初始化参数的比较。设置与主要消融结果表6相同。

Model	ViT during finetuning	ViT during pretraining	COCO (Karp. test)
PaLI-3B	Fine-tuned	Frozen	139.3
		Fine-tuned	138.8
PaLI-15B	Fine-tuned	Frozen	141.4
		Fine-tuned	140.1

表 15：短时间预训练期间，冻结的 ViT 与微调的 ViT 在 COCO 上的性能比较。在此比较中，COCO 的微调在 224×224 分辨率下进行。

C.4 PALI 的视觉组件评估：VIT-E

表17比较了 ViT-e 架构与较小的 ViT-G 和 ViT-g 架构在纯视觉任务和视觉-语言任务上的表现。结果表明，即使在高端模型范围内，V&L 任务也可能从扩大视觉骨干网络规模中获得更多收益。在表18中，我们在 ImageNet 数据集上对预训练的 ViT-e 模型进行微调，然后报告其在若干分布外测试变体上的评估分数：ImageNet-v2、ObjectNet 和 ReaL（Beyer 等，2020）。我们遵循 Zhai 等（2022a）的微调协议，但使用 560×560 的分辨率。我们以 644×644（Touvron 等，2019）的分辨率评估微调后的模型（根据留出的 2% 训练集选择），结果见表18。ViT-e 在 ImageNet 上达到 90.9% 的 top-1 准确率，并在 OOD 基准上显示出明显优势。

由于 ViT-e 是新的，且此前的工作尚未对其进行评估，因此我们评估其独立性能。为此，我们在标准分类任务上进行监督式微调。此外，我们还进行 LiT 迁移（Zhai 等，2022b），以在零样本设置下评估冻结表征的质量。

我们遵循 LiT（Zhai 等，2022b）为（冻结的）ViT-e 模型（PaLI 的视觉组件）添加零样本迁移能力。更具体地说，我们在冻结 ViT 图像编码器的情况下调优文本编码器。由于表 19 中的所有评估任务均为英语，我们使用 WebLI 数据集的英语子集来训练文本编码器。这些结果表明，从 ViT-g 转向 ViT-e 能够持续带来更好的结果。值得注意的是，采用 ViT-e 的 LiT 在具有挑战性的分布外 ObjectNet 测试集上取得了 84.9% 的零样本准确率，创下了新的最先进水平。VTAB-Natural 基准（Zhai 等，2019）由七个多样化的自然图像数据集组成，在这些数据集上，LiT 使用 ViT-e 相比 ViT-g 同样受益。每个 VTAB-Natural 任务的详细结果见附录 C.6。

我们还在该设置下使用 WebLI 测试多语言性能。我们进一步使用与训练 PaLI 相同的多语言 WebLI 数据集进行 LiT 迁移，并使用 Crossmodal-3600 评估跨语言图像-文本检索性能。图 6 显示，在英语子集上预训练的 LiT ViT-e 显著优于在多语言数据集上预训练的同一模型。同样的观察也适用于一些与英语相似的语言，例如西班牙语（es）、法语（fr）、意大利语（it）。然而，多语言模型在大多数其他语言上表现要好得多，尤其是那些使用非拉丁文字的语言，如中文（zh）、日语（ja）、韩语（ko）和希伯来语（iw）。平均而言（avg），多语言 LiT ViT-e 大幅优于仅英语模型。更多结果见表 23。这些结果突显了拥有良好多语言基准的重要性，以衡量在 WebLI 等多样化数据集上训练模型所带来的收益。

C.5 在不将检测到的 OCR 作为输入的情况下，在 TEXTCAPS、TEXTVQA 和 VIZWIZ-QA 上的结果

在正文中，我们展示了在 TextCaps、TextVQA、VizWiz-Cap、VizWiz-QA 和 ST-VQA 上以检测到的 OCR 字符串作为输入的结果。按照 Kil 等人（2022）的做法，我们对 OCR 项进行排序，依据

Pretraining Data	XM-3600 (FT on COCO-35L)	COCO-Cap	xGQA (FT on VQAv2-13L)
only WebLI-en	86.0 (en) / 8.2 (6L)	132.2	40.6 (en) / 34.0 (7L)
only WebLI	87.2 (en) / 30.0 (6L)	134.3	42.8 (en) / 38.6 (7L)
WebLI-en & rest of PaLI mix	91.2 (en) / 39.0 (6L)	135.3	44.9 (en) / 40.9 (7L)
Full PaLI mix	92.2 (en) / 41.9 (6L)	135.4	45.1 (en) / 41.3 (7L)

Table 16: Ablation studies on the effect of including the multilingual examples of WebLI on multi-(cross-)lingual benchmarks XM-3600 and xGQA. We also included the English benchmark COCO-Captions in the comparison. This set of comparison results is performed with a 1.5B version of PaLI model, consisting of mT5-Large and ViT-L (with 300M parameters).

	INet-10	INet-25	COCO	VQAv2
ViT-g	84.5	85.4	-	-
ViT-G	84.9	85.6	146.2	82.9
ViT-e	85.2	85.8	149	83.4

Table 17: Impact of scaling ViT. For vision-only tasks, we report 10-shot and 25-shot accuracy on ImageNet. For vision-language tasks, ViT models are paired with the mT5-XXL model in PaLI and we report captioning (COCO) and VQA (VQAv2). For direct comparison, results with ViT-e on COCO and VQAv2 do not include the high resolution phase of pretraining.

Model	INet	INet-v2	ObjNet	ReaL
ViT-G	90.5	83.3	70.5	90.8
CoCa	91.0	-	-	-
ViT-e	90.9	84.3	72.0	91.1

Table 18: ViT-e on ImageNet and OOD test sets.

on their locations in the image, from top left to bottom right. We only include the OCR strings themselves, without the OCR-item locations provided by the API. GIT2 (Wang et al., 2022a) has demonstrated strong performance without the OCR input, while PaLI-17B shows the superiority of levaraging a specialized OCR system for a better recipe to solve these tasks.

Table 20 shows the results on TextCaps, TextVQA and VizWiz-QA without the detected OCR strings as input. PaLI slightly suffers without OCR input, while its performance remains close to the first version of GIT. This result may suggest that the significantly larger vocab of PaLI adds further difficulty to OCR string generation.

However, for VizWiz-QA, PaLI establishes SOTA performance without OCR input.

C.6 Detailed VTAB Results

For the VTAB benchmark (Zhai et al., 2019), we follow the methodology outlined in (Zhai et al., 2022b). PaLI sets a new state-of-the-art zero-shot performance for the "natural" subset (see Table 21).

C.7 Top 5 Accuracy on Zero-shot ImageNet Datasets

Table 22 shows the Top 5 Accuracy results on Zero-shot evaluation on ImageNet Datasets.

C.8 More Zero-shot Image-text Retrieval Results on Crossmodal-3600

Table 23 shows more zero-shot image-text retrieval results on Crossmodal-3600.

D Model Fairness, Biases, and Other Potential Issues

Models trained on web data are at risk of being biased or unfair due to biases in that data. A first step towards addressing those risks is being transparent about their existence, and then measuring them.

Pretraining Data	XM-3600 (FT on COCO-35L)	COCO-Cap	xGQA (FT on VQAv2-13L)
only WebLI-en	86.0 (en) / 8.2 (6L)	132.2	40.6 (en) / 34.0 (7L)
only WebLI	87.2 (en) / 30.0 (6L)	134.3	42.8 (en) / 38.6 (7L)
WebLI-en & rest of PaLI mix	91.2 (en) / 39.0 (6L)	135.3	44.9 (en) / 40.9 (7L)
Full PaLI mix	92.2 (en) / 41.9 (6L)	135.4	45.1 (en) / 41.3 (7L)

表16：关于纳入 WebLI 多语言样本对多（跨）语言基准 XM-3600 和 xGQA 影响的消融研究。我们还在比较中加入了英语基准 COCO-Captions。这组比较结果使用 PaLI 模型的 1.5B 版本完成，该模型由 mT5-Large 和 ViT-L（含 300M 参数）组成。

	INet-10	INet-25	COCO	VQAv2
ViT-g	84.5	85.4	-	-
ViT-G	84.9	85.6	146.2	82.9
ViT-e	85.2	85.8	149	83.4

表 17：扩展 ViT 的影响。对于纯视觉任务，我们报告 ImageNet 上的 10-shot 和 25-shot 准确率。对于视觉-语言任务，ViT 模型在 PaLI 中与 mT5-XXL 模型配对，并报告图像描述（COCO）和 VQA（VQAv2）结果。为便于直接比较，ViT-e 在 COCO 和 VQAv2 上的结果不包括预训练的高分辨率阶段。

Model	INet	INet-v2	ObjNet	ReaL
ViT-G	90.5	83.3	70.5	90.8
CoCa	91.0	-	-	-
ViT-e	90.9	84.3	72.0	91.1

表 18：ViT-e 在 ImageNet 和 OOD 测试集上。

根据它们在图像中的位置，从左上到右下排列。我们只包含 OCR 字符串本身，而不包含 API 提供的 OCR 项位置。GIT2（Wang 等，2022a）在没有 OCR 输入的情况下表现出强劲性能，而 PaLI-17B 则显示出利用专门的 OCR 系统作为更好方案来解决这些任务的优越性。

表 20 展示了在不将检测到的 OCR 字符串作为输入的情况下，在 TextCaps、TextVQA 和 VizWiz-QA 上的结果。PaLI 在没有 OCR 输入时性能略有下降，但其表现仍接近第一版 GIT。该结果可能表明，PaLI 显著更大的词汇表进一步增加了 OCR 字符串生成的难度。

然而，对于 VizWiz-QA，PaLI 在没有 OCR 输入的情况下取得了 SOTA 性能。

C.6 详细 VTAB 结果

对于 VTAB 基准（Zhai 等，2019），我们遵循（Zhai 等，2022b）中概述的方法。PaLI 在“natural”子集上创下了新的零样本性能最新水平（见表 21）。

C.7 零样本 ImageNet 数据集上的 Top-5 准确率 TS

表 22 展示了在 ImageNet 数据集上进行零样本评估的 Top-5 准确率结果。

C.8 在 CROSSMODAL-3600 上更多零样本图像-文本检索结果

表23展示了在 Crossmodal-3600 上更多的零样本图像-文本检索结果。

D MOD EL 的公平性、偏见和其他潜在 I 问题

基于网络数据训练的模型可能会因这些数据中的偏见而存在偏向或不公平的风险。应对这些风险的第一步是透明地承认其存在，然后对其进行衡量。

Model	INet	INet-v2	INet-R	INet-A	ObjNet	ReaL	VTAB-N
CLIP (Radford et al., 2021)	76.2	70.1	88.9	77.2	72.3	-	73.9
ALIGN (Jia et al., 2021)	76.4	70.1	92.2	75.8	72.2	-	-
BASIC (Pham et al., 2021)	85.7	80.6	95.7	85.6	78.9	-	-
CoCa (Yu et al., 2022)	86.3	80.7	96.5	90.2	82.7	-	-
LiT ViT-g (Zhai et al., 2022b)	85.2	79.8	94.9	81.8	82.5	88.6	74.7
LiT ViT-e (ours)	85.4	80.6	96.1	88.0	84.9	88.4	76.9

Table 19: Zero-shot transfer results of ViT-e on ImageNet, OOD test sets and VTAB-Natural datasets.

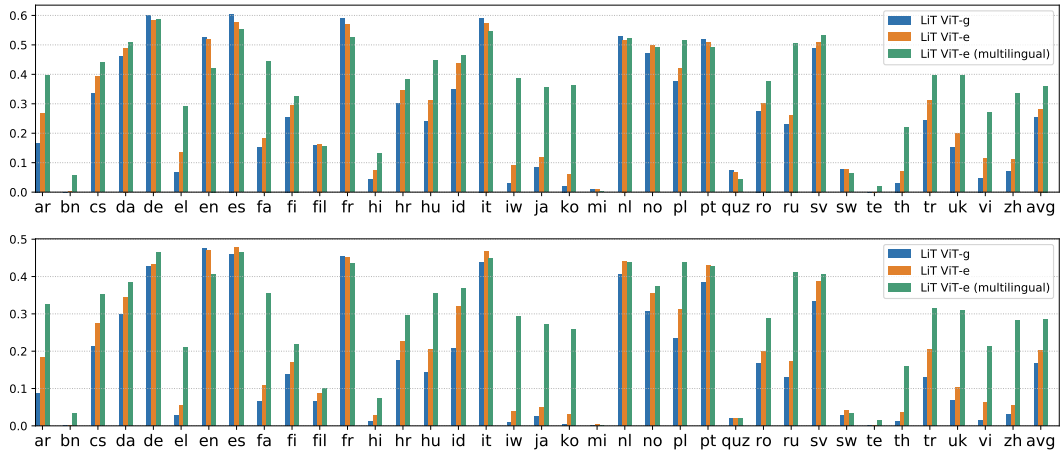


Figure 6: Zero-shot image-text retrieval results on all 36 languages of Crossmodal-3600. Top: image-to-text retrieval accuracy; bottom: text-to-image retrieval accuracy.

		TextCaps	TextVQA	VizWiz-QA	
Method	OCR input?	test	test	test-dev	test-std
TAP (Yang et al., 2021)	Yes	103.2	53.97	-	-
GIT	No	138.2	59.75	68.0	67.5
GIT2	No	145.0	67.27	71.0	70.1
PaLI	No	135.4	58.80	71.6	70.7
PaLI	Yes	160.4	73.06	74.4	73.3

Table 20: Results on TextCaps, TextVQA and VizWiz-QA with and without detected OCR as input for PaLI

	Caltech101	CIFAR-100	DTD	Flowers102	Pets	Sun397	SVHN	Mean
CLIP	**92.8**	77.5	55.7	78.3	93.5	68.4	**51.0**	73.9
LiT *ViT-g*	79.2	83.6	66.6	**92.3**	97.7	76.0	27.5	74.7
LiT *ViT-e*	79.8	**90.4**	**68.8**	91.2	**98.1**	**76.3**	33.8	**76.9**

Table 21: Accuracies for zero-shot evaluation of different VTAB "natural" tasks, and the average over these tasks. Note that CLIP is using OCR for the SVHN task (as opposed to LiT and PaLI, which do not use OCR).

To this end, we add a data card (Pushkarna et al., 2022) for WebLI and model card (Mitchell et al., 2019) for PaLI in Appendix G and F.

To understand the demographic properties of the data, we sample 112,782 (0.001% of the full data set, randomly sampled due to the limitations of the labeling tool, described next) examples and analyze both images and texts of the sampled data with the Know Your Data (KYD) tool. We use KYD to

Model	INet	INet-v2	INet-R	INet-A	ObjNet	ReaL	VTAB-N
CLIP (Radford et al., 2021)	76.2	70.1	88.9	77.2	72.3	-	73.9
ALIGN (Jia et al., 2021)	76.4	70.1	92.2	75.8	72.2	-	-
BASIC (Pham et al., 2021)	85.7	80.6	95.7	85.6	78.9	-	-
CoCa (Yu et al., 2022)	86.3	80.7	96.5	90.2	82.7	-	-
LiT ViT-g (Zhai et al., 2022b)	85.2	79.8	94.9	81.8	82.5	88.6	74.7
LiT ViT-e (ours)	85.4	80.6	96.1	88.0	84.9	88.4	76.9

表 19：ViT-e 在 ImageNet、OOD 测试集和 VTAB-Natural 数据集上的零样本迁移结果。

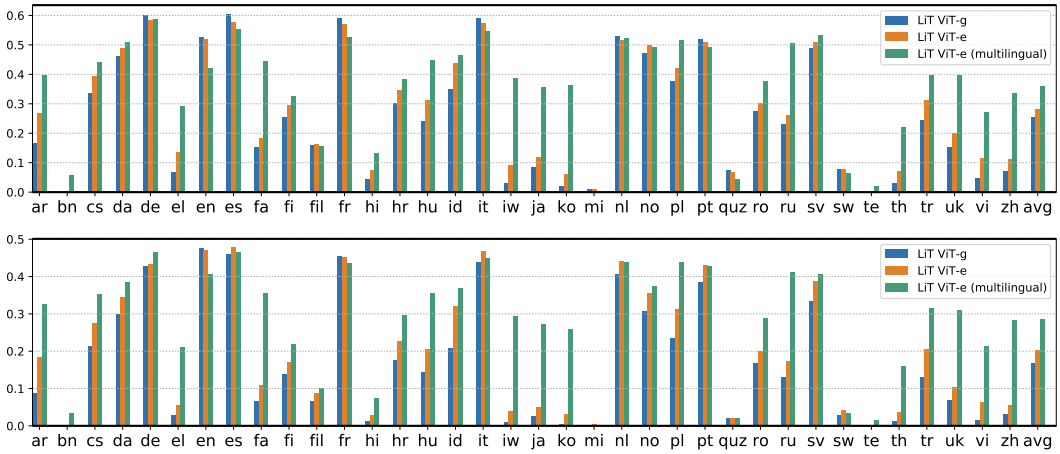


图 6：在 Crossmodal-3600 的全部 36 种语言上的零样本图文检索结果。上：图像到文本检索准确率；下：文本到图像检索准确率。

		TextCaps	TextVQA	VizWiz-QA	
Method	OCR input?	test	test	test-dev	test-std
TAP (Yang et al., 2021)	Yes	103.2	53.97	-	-
GIT	No	138.2	59.75	68.0	67.5
GIT2	No	145.0	67.27	71.0	70.1
PaLI	No	135.4	58.80	71.6	70.7
PaLI	Yes	160.4	73.06	74.4	73.3

表 20：PaLI 在 TextCaps、TextVQA 和 VizWiz-QA 上以检测到的 OCR 作为输入和不作为输入的结果

	Caltech101	CIFAR-100	DTD	Flowers102	Pets	Sun397	SVHN	Mean
CLIP	**92.8**	77.5	55.7	78.3	93.5	68.4	**51.0**	73.9
LiT *ViT-g*	79.2	83.6	66.6	**92.3**	97.7	76.0	27.5	74.7
LiT *ViT-e*	79.8	**90.4**	**68.8**	91.2	**98.1**	**76.3**	33.8	**76.9**

表 21：不同 VTAB “natural” 任务的零样本评估准确率，以及这些任务的平均值。请注意，CLIP 在 SVHN 任务中使用了 OCR（不同于 LiT 和 PaLI，它们不使用 OCR）。

为此，我们在附录 G 和 F 中分别为 WebLI 添加了数据卡（Pushkarna 等，2022），并为 PaLI 添加了模型卡（Mitchell 等，2019）。

为了解数据的人口统计属性，我们抽取了 112,782 个样本（占完整数据集的 0.001%，由于接下来将描述的标注工具限制而随机抽样），并使用 Know Your Data（KYD）工具分析抽样数据中的图像和文本。我们使用 KYD 来

Model	INet	INet-R	INet-A	INet-sketch	INet-v2	ObjNet
PaLI-3B	84.31	90.05	55.04	76.47	78.49	53.71
PaLI-15B	84.78	90.91	59.00	76.81	79.54	55.29
PaLI-17B	86.18	91.51	62.72	79.30	80.71	58.35

Table 22: Top 5 accuracy results of Zero-shot image classification on ImageNet (Deng et al., 2009), ImageNet-R (Hendrycks et al., 2021a), ImageNet-A (Hendrycks et al., 2021b), ImageNet-Sketch (Wang et al., 2019b), ImageNet-v2 (Recht et al., 2019) and ObjectNet (Barbu et al., 2019).

Language	Image-to-text			Text-to-image		
	LiT ViT-g	LiT ViT-e	LiT ViT-e (multilingual)	LiT ViT-g	LiT ViT-e	LiT ViT-e (multilingual)
ar	5.28	26.58	39.69	2.80	18.46	32.60
bn	0.00	0.11	5.67	0.00	0.06	3.31
cs	18.19	39.25	44.03	11.24	27.35	35.24
da	26.44	48.92	50.75	14.07	34.43	38.48
de	37.83	58.42	58.53	23.61	43.25	46.50
el	1.56	13.47	29.03	0.39	5.46	20.92
en	51.22	51.78	42.11	46.24	47.07	40.63
es	41.81	57.50	55.22	30.29	47.71	46.55
fa	3.78	18.39	44.50	1.57	10.74	35.58
fi	14.14	29.42	32.64	6.59	16.91	21.80
fil	10.94	16.39	15.53	4.18	8.66	10.04
fr	38.28	57.06	52.61	28.02	45.20	43.47
hi	0.47	7.33	13.14	0.08	2.90	7.42
hr	15.86	34.47	38.31	8.80	22.72	29.55
hu	15.11	31.17	44.67	8.45	20.52	35.49
id	24.11	43.72	46.33	12.99	32.08	36.75
it	39.69	57.47	54.53	27.07	46.79	44.76
iw	1.75	9.11	38.67	0.86	3.99	29.39
ja	3.61	11.67	35.47	1.20	4.91	27.24
ko	1.78	6.00	36.11	0.35	3.14	25.95
mi	0.58	0.92	0.33	0.19	0.30	0.22
nl	37.47	51.67	52.14	27.26	44.08	43.79
no	26.53	49.69	49.17	14.61	35.59	37.35
pl	19.67	42.03	51.42	12.00	31.13	43.72
pt	33.92	50.81	49.19	23.58	42.97	42.73
quz	5.08	6.83	4.31	1.85	1.89	1.90
ro	17.94	30.08	37.75	10.15	20.06	28.82
ru	12.00	26.22	50.64	5.76	17.19	41.11
sv	25.50	51.00	53.22	15.11	38.80	40.66
sw	4.47	7.75	6.42	1.58	4.17	3.41
te	0.06	0.03	1.92	0.03	0.03	1.42
th	1.89	7.22	22.00	0.79	3.71	16.06
tr	10.72	31.28	39.50	4.73	20.42	31.47
uk	7.67	19.94	39.53	3.38	10.40	30.81
vi	3.08	11.44	27.08	0.98	6.22	21.28
zh	4.53	11.11	33.61	1.67	5.60	28.24
avg	15.64	28.23	35.99	9.79	20.14	28.46

Table 23: Image-to-text and text-to-image zero-shot retrieval results on all 36 languages of Crossmodal-3600. Models are trained following LiT (Zhai et al., 2022b) method with diverse visual backbones (ViT-g or ViT-e) and datasets (English or multilingual).

analyze the perceived gender presentation of image subjects (Schumann et al., 2021) along with gender expressed through pronouns in text. In the sampled images, 54% of people appear feminine presenting with 46% masculine presenting. In the sampled text, female pronouns (e.g., she, her) are used 30% of the time, male pronouns (e.g., he, him) 38% of the time, and they or them (either singular or plural) 31% of the time. We also analyze the perceived age of individuals appearing in the sampled images, resulting in the distribution displayed in Figure 7.

We consider all the effort above a first step, and know that it will be important to continue to measure and mitigate bias as we apply our model to new tasks. Deeper analysis will include the study of the model's recognition capabilities and potential biases observed towards specific attributes, e.g. related to gender, age, etc. and how scaling affects these observations.

Model	INet	INet-R	INet-A	INet-sketch	INet-v2	ObjNet
PaLI-3B	84.31	90.05	55.04	76.47	78.49	53.71
PaLI-15B	84.78	90.91	59.00	76.81	79.54	55.29
PaLI-17B	86.18	91.51	62.72	79.30	80.71	58.35

表 22：在 ImageNet（Deng 等，2009）、ImageNet-R（Hendrycks 等，2021a）、ImageNet-A（Hendrycks 等，2021b）、ImageNet-Sketch（Wang 等，2019b）、ImageNet-v2（Recht 等，2019）和 ObjectNet（Barbu 等，2019）上进行零样本图像分类的前 5 个准确率结果。

Language	Image-to-text			Text-to-image		
	LiT ViT-g	LiT ViT-e	LiT ViT-e (multilingual)	LiT ViT-g	LiT ViT-e	LiT ViT-e (multilingual)
ar	5.28	26.58	39.69	2.80	18.46	32.60
bn	0.00	0.11	5.67	0.00	0.06	3.31
cs	18.19	39.25	44.03	11.24	27.35	35.24
da	26.44	48.92	50.75	14.07	34.43	38.48
de	37.83	58.42	58.53	23.61	43.25	46.50
el	1.56	13.47	29.03	0.39	5.46	20.92
en	51.22	51.78	42.11	46.24	47.07	40.63
es	41.81	57.50	55.22	30.29	47.71	46.55
fa	3.78	18.39	44.50	1.57	10.74	35.58
fi	14.14	29.42	32.64	6.59	16.91	21.80
fil	10.94	16.39	15.53	4.18	8.66	10.04
fr	38.28	57.06	52.61	28.02	45.20	43.47
hi	0.47	7.33	13.14	0.08	2.90	7.42
hr	15.86	34.47	38.31	8.80	22.72	29.55
hu	15.11	31.17	44.67	8.45	20.52	35.49
id	24.11	43.72	46.33	12.99	32.08	36.75
it	39.69	57.47	54.53	27.07	46.79	44.76
iw	1.75	9.11	38.67	0.86	3.99	29.39
ja	3.61	11.67	35.47	1.20	4.91	27.24
ko	1.78	6.00	36.11	0.35	3.14	25.95
mi	0.58	0.92	0.33	0.19	0.30	0.22
nl	37.47	51.67	52.14	27.26	44.08	43.79
no	26.53	49.69	49.17	14.61	35.59	37.35
pl	19.67	42.03	51.42	12.00	31.13	43.72
pt	33.92	50.81	49.19	23.58	42.97	42.73
quz	5.08	6.83	4.31	1.85	1.89	1.90
ro	17.94	30.08	37.75	10.15	20.06	28.82
ru	12.00	26.22	50.64	5.76	17.19	41.11
sv	25.50	51.00	53.22	15.11	38.80	40.66
sw	4.47	7.75	6.42	1.58	4.17	3.41
te	0.06	0.03	1.92	0.03	0.03	1.42
th	1.89	7.22	22.00	0.79	3.71	16.06
tr	10.72	31.28	39.50	4.73	20.42	31.47
uk	7.67	19.94	39.53	3.38	10.40	30.81
vi	3.08	11.44	27.08	0.98	6.22	21.28
zh	4.53	11.11	33.61	1.67	5.60	28.24
avg	15.64	28.23	35.99	9.79	20.14	28.46

表23：在 Crossmodal-3600 的全部 36 种语言上的图像到文本和文本到图像零样本检索结果。模型按照 LiT（Zhai 等，2022b）方法训练，采用不同的视觉骨干网络（ViT-g 或 ViT-e）和数据集（英语或多语言）。

分析图像主体的感知性别呈现（Schumann 等，2021），以及文本中通过代词表达的性别。在抽样图像中，54%的人呈现为女性化，46%呈现为男性化。在抽样文本中，女性代词（例如 she、her）的使用比例为 30%，男性代词（例如 he、him）的使用比例为 38%，they 或 them（单数或复数）使用比例为 31%。我们还分析了抽样图像中出现个体的感知年龄，其分布如图 7 所示。

我们将上述所有努力视为第一步，并认识到，随着我们将模型应用于新任务，持续衡量和缓解偏见将非常重要。更深入的分析将包括研究模型的识别能力以及针对特定属性（例如与性别、年龄等相关的属性）观察到的潜在偏见，以及规模扩展如何影响这些观察结果。

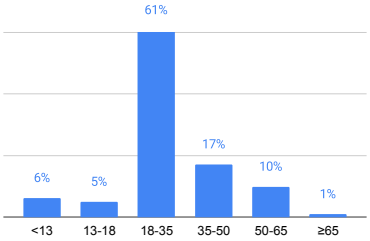


Figure 7: The distribution of ages recognized from the sampled images of WebLI.

E LIMITATIONS

Despite good performance, our model has a number of limitations. For example, the model might not describe very thoroughly a complex scene with many objects because most of the source data does not have complex annotations. We have tried to mitigate this with the object-aware and localization aware queries, added to the data.

We also noticed that some of the multilingual capabilities are lost when fine-tuned on English-only data, which is consistent with other model fine-tuning behavior. Ideally these models should be fine-tuned on a mix of multiple datasets including multilingual ones.

There are limitations related to the evaluation procedures of the benchmarks. Since we are evaluating in the open-vocabulary generative setting, for example in VQA, the model might generate a correct response which is a synonym or a paraphrase of the target response and does not match the target exactly. In these cases the answer is counted as incorrect. Fixed-vocabulary approaches do not suffer from these issues, but are limited in generalization beyond the answers of a specific dataset. Further, in terms of evaluation, some benchmarks might need more comprehensive strategies to avoid evaluations with Western-centric bias. Multilingual models and benchmarks are a first step in that direction.

F PALI MODEL CARD

Following Mitchell et al. (2019), we present the PaLI model card in Table 24.

Model Summary	
Model Architecture	PaLI is a multimodal sequence-to-sequence Transformer (Vaswani et al., 2017) model derived from the T5 (Raffel et al., 2020) encoder-decoder architecture. It takes text tokens and ViT (Dosovitskiy et al., 2021) dense image embeddings as inputs to an encoder and autoregressively predicts discrete text tokens with a decoder.
Input(s)	A pair of image and text.
Output(s)	Generated text.
Usage	
Application	The model is for research prototype and the current version is not available for the public.
Known Caveats	No.
System Type	
System Description	This is a standalone model.
Upstream Dependencies	No.
Downstream Dependencies	No.
Implementation Frameworks	

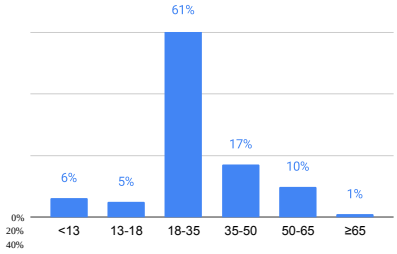


图 7：从 WebLI 抽样图像中识别出的年龄分布。

E 局限性

尽管性能良好，我们的模型仍存在一些局限性。例如，对于包含许多物体的复杂场景，模型可能无法非常详尽地描述，因为大多数源数据并没有复杂的标注。我们已尝试通过在数据中加入对象感知和定位感知查询来缓解这一问题。

我们还注意到，当仅使用英文数据进行微调时，部分多语言能力会丧失，这与其他模型的微调行为一致。理想情况下，这些模型应在包含多语言数据集在内的多个数据集混合数据上进行微调。

基准测试的评估流程存在一些局限性。由于我们是在开放词汇生成式设定下进行评估，例如在 VQA 中，模型可能会生成一个正确回答，该回答是目标回答的同义表达或改述，但与目标并不完全匹配。在这些情况下，答案会被判为错误。固定词汇方法不会受到这些问题的影响，但其泛化能力受限于特定数据集中的答案。此外，就评估而言，一些基准测试可能需要更全面的策略，以避免带有西方中心主义偏见的评估。多语言模型和基准测试是朝这一方向迈出的第一步。

F PALI 模型卡

Followi根据 Mitchell 等人（2019），我们在 T 中展示 PaLI 模型 能够 24。

Model Summary	
Model Architecture	PaLI is a multimodal sequence-to-sequence Transformer (Vaswani et al., 2017) model derived from the T5 (Raffel et al., 2020) encoder-decoder architecture. It takes text tokens and ViT (Dosovitskiy et al., 2021) dense image embeddings as inputs to an encoder and autoregressively predicts discrete text tokens with a decoder.
Input(s)	A pair of image and text.
Output(s)	Generated text.
Usage	
Application	The model is for research prototype and the current version is not available for the public.
Known Caveats	No.
System Type	
System Description	This is a standalone model.
Upstream Dependencies	No.
Downstream Dependencies	No.
Implementation Frameworks	

Hardware & Software	Hardware: TPU v4 (Jouppi et al., 2020). Software: T5X (Roberts et al., 2022), JAX (Bradbury et al., 2018), Flaxformer (Heek et al., 2020) Details are reported in Section A.1.
Compute Requirements	Reported in Section A.1.
Model Characteristics	
Model Initialization	The model is initialized from pre-trained language (mT5) (Xue et al., 2021) and Vision Transformer (ViT) (Zhai et al., 2022a; Dosovitskiy et al., 2021) checkpoints.
Model Status	This is a static model trained on an offline dataset.
Model Stats	The largest PaLI model has 17B parameters, which consists of a 13B parameter mT5-XXL model and a 4B parameter ViT-e model. We have also trained 3B and 15B parameter models.
Data Overview	
Training dataset	The model is pre-trained on the following mixture of datasets: WebLI (Table 25), CC3M-35L (Sharma et al., 2018), VQ^2A-CC3M-35L (Changpinyo et al., 2022a), Open Images (Kuznetsova et al., 2020), Visual Genome (Krishna et al., 2017) and Object365 (Shao et al., 2019). Details are reported in Section A.2.

Hardware & Software	Hardware: TPU v4 (Jouppi et al., 2020). Software: T5X (Roberts et al., 2022), JAX (Bradbury et al., 2018), Flaxformer (Heek et al., 2020) Details are reported in Section A.1.
Compute Requirements	Reported in Section A.1.
Model Characteristics	
Model Initialization	The model is initialized from pre-trained language (mT5) (Xue et al., 2021) and Vision Transformer (ViT) (Zhai et al., 2022a; Dosovitskiy et al., 2021) checkpoints.
Model Status	This is a static model trained on an offline dataset.
Model Stats	The largest PaLI model has 17B parameters, which consists of a 13B parameter mT5-XXL model and a 4B parameter ViT-e model. We have also trained 3B and 15B parameter models.
Data Overview	
Training dataset	The model is pre-trained on the following mixture of datasets: WebLI (Table 25), CC3M-35L (Sharma et al., 2018), VQ^2A-CC3M-35L (Changpinyo et al., 2022a), Open Images (Kuznetsova et al., 2020), Visual Genome (Krishna et al., 2017) and Object365 (Shao et al., 2019). Details are reported in Section A.2.

Evaluation and Fine-tuning Dataset	• **Vision + language tasks** – **Image captioning (English)**: COCO (Chen et al., 2015), NoCaps (Agrawal et al., 2019), TextCaps (Sidorov et al., 2020) – **Image captioning (multilingual)**: Crossmodal-3600 (Thapliyal et al., 2022) – **Visual question answering (English)**: VQAv2 (Goyal et al., 2017), OKVQA (Gui et al., 2021), TextVQA (Singh et al., 2019), VizWiz-QA (Gurari et al., 2018) – **Visual question answering (multilingual)**: xGQA (Pfeiffer et al., 2022), MaXM (Changpinyo et al., 2022b) • **Vision-only tasks** – **Image classification (fine-tuning)**: ImageNet (Deng et al., 2009), ImageNet-V2 (Recht et al., 2019), ObjectNet (Barbu et al., 2019), ReaL (Beyer et al., 2020) – **Image classification (zero-shot)**: ImageNet (Deng et al., 2009), ImageNet-V2 (Recht et al., 2019), ImageNet-R (Hendrycks et al., 2021a), ImageNet-A (Hendrycks et al., 2021b), ImageNet-Sketch (Wang et al., 2019b), ObjectNet (Barbu et al., 2019), ReaL (Beyer et al., 2020), VTAB (Zhai et al., 2019) • **Language-only tasks** – **Natural language inference (English)**: SuperGLUE (Wang et al., 2019a) – Natural language inference (multilingual): XNLI (Conneau et al., 2018) – **Question Answering (multilingual)**: XQuAD (Artetxe et al., 2020), TyDiQA (Clark et al., 2020)
Evaluation Results	
Evaluation Results	Reported in Section 4.
Model Usage & Limitations	
Sensitive Use	The model is capable of open-ended text generations. This model should not be used for any of the unacceptable language model use cases, e.g., generation of toxic speech.
Known Limitations	Reported in Section E.
Ethical Considerations & Risks	Reported in Section D.

Table 24: PaLI model card.

评估和微调数据集	• 视觉 + 语言任务 – 图像描述（英语）：COCO（Chen 等，2015）、NoCaps（Agrawal 等，2019）、TextCaps（Sidorov 等，2020）– 图像描述（多语言）：Crossmodal-3600（Thapliyal 等，2022）– 视觉问答（英语）：VQAv2（Goyal 等，2017）、OKVQA（Gui 等，2021）、TextVQA（Singh 等，2019）、VizWiz-QA（Gurari 等，2018）– 视觉问答（多语言）：xGQA（Pfeiffer 等，2022）、MaXM（Chang-pinyo 等，2022b） • 纯视觉任务 – 图像分类（微调）：Ima- geNet（Deng 等，2009）、ImageNet-V2（Recht 等，2019）、ObjectNet（Barbu 等，2019）、ReaL（Beyer 等，2020） – 图像分类（零样本）：Ima- geNet（Deng 等，2009）、ImageNet-V2（Recht 等，2019）、ImageNet-R（Hendrycks 等，2021a）、ImageNet-A（Hendrycks 等，2021b）、ImageNet-Sketch（Wang 等，2019b）、Object- Net（Barbu 等，2019）、ReaL（Beyer 等，2020）、VTAB（Zhai 等，2019） • 纯语言任务 – 自然语言推理（英语）：Super- GLUE（Wang 等，2019a）– 自然语言推理（多语言）：XNLI（Conneau 等，2018）– 问答（多语言）：XQuAD（Artetxe 等，2020），TyDiQA（Clark 等，2020）
Evaluation Results	
Evaluation Results	Reported in Section 4.
Model Usage & Limitations	
Sensitive Use	The model is capable of open-ended text generations. This model should not be used for any of the unacceptable language model use cases, e.g., generation of toxic speech.
Known Limitations	Reported in Section E.
Ethical Considerations & Risks	Reported in Section D.

表 24：PaLI 模型卡。

G WEBLI DATASHEET

Following Gebru et al. (2021), we present the WebLI datasheet in Table 25.

Motivation	
For what purpose was the dataset created? Who created the dataset? Who funded the creation of the dataset?	The dataset was created to support vision-language research, such as the large-scale pre-training for image understanding, image captioning, visual question answering, object detection etc.
Any other comments?	No user data is included in the data source. Personally identifiable and privileged data are filtered out during the dataset construction.
Composition	
What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)?	Each instance is presented as an image and associated texts (alt-text, page title and OCR) collected from the web.
How many instances are there in total (of each type, if appropriate)?	There are 9,624,017,440 instances in total (about 260 TB in size).
Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set?	The dataset is built from the public web pages. It is not a complete set but rather a subset of the publicly available image-text pairs.
What data does each instance consist of?	Each instance consists of 20+ features. Most features are from public web pages; a few are from publicly available automatic services. The primary features are image pixels and the associated texts, including alt-text, page title and OCR. Other features include rich image and page meta information (e.g. URL, MIME type) and filter signals (attached to alt-text only).
Is there a label or target associated with each instance?	No.
Is any information missing from individual instances?	No.
Are relationships between individual instances made explicit?	There are no relationships between individual instances.
Are there recommended data splits?	There is only one split containing all the instances of the dataset.
Are there any errors, sources of noise, or redundancies in the dataset?	The dataset is built from the web and only applied a few filters. The data is noisy and redundant images or texts may exist.
Is the dataset self-contained, or does it link to or otherwise rely on external resources?	The dataset is self-contained.
Does the dataset contain data that might be considered confidential?	No.

G WEBLI 数据表

继 Gebru 等人（2021）之后，我们在表 25 中展示了 WebLI 数据表。

Motivation	
For what purpose was the dataset created? Who created the dataset? Who funded the creation of the dataset?	The dataset was created to support vision-language research, such as the large-scale pre-training for image understanding, image captioning, visual question answering, object detection etc.
Any other comments?	No user data is included in the data source. Personally identifiable and privileged data are filtered out during the dataset construction.
Composition	
What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)?	Each instance is presented as an image and associated texts (alt-text, page title and OCR) collected from the web.
How many instances are there in total (of each type, if appropriate)?	There are 9,624,017,440 instances in total (about 260 TB in size).
Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set?	The dataset is built from the public web pages. It is not a complete set but rather a subset of the publicly available image-text pairs.
What data does each instance consist of?	Each instance consists of 20+ features. Most features are from public web pages; a few are from publicly available automatic services. The primary features are image pixels and the associated texts, including alt-text, page title and OCR. Other features include rich image and page meta information (e.g. URL, MIME type) and filter signals (attached to alt-text only).
Is there a label or target associated with each instance?	No.
Is any information missing from individual instances?	No.
Are relationships between individual instances made explicit?	There are no relationships between individual instances.
Are there recommended data splits?	There is only one split containing all the instances of the dataset.
Are there any errors, sources of noise, or redundancies in the dataset?	The dataset is built from the web and only applied a few filters. The data is noisy and redundant images or texts may exist.
Is the dataset self-contained, or does it link to or otherwise rely on external resources?	The dataset is self-contained.
Does the dataset contain data that might be considered confidential?	No.

Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?	The dataset likely contains data that might be considered offensive, insulting or threatening as the data is collected from the web. We use algorithmic methods and classifiers to remove sensitive / personal identifiable information (PII) / pornographic images.
Collection Process	
How was the data associated with each instance acquired?	Images, alt-text and meta information are from the public web. Text language identification and OCR annotation are done via publicly available automatic services.
What mechanisms or procedures were used to collect the data?	The data was collected using a variety of pipelines, software programs and publicly available automatic services to extract and filter images and texts.
If the dataset is a sample from a larger set, what was the sampling strategy?	The dataset is built from a subset of public web pages.
Over what timeframe was the data collected?	2021-2022
Were any ethical review processes conducted?	No.
Preprocessing, cleaning, and labeling	
Was any preprocessing, cleaning, or labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)?	The dataset is not annotated. Images which are identified as having adult content are excluded. Empty texts and texts (alt-text, page title and OCR) which are identified as PII are excluded. Images identified as having adult content, with improper shape, or with too many paired-texts are excluded.
Is the software used to preprocess, clean, or label the instances available?	No.
Uses	
Has the dataset been used for any tasks already?	Yes, we use the dataset for pre-training PaLI models.
Is there a repository that links to any or all papers or systems that use the dataset?	No.
What (other) tasks could the dataset be used for?	Vision-only tasks (image classification, object detection etc.), language-only tasks (question answering, natural language inference etc.) and vision+Language tasks (image captioning, visual question answering, image-text retrieval etc.).
Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?	The dataset is in a stable version and will be refreshed in the future to follow data policies.
Are there tasks for which the dataset should not be used?	The dataset should not be used for training any of the unacceptable vision, language or vision-language model use cases, e.g., generation of toxic captions or inappropriate images.
Distribution	

Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?	The dataset likely contains data that might be considered offensive, insulting or threatening as the data is collected from the web. We use algorithmic methods and classifiers to remove sensitive / personal identifiable information (PII) / pornographic images.
Collection Process	
How was the data associated with each instance acquired?	Images, alt-text and meta information are from the public web. Text language identification and OCR annotation are done via publicly available automatic services.
What mechanisms or procedures were used to collect the data?	The data was collected using a variety of pipelines, software programs and publicly available automatic services to extract and filter images and texts.
If the dataset is a sample from a larger set, what was the sampling strategy?	The dataset is built from a subset of public web pages.
Over what timeframe was the data collected?	2021-2022
Were any ethical review processes conducted?	No.
Preprocessing, cleaning, and labeling	
Was any preprocessing, cleaning, or labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)?	The dataset is not annotated. Images which are identified as having adult content are excluded. Empty texts and texts (alt-text, page title and OCR) which are identified as PII are excluded. Images identified as having adult content, with improper shape, or with too many paired-texts are excluded.
Is the software used to preprocess, clean, or label the instances available?	No.
Uses	
Has the dataset been used for any tasks already?	Yes, we use the dataset for pre-training PaLI models.
Is there a repository that links to any or all papers or systems that use the dataset?	No.
What (other) tasks could the dataset be used for?	Vision-only tasks (image classification, object detection etc.), language-only tasks (question answering, natural language inference etc.) and vision+Language tasks (image captioning, visual question answering, image-text retrieval etc.).
Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?	The dataset is in a stable version and will be refreshed in the future to follow data policies.
Are there tasks for which the dataset should not be used?	The dataset should not be used for training any of the unacceptable vision, language or vision-language model use cases, e.g., generation of toxic captions or inappropriate images.
Distribution	

Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created?	No.

Table 25: WebLI datasheet.

Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created?	No.

Table 25: WebLI datasheet.